

MATHEMATICS



Resonance
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TARGET: JEE (ADVANCED) 2014

MEDIUM : ENGLISH

CRITICAL QUESTIONS BANK

SUBJECT : MATHEMATICS

MEDIUM : ENGLISH

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QUESTION FORMAT & MARKING CRITERIA

SUBJECT : MATHEMATICS

A. Questions Format

In the booklet check that it contains all the **180 questions** and corresponding answer choices are legible. Read carefully the Instructions printed at the beginning of each section.

1. **Section 1** contains **53 multiple choice questions**. Each question has Four choices (A), (B), (C) and (D) out of which **only ONE is correct**.
2. **Section 2** contains **37 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.
3. **Section 3** contains **2 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **Only ONE is correct**.
4. **Section 4** contains **10 paragraphs** each describing theory, experiment, data etc. **Twenty questions** related to Ten paragraphs with one or two or three questions on each paragraph. Each question of a paragraph has **ONLY ONE correct answer** among the four choices (A), (B), (C) and (D).
5. **Section 5** contains **4 question**. Each question contains statements given in two columns which have to be matched. Statements in **Column I** are labelled as A,B,C and D whereas statements in **Column II** are labelled as p,q,r,s and t. The answers to these questions have to be appropriately bubbled as illustrated in the following example.
6. **Section 6** contains **1 multiple choice questions**. Each questions has matching lists. The codes for the lists have coices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.
7. **Section 7** contains **43 questions**. The answer to each question is a single-digit integer, ranging from 0 to 9 (both inclusive).
8. **Section 8** contains **17 questions**. The answer to each question is a double-digit integer, ranging from 00 to 99 (both inclusive).

B. Marking Scheme

9. For each question in **Section 1**, you will be awarded **3 marks** if you darken the bubble corresponding to only the correct answer and **zero mark** if no bubbles are darkened. In all other cases, **minus one (–1) mark** will be awarded.
10. For each question in **Section 2**, you will be awarded **4 marks** if you darken the bubble(s) corresponding to only the correct answer and **zero mark** if no bubbles are darkened. **No negative** marks will be awarded for incorrect answers in this section.
11. For each question in **Section 3**, you will be awarded **4 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened.
12. For each question in **Section 4** contains **10 paragraphs** each describing theory, experiment, data etc. **Twenty questions** related to Ten paragraphs with one or two or three questions on each paragraph. Each question of a paragraph has **ONLY ONE correct answer** among the four choices (A), (B), (C) and (D).
13. For each question in **Section–5**, you will be **awarded 2 marks** for each row in which you have darkened the bubble corresponding to the correct answer. Thus, each question in this section carries a maximum of **8 marks**. There is **no negative marking** for incorrect answer(s) in this section.
14. For each question in **Section 6**, you will be awarded **3 marks** if you darken **all** the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. In all other cases, **minus one (–1) mark** will be awarded.
15. For each question in **Section 7**, you will be awarded **4 marks** if you darken the bubble corresponding to only the correct answer and **zero mark** if no bubbles are darkened. **No negative** marks will be awarded for incorrect answers in this section.
16. For each question in **Section 8**, you will be awarded **4 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and zero mark if no bubbles are darkened. **No negative** marks will be awarded for incorrect answers in this section.

SECTION – 1 : (Only One option correct Type)

This section contains 53 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

- Let $f(x) = \sin^2(x + \alpha) + \sin^2(x + \beta) - 2\cos(\alpha - \beta) \sin(x + \alpha) \sin(x + \beta)$. Which of the following is **TRUE**?
 (A) $f(x)$ is strictly increasing in $x \in (\alpha, \beta)$
 (B) $f(x)$ is strictly decreasing in $x \in (\alpha, \beta)$
 (C) $f(x)$ is strictly increasing in $x \in \left(\alpha, \frac{\alpha + \beta}{2}\right)$ and strictly decreasing in $x \in \left(\frac{\alpha + \beta}{2}, \beta\right)$
 (D) $f(x)$ is a constant function
- If the roots of the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ (where a, b, c are unequal real numbers) are real and equal and α, β be the roots of equation $ax^2 + bx + c = 0, a \neq 0$ then harmonic mean of α, β is
 (A) $1 - \alpha\beta$ (B) $1 + \alpha\beta$ (C) $\alpha\beta - 1$ (D) $-1 - \alpha\beta$
- If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ are two matrices such that $AB = BA$ and $c \neq 0$, then value of $\frac{3a - 3d}{3b + c}$ is
 (A) 0 (B) 2 (C) -2 (D) -1
- $\lim_{x \rightarrow 0^+} \frac{e^{(x^x - 1)} - x^x}{((x^x)^x - 1)^2} =$
 (A) 1 (B) $\frac{1}{8}$ (C) -1 (D) does not exist
- If the equation $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ represent three straight lines, two of them are perpendicular then sum of all possible values of a is
 (A) $-\frac{3}{4}$ (B) 5 (C) $\frac{4}{5}$ (D) -3
- If $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$ terms of an AP are P, Q, R respectively, then which of the following must be **CORRECT**?
 (A) $pQ + qR + rP = pR + rQ + qP$
 (B) $(p + q + r)^{\text{th}}$ term will be $P + Q + R$.
 (C) If $p < q < r$, then $P < Q < R$
 (D) If $P, Q, R \in \mathbb{N}$ then common difference of AP will be an integer.
- The set of values of 'a' for which $x^2 + ax + \sin^{-1}(x^2 - 4x + 5) + \cos^{-1}(x^2 - 4x + 5) = 0$ has at least one solution is
 (A) $(-\infty, -\sqrt{2\pi}] \cup [\sqrt{2\pi}, \infty)$ (B) $(-\infty, -\sqrt{2\pi}) \cup (\sqrt{2\pi}, \infty)$
 (C) $\mathbb{R}$ (D) $-\left(\frac{\pi + 8}{4}\right)$
- Let $\alpha = \log_{10} 15$ and $\beta = \log_{10} 16$, a set $A = \{\log_{10} 1, \log_{10} 2, \dots, \log_{10} 50\}$ the number of elements in set A which can be written in the form of $a\alpha + b\beta + c$ where a, b, c are rational numbers, is
 (A) 23 (B) 24 (C) 25 (D) none of these
- If both the roots of the equation $x^2 + (3 - 2k)x - 6k = 0$ belongs to the interval $(-6, 10)$, then largest value of k is
 (A) -1 (B) 3 (C) 5 (D) none of these

10. The domain of the function $f(x) = \sqrt{\frac{(2x-1)}{(2x^3+3x^2+x)}} + \sqrt{\sin^{-1}(\log_2 x)}$ is
 (A) $\left(\frac{1}{2}, \infty\right)$ (B) $\left[\frac{1}{2}, 2\right]$ (C) $[1, 2]$ (D) $(1, \infty)$
11. Let G, S, I be respectively centroid, circumcentre, incentre of triangle ABC. If R, r are circumradius and inradius respectively then which of the following is **INCORRECT** ?
 (A) $SI^2 = R^2(1 - \cos A \cos B \cos C)$; A, B, C being angles of triangle.
 (B) $SI^2 = R^2 - 2Rr$
 (C) $SG^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$; a, b, c being sides of triangle.
 (D) $SG \leq SI$
12. Two circles are given as $x^2 + y^2 + 14x - 6y + 40 = 0$ and $x^2 + y^2 - 2x + 6y + 7 = 0$ with their centres as C_1 and C_2 . If equation of another circle whose centre C_3 lies on the line $3x + 4y - 16 = 0$ and touches the circle C_1 externally and also $C_1C_2 + C_2C_3 + C_3C_1$ is minimum, is $x^2 + y^2 + ax + by + c = 0$ then the value of $(a + b + c)$ is
 (A) 2 (B) 0 (C) 16 (D) None of these
13. If $g(x) = \begin{cases} [f(x)], & x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right) \\ 3, & x = \pi/2 \end{cases}$ where $[x]$ denotes the greatest integer function and
 $f(x) = \frac{2(\sin x - \sin^n x) + |\sin x - \sin^n x|}{2(\sin x - \sin^n x) - |\sin x - \sin^n x|}, n \in \mathbb{R} - \{1\}$, then
 (A) $g(x)$ is continuous and differentiable at $x = \pi/2$, when $0 < n < 1$
 (B) $g(x)$ is continuous and differentiable at $x = \pi/2$, when $n > 1$
 (C) $g(x)$ is continuous but not differentiable at $x = \pi/2$, when $0 < n < 1$
 (D) $g(x)$ is neither continuous nor differentiable, at $x = \pi/2$, when $n > 1$
14. If $\int \frac{\sec^2 x - 2010}{\sin^{2010} x} dx = \frac{P(x)}{(\sin x)^{2010}} + C$, then value of $P\left(\frac{\pi}{3}\right)$ is
 (A) 0 (B) $\frac{1}{\sqrt{3}}$ (C) $\sqrt{3}$ (D) $\frac{3\sqrt{3}}{2}$
15. If $\hat{i} + \hat{j}$ bisects the angle between $\hat{c}$ and $\hat{j} + \hat{k}$, then value of $\hat{c} \cdot \hat{j}$ is equal to
 (A) 0 (B) $\frac{1}{\sqrt{2}}$ (C) $-\frac{1}{\sqrt{2}}$ (D) 1
16. If A and B are two square matrices such that $B = -A^{-1}BA$, then $(A + B)^2$ is equal to
 (A) O (B) $A^2 + B^2$ (C) $A^2 + 2AB + B^2$ (D) $A + B$
17. The value of θ where $\theta = \sin^{-1} \sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1} \frac{\sqrt{12}}{4} + \sec^{-1}(\sqrt{2})$ is equal to
 (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$
18. $f(x) = \begin{cases} x, & x \in \mathbb{Q} \\ \cos x, & x \in \mathbb{Q}' \end{cases}$. The number of points of continuity of $f(x)$ is
 (A) 0 (B) 1 (C) 2 (D) infinitely many

19. Let $f(x) = \int_x^2 \frac{dy}{\sqrt{1+y^3}}$. The value of the integral $\int_0^2 xf(x) dx$ is equal to
 (A) 1 (B) $\frac{4}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$
20. Let P be a point on ellipse $4x^2 + y^2 = 8$ with eccentric angle $\frac{\pi}{4}$. If tangent at P intersects the x-axis at A and y-axis at B and normal at P intersect the x-axis at A' and y-axis at B'. The ratio of area of triangle APA' to area of triangle BPB' is
 (A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 4 : 1
21. $\int \frac{\sqrt{1-\sin x}}{1+\cos x} e^{-x/2} dx, x \in \left(\frac{5\pi}{2}, \frac{9\pi}{2}\right)$
 (A) $e^{-x/2} \sin \frac{x}{2} + C$ (B) $-e^{-x/2} \sec \frac{x}{2} + C$ (C) $-e^{-x/2} \sin \frac{x}{2} + C$ (D) $-e^{-x/2} \cos \frac{x}{2} + C$
22. $\lim_{x \rightarrow 0^-} \frac{e^{[x]+|x|} - 2}{[x] + |x|} =$ (where [.] denotes greatest integer function)
 (A) $2 - \frac{1}{e}$ (B) -1 (C) $2 - e$ (D) $\frac{1}{e} - 2$
23. If $\alpha = \tan^{-1}(1) + \frac{1}{2}\tan^{-1}(2) + \frac{1}{3}\tan^{-1}(3)$, $\beta = \tan^{-1}(1) + 2\tan^{-1}\left(\frac{1}{2}\right) + 3\tan^{-1}\left(\frac{1}{3}\right)$ and
 $|\alpha - \beta| = \frac{p\pi}{q} + \frac{r}{s}\cot^{-1}(3)$,
 where p, q, r, s $\in \mathbb{N}$ and are in their lowest form then which of the following is **INCORRECT** ?
 (A) $p - r = 0$ (B) $q = 4s$ (C) $p + q + r + s = 42$ (D) $pr = 1 + q$
24. The length of sub-normal at any point P(x,y) on the curve, which (curve) is passing through Q(0, 1) is unity. The area bounded by the curves satisfying this condition is equal to
 (A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{1}{3}$ (D) $\frac{8}{3}$
25. Let $f(x) = x + x^2 + x^4 + x^8 + x^{16} + x^{32} + \dots$ the coefficient of x^{10} in $f(f(x))$ is
 (A) 28 (B) 40 (C) 52 (D) none of these
26. If $1, \alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{2008}$ are $(2009)^{\text{th}}$ roots of unity, then the value of $\sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$ is equal to
 (A) 2009 (B) 2008 (C) 0 (D) -2009
27. Let $\lim_{x \rightarrow \frac{\pi}{2}} \frac{(\cos x)^\alpha \cdot \sin(\sec x)}{\left(x - \frac{\pi}{2}\right)^2} = \beta$, where $\alpha, \beta \in \mathbb{R}$
 (A) $\alpha = 2, \beta = 0$ (B) $\alpha = 0, \beta = 2$ (C) $\alpha = \frac{7}{3}, \beta = 0$ (D) $\alpha = \frac{5}{3}, \beta = 0$

28. Let $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ be vertices of triangle ABC with $BC = a$, $AB = c$, $AC = b$. If algebraic sum of perpendicular distances from $L\left(\frac{3ax_1}{a+b+c}, \frac{3ay_1}{a+b+c}\right)$, $M\left(\frac{3bx_2}{a+b+c}, \frac{3by_2}{a+b+c}\right)$, $N\left(\frac{3cx_3}{a+b+c}, \frac{3cy_3}{a+b+c}\right)$ to a variable line is zero then all such lines passes through
 (A) orthocentre of $\triangle ABC$ (B) centroid of $\triangle ABC$
 (C) circumcentre of $\triangle ABC$ (D) incentre of $\triangle ABC$
29. If $|f''(x)| \leq 1 \forall x \in \mathbb{R}$ and $f(0) = 0 = f'(0)$ then which of the following can be **TRUE**?
 (A) $f\left(\frac{1}{3}\right) = \frac{1}{5}$ (B) $f\left(-\frac{1}{3}\right) = \frac{1}{4}$ (C) $f(-3) = \frac{1}{3}$ (D) none of these
30. If $f(x) = \cos 8\pi\{x\} + \sin 2\pi x \operatorname{cosec} 2\pi x$ (where $\{.\}$ represents fractional part function), then fundamental period of $f(x)$ is
 (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{7}{8}$
31. The principle argument of $z = x + iy$, if it lies in second quadrant is equal to
 (A) $\frac{\pi}{2} + \tan^{-1}\left|\frac{y}{x}\right|$ (B) $\pi - \tan^{-1}\left|\frac{y}{x}\right|$ (C) $\pi + \tan^{-1}\left|\frac{y}{x}\right|$ (D) $\frac{\pi}{2} - \tan^{-1}\left|\frac{y}{x}\right|$
32. If the function $f(x) = \begin{cases} x^2\{e^{1/x}\} & , x \neq 0 \\ k & , x = 0 \end{cases}$ where $\{.\}$ denotes fractional part function, is continuous at $x = 0$, then
 (A) $k = 1$ (B) $f(x)$ is non-derivable at $x = 0$
 (C) $f(x)$ is derivable at $x = 0$ (D) $f(x)$ is continuous at every point in its domain.
33. $f(x) = -x^2 + 1$, $g(x) = -\sqrt[3]{x}$. Then $(g \circ f \circ g \circ f \circ g \circ f)(x)$ is
 (A) odd function (B) even function (C) polynomial function (D) identity function
34. If $a = 2$, then sum of series $\cot^{-1}(2a^{-1} + a) + \cot^{-1}(2a^{-1} + 3a) + \cot^{-1}(2a^{-1} + 6a) + \cot^{-1}(2a^{-1} + 10a) + \dots$ upto infinite terms, is
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$
35. The normal to the curve $5x^5 - 10x^3 + x + 2y + 6 = 0$ at $P(0, -3)$ meets the curve again at two points, then the equations of the tangents to the curve at these points is/are
 (A) $2x - y - 3 = 0$ (B) $2x + y - 3 = 0$ (C) $2x + y + 3 = 0$ (D) $2x - y + 3 = 0$
36. If $f(x + y + 1) = (\sqrt{f(x)} + \sqrt{f(y)})^2$ and $f(0) = 1, \forall x, y \in \mathbb{R}$ then $f(x)$ can be
 (A) $1 - x^2$ (B) $1 - x$ (C) $(x + 1)^2$ (D) $x^2 - 1$
37. $f(x) = \sqrt{\cos^{-1}(e^{2x} + \sin^{-1} x)}$
 (A) domain of f is $[-1, 1]$ (B) f is injective
 (C) range of f is $[0, 5\pi]$ (D) $f^{-1}(x) = f(x)$ have no solution.
38. A normal chord drawn to a parabola $y^2 = 4x$ at any point P intersects the parabola again at a point Q , then the minimum distance of Q from the vertex of the parabola is equal to
 (A) $4\sqrt{2}$ (B) $4\sqrt{3}$ (C) $4\sqrt{5}$ (D) $4\sqrt{6}$
39. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x = 2$ and a local minimum at $x = 4$. If $p(2) = 8$, $p(4) = 1$, then $p'(0)$ is
 (A) $\frac{168}{5}$ (B) 42 (C) 43 (D) 45

40. Let $f(x) = (3x + 2)^2 - 1$, $-\infty < x \leq -\frac{2}{3}$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to line $y = x$, then $g(x)$ equals to
- (A) $\frac{1}{3}(-2 + \sqrt{x+1})$, $x \geq -1$ (B) $\frac{1}{3}(-2 - \sqrt{x+1})$, $x \geq -1$
 (C) $\frac{1}{3}(-1 - \sqrt{x+2})$, $x \geq -2$ (D) $\frac{1}{3}(-1 + \sqrt{x+2})$, $x \geq -2$
41. $\cot^{-1}\left(\frac{9}{2}\right) + \cot^{-1}\left(\frac{33}{4}\right) + \cot^{-1}\left(\frac{129}{8}\right) \dots \infty$ is
- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{4}$ (D) none of these
42. The sum $\binom{100}{98} + \binom{99}{97} + \binom{98}{96} + \dots + \binom{3}{1} + \binom{2}{0}$ is equal to, where $\binom{n}{r} = {}^nC_r$
- (A) $\binom{100}{98}$ (B) $\binom{101}{99}$ (C) $\binom{100}{99}$ (D) $\binom{101}{98}$
43. A circle with centre $(3\alpha, 3\beta)$ and of variable radius cuts the hyperbola $x^2 - y^2 = 36$ at the points P, Q, R and S. If the locus of centroid of ΔPQR is $(x - 2\alpha)^2 - (y - 2\beta)^2 = \lambda$, then the value of λ is
- (A) 3 (B) -2 (C) 4 (D) -3
44. Let $f(x) = \begin{cases} 4x - x^3 + \ln(b^2 - 3b + 3), & 2 \leq x < 3 \\ x - 18, & x \geq 3 \end{cases}$. Find all the possible real values of b such that $f(x)$ has the smallest value at $x = 3$.
- (A) $(-\infty, 2] \cup [3, \infty)$ (B) $(-\infty, 1] \cup [2, \infty)$ (C) $(1, 2]$ (D) $(-\infty, 2]$
45. The solution set of the inequality $(\operatorname{cosec}^{-1}x)^2 - 2(\operatorname{cosec}^{-1}x) \geq \frac{\pi}{6} \operatorname{cosec}^{-1}x - \frac{\pi}{3}$ is $(-\infty, a] \cup [b, \infty)$, then $(a + b)$ is equal to
- (A) 0 (B) 2 (C) -3 (D) 1
46. $\arg\left(\frac{1 + Z_1 + Z_2 + Z_3 + \dots + Z_7}{1 + Z_8 + Z_9 + Z_{10} + \dots + Z_{14}}\right) =$
- (A) $\frac{1}{2} \arg(Z_1 Z_2 \dots Z_7)$ (B) $\arg(Z_1 Z_2 \dots Z_7)$
 (C) $\frac{1}{4} \arg(Z_1 Z_2 \dots Z_7)$ (D) $\frac{1}{8} \arg(Z_1 Z_2 \dots Z_7)$
47. Image of the curve $xy = 1$ in the curve $\left(x + \sqrt{x^2 + 1}\right)\left(y + \sqrt{y^2 + 1}\right) = 1$ (on coordinate plane) is
- (A) $xy = 1$ (B) $xy + 1 = 0$ (C) $xy = 0$ (D) $x^2 + y^2 = 1$
48. $|1 + Z_1 + Z_2 + \dots + Z_7| =$
- (A) $\sec\left(\frac{7\pi}{15}\right)$ (B) $\frac{1}{2} \sec\left(\frac{7\pi}{15}\right)$ (C) $2 \sec\left(\frac{7\pi}{15}\right)$ (D) None of these
49. Let O be centre, S, S' be foci of hyperbola. If tangent at any point P on hyperbola cuts asymptotes at M and N then $OM + ON =$
- (A) $|SP - S'P|$ (B) $SP + S'P$ (C) SS' (D) distance between vertices

50. DABC be a tetrahedron such that AD is perpendicular to the base ABC and $\angle ABC = 30^\circ$. The volume of tetrahedron is 18. If value of $AB + BC + AD$ is minimum then length of AC is
 (A) $3(\sqrt{6} - \sqrt{3})$ (B) $3(\sqrt{6} - \sqrt{2})$ (C) $2(\sqrt{6} - \sqrt{3})$ (D) $2(\sqrt{6} - \sqrt{2})$
51. The solution set of values of x satisfying equation $2\cot^{-1}x + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \pi$, is
 (A) all real numbers (B) $(-\infty, 0]$ (C) $[0, \infty)$ (D) $(-\infty, -1) \cup (1, \infty)$
52. If $g(x) = 2f(2x^3 - 3x^2) + f(6x^2 - 4x^3 - 3)$, $\forall x \in \mathbb{R}$ and $f''(x) > 0$, $\forall x \in \mathbb{R}$, then $g(x)$ is increasing on the interval
 (A) $\left(-\infty, -\frac{1}{2}\right) \cup (0, 1)$ (B) $\left(-\frac{1}{2}, 0\right) \cup (1, \infty)$ (C) $(0, \infty)$ (D) $(-\infty, 1)$
53. Let a circle with centre at C be made to pass through the point P(1,2), touching the straight lines $7x - y = 5$ and $x + y + 13 = 0$ at A and B respectively. If tangents at A and B intersect at Q, then
 (A) area of quadrilateral ACBQ can be 100 sq. units (B) radius of circle can be $\sqrt{40}$
 (C) area of quadrilateral ACBQ can be 200 sq. units (D) radius of circle can be 10

SECTION - 2 : (One or more options correct Type)

This section contains 37 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE** or **MORE** are correct.

54. Consider the locus of the complex number z in the Argand plane is given by $\operatorname{Re}(z) - 2 = |z - 7 + 2i|$. Let $P(z_1)$ and $Q(z_2)$ be two complex number satisfying the given locus and also satisfying
 $\arg\left(\frac{z_1 - (2 + \alpha i)}{z_2 - (2 + \alpha i)}\right) = \frac{\pi}{2}$ ($\alpha \in \mathbb{R}$) then the minimum value of PQ is divisible by
 (A) 3 (B) 5 (C) 7 (D) 2
55. Let $f(x^2 + y) = (f(x))^2 + f(y)$ for all $x, y \in \mathbb{R}$, then
 (A) $f(x)$ is odd
 (B) $f(x)$ is even $\Rightarrow f(x) = 0 \forall x \in \mathbb{R}$
 (C) $f(x)$ is continuous at $x = 0 \Rightarrow$ it is continuous everywhere
 (D) $f(x)$ is differentiable at $x = 0 \Rightarrow f(x) = x f'(0)$, $\forall x \in \mathbb{R}$
56. Let $f(x) = x^3 + px^2 + qx + r$, where p, q and r are integers, $f(0)$ and $f(-1)$ are odd integers. Which of the following is/are CORRECT?
 (A) $f(1)$ is an even integer (B) $f(1)$ is an odd integer
 (C) $f(x) = 0$ has three distinct integer roots (D) $f(x) = 0$ cannot have three integer roots.
57. Let $a, b, c \in \mathbb{R}$. If $ax^2 + bx + c = 0$ have one root < -1 and other root > 1 , then
 (A) $1 - \frac{b}{a} + \frac{c}{a} > 0$ (B) $1 + \frac{b}{a} + \frac{c}{a} < 0$ (C) $1 - \left|\frac{b}{a}\right| + \frac{c}{a} > 0$ (D) $1 - \left|\frac{b}{a}\right| + \frac{c}{a} < 0$
58. The value(s) of x satisfying $1 - \log_9(x+1)^2 = \frac{1}{2} \log_{\sqrt{3}}\left(\frac{x+5}{x+3}\right)$ is/are
 (A) 1 (B) -2 (C) -7 (D) -4
59. If $g(x) = \lim_{m \rightarrow \infty} \frac{x^m f(x) + h(x) + 3}{2x^m + 4x + 1}$ when $x \neq 1$ and $g(1) = e^3$ such that $f(x)$, $g(x)$ and $h(x)$ are continuous functions at $x = 1$, then
 (A) $f(1) = 2e^3$ (B) $h(1) = 5e^3 - 3$ (C) $f(1) + h(1) = 7e^3 + 5$ (D) $f(1) - h(1) = 7e^3 + 5$

60. A parabola C whose focus is S(0,0) and passing through P(3, 4). Equation of tangent at P to parabola is $3x + 4y - 25 = 0$. A chord through S parallel to tangent at P intersects the parabola at A and B. Which of the following are **CORRECT** ?
 (A) Length of AB is 20 units
 (B) Latus rectum of parabola is 20 units
 (C) Only one real normal can be drawn from the point $(-3, -4)$
 (D) Only one real normal can be drawn from the point $(-6, -8)$
61. If $\sqrt{4 + \sqrt{8 - \sqrt{32 + \sqrt{768}}}} = a\sqrt{2} \cos\left(\frac{11\pi}{b}\right)$, where a and b are natural numbers then $\frac{b}{a}$ is divisible by
 (A) 2 (B) 4 (C) 3 (D) 6
62. For the function $f(x) = (x^2 + bx + c)e^x$ and $g(x) = (x^2 + bx + c)e^x + e^x(2x + b)$, which of the following holds good ?
 (A) $f(x) > 0$ for all real $x \not\Rightarrow g(x) > 0$ (B) $f(x) > 0$ for all real $x \Rightarrow g(x) > 0$
 (C) $g(x) > 0$ for all real $x \Rightarrow f(x) > 0$ (D) $g(x) > 0$ for all real $x \not\Rightarrow f(x) > 0$
63. If the equation $\left(2^{\frac{1}{\cos^{-1}x}}\right)^{2\pi} - \left(a + \frac{1}{2}\right)\left(2^{\frac{1}{\cos^{-1}x}}\right)^{\pi} - a^2 = 0$ has only one real solution then subsets of values of 'a' are
 (A) $(-3, 1)$ (B) $(-\infty, -3]$ (C) $[1, \infty)$ (D) $[-3, \infty)$
64. $f(x) = \begin{cases} x+2, & x \leq 2 \\ x^2, & 2 < x \leq 3 \\ x^3 - 18, & x > 3 \end{cases}$
 (A) $f(x)$ is continuous in R. (B) $f(x)$ is discontinuous at 'a' for some $a \in R$.
 (C) $f(x)$ is continuous at infinitely many real x. (D) $f(x)$ is discontinuous at infinitely many real x.
65. Which of the following must hold good for the expansion of the binomial $\left(x^4 + \frac{1}{x^3}\right)^{15}$?
 (A) There exist a term which is independent of x.
 (B) 8th and 9th terms of the expansion have the greatest binomial coefficient
 (C) Coefficients of x^{32} and x^{-17} are equal
 (D) If $x = \sqrt{2}$, then number of rational terms in the expansion is 5
66. Which of the following is NOT in the solution set of the inequality $|x^2 - 1| + |2x^2 - 3| < |2 - x^2|$?
 (A) $(-1, 1)$ (B) $(-2, 2)$ (C) $\left[-1, \frac{1}{2}\right)$ (D) $(3, \infty)$
67. Let a function $f(x)$, $x \neq 0$ be such that $f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right)$ then $f(x)$ can be :
 (A) $1 - x^{2013}$ (B) $\sqrt{|x|} + 1$
 (C) $\frac{\pi}{2 \tan^{-1}|x|}$ (D) $\frac{2}{1 + k \ln|x|}$, k being arbitrary constant
68. TP and TQ are tangents to parabola $y^2 = 8x$ and normals at P and Q intersect at a point R on the parabola. The locus of circumcentre of $\triangle TPQ$ is a parabola whose -
 (A) Vertex is (2, 0) (B) foot of perpendicular from focus on the directrix is $\left(\frac{7}{4}, 0\right)$
 (C) length of latus rectum is 1 (D) focus is $\left(\frac{9}{4}, 0\right)$

69. The function $f(x) = 2|x| + |x + 1| - ||x - 3| - 3|x||$ has a local minimum or a local maximum at $x =$
- (A) 0 (B) $-\frac{3}{2}$ (C) $\frac{3}{4}$ (D) 3
70. If z_1, z_2, z_3, z_4 are roots of the equation $a_0z^4 + a_1z^3 + a_2z^2 + a_3z + a_4 = 0$, where a_0, a_1, a_2, a_3 and a_4 are real, then
- (A) $\bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4$ are also roots of the equation
 (B) z_1 is equal to at least one of $\bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4$
 (C) $-\bar{z}_1, -\bar{z}_2, -\bar{z}_3, -\bar{z}_4$ are also roots of the equation
 (D) None of the above.
71. $\lim_{x \rightarrow 3} \frac{\sin(e^{x-3} - 1)}{\ln(x - 2)} = c$, which of the following is/are **CORRECT** ?
- (A) $\sin^{-1}c = -\frac{\pi}{2}$ (B) $c = 0$
 (C) c is a positive integer (D) c is neither prime nor composite
72. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{3}$ and $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = \lambda$. Which of the following are **CORRECT** ?
- (A) The maximum value of λ is 0
 (B) If λ is maximum then the volume of parallelepiped determined by $\vec{a}, \vec{b}$ and $\vec{c}$ is $\sqrt{3}$
 (C) If λ is maximum then the value of $|(2\vec{a} + 3\vec{b} + 4\vec{c}) \cdot (\vec{a} \times \vec{b} + 5\vec{b} \times \vec{c} + 6\vec{c} \times \vec{a})|$ is 32.
 (D) None of these
73. Let g be the inverse of the continuous function f , Let there be a point (α, β) , where $\alpha \neq \beta$, is such that it satisfies each of $y = f(x)$ and $y = g(x)$ then
- (A) the equation $f(x) = g(x)$ has infinitely many solutions
 (B) the equation $f(x) = g(x)$ has atleast 3 solutions
 (C) f must be a decreasing function of x
 (D) g can be an increasing function of x
74. The maximum and minimum value of $a \sin x + b \sqrt{1 - a^2} \cos x + c$ ($|a| < 1, b > 0$), lie in interval
- (A) $[c - b, b + c]$ (B) $(b - c, b + c)$
 (C) $[c - 2b, c + 2b]$ (D) $[-c, c]$
75. If $\lim_{x \rightarrow 2} (\log_3(ax^2 + 3x + 1))^{\log_{(x-1)} 3} = \ell$, where ℓ is a finite real number then
- (A) $a = -1$ (B) 'a' can have more than one values
 (C) $\ell = e^{-2/3}$ (D) $\ell = e^{-1/3}$
76. Which of the following functions is an injective (one-one) function in their respective domain?
- (A) $f(x) = 2x + \sin 3x$ (B) $f(x) = x \cdot [x]$, (where $[.]$ denotes the G.I.F.)
 (C) $f(x) = \frac{2^x + 1}{4^x - 1}$ (D) None of these

77. The centre of a circle $S = 0$ lies on $2x - 2y + 9 = 0$ and $S = 0$ cuts orthogonally the circle $x^2 + y^2 = 4$. Then the circle must pass through the point
 (A) (1, 1) (B) $(-1/2, 1/2)$ (C) (5, 5) (D) $(-4, 4)$
78. If $2f(x) + xf\left(\frac{1}{x}\right) - 2f\left(\sqrt{2}\sin\pi\left(x + \frac{1}{4}\right)\right) = 4\cos^2\left(\frac{\pi x}{2}\right) + x\cos\frac{\pi}{x}$, $\forall x \in \mathbb{R} - \{0\}$ then which of the following statements(s) is/are true?
 (A) $f(2) + f\left(\frac{1}{2}\right) = 1$ (B) $f(2) + f(1) = 0$
 (C) $f(2) + f(1) = f\left(\frac{1}{2}\right)$ (D) $f(1) + f\left(\frac{1}{2}\right) = 1$
79. If $\sqrt{4 + \sqrt{8 - \sqrt{32 + \sqrt{768}}}} = a\sqrt{2}\cos\left(\frac{11\pi}{b}\right)$, where a and b are natural numbers then $(b - a)$ is divisible by
 (A) 2 (B) 23 (C) 69 (D) 46
80. Equation of the plane passing through the line of intersection of the two planes $\vec{r} \cdot \vec{n}_1 = q_1$ and $\vec{r} \cdot \vec{n}_2 = q_2$ and parallel to the line of intersection of $\vec{r} \cdot \vec{n}_3 = q_3$ and $\vec{r} \cdot \vec{n}_4 = q_4$ is
 (A) dependent on $\vec{n}_1 \cdot \vec{n}_3$ (B) dependent on $\vec{n}_3 \cdot \vec{n}_4$
 (C) independent of q_1 and q_2 (D) independent of q_3 and q_4
81. Let ABCD be a tetrahedron, where $A = (2, 0, 0)$, $B = (0, 4, 0)$. If edge CD lies on line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$, such that centroid (α, β, γ) of tetrahedron satisfies $\frac{2\alpha - \frac{5}{2}}{1} = \frac{\beta - y_1}{a} = \frac{\gamma - z_1}{b}$ then which of the following are CORRECT?
 (A) $a + b = \frac{5}{2}$ (B) $y_1 + z_1 = \frac{19}{4}$ (C) $y_1 - z_1 = \frac{1}{4}$ (D) $a + b + y_1 = 5$
82. The line $3x + 6y = k$ intersects the curve $2x^2 + 2xy + 3y^2 = 1$ at points A and B. The circle on AB as diameter passes through the origin. The possible value of k is -
 (A) 3 (B) 4 (C) -4 (D) -3
83. A rod of length 2 units whose one end is $(1, 0, -1)$ and other end touches the plane $x - 2y + 2z + 4 = 0$, is rotated on this plane, then
 (A) the rod sweeps a solid structure whose volume is π cubic units
 (B) the area of the region which the rod traces on the plane is 2π
 (C) the length of projection of the rod on the plane is $\sqrt{3}$ units
 (D) the centre of the region which the rod traces on the plane is $\left(\frac{2}{3}, \frac{2}{3}, -\frac{5}{3}\right)$
84. Let $f : \left[0, \frac{\pi}{2}\right] \rightarrow [0, 1]$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 1$, then
 (A) $f'(\alpha) = \sqrt{1 - (f(\alpha))^2}$ for all $\alpha \in \left(0, \frac{\pi}{2}\right)$ (B) $f'(\alpha) = \frac{2}{\pi}$ for all $\alpha \in \left(0, \frac{\pi}{2}\right)$
 (C) $f(\alpha) f'(\alpha) = \frac{1}{\pi}$ for at least one $\alpha \in \left(0, \frac{\pi}{2}\right)$ (D) $f'(\alpha) = \frac{8\alpha}{\pi^2}$ for at least one $\alpha \in \left(0, \frac{\pi}{2}\right)$

85. Two equal sides of an isosceles triangle are given by the equation $7x - y + 3 = 0$ and $x + y - 3 = 0$ and its third side passes through the point $(1, -10)$. The equation of the third side can be
 (A) $x + 3y + 29 = 0$ (B) $x - 3y = 31$ (C) $3x + y = 3$ (D) $3x + y + 7 = 0$
86. Let $f(x) = \cos^{-1}(\cos 2x)$ and $g(x) = |\cos x|$ then
 (A) number of solution of $f(x) = g(x)$ in $[0, 2\pi]$ is 4.
 (B) $\max\{f(x), g(x)\}$ is a periodic function
 (C) $\max\{f(x), g(x)\}$ is a non differentiable function for some x ,
 (D) $\min\{f(x), g(x)\}$ is an even function
87. Consider $f(x) = 2\sin x + \sin 2x$, $x \in \left[0, \frac{3\pi}{2}\right]$
 (A) Greatest value of $f(x)$ is 3 (B) Greatest value of $f(x)$ is $\frac{3\sqrt{3}}{2}$
 (C) Least value of $f(x)$ is zero (D) Least value of $f(x)$ is -2
88. For a Function $f : A \rightarrow B$ such that $n(A) = a$, $n(B) = b$ ($a, b \in \mathbb{N}$) then which of the following statements must be **CORRECT** ?
 (A) If function is one - one, onto, then $a > b$
 (B) If function is one - one, into, then $a < b$
 (C) If function is many - one, onto, then $a > b$
 (D) If function is many - one, into, then $a < b$
89. Which of the following statement(s) is/are '**CORRECT**' ?
 (A) Rolle's theorem is applicable to the function $f(x) = 1 - \sqrt[5]{x^6}$ in the interval $[-1, 1]$
 (B) The value of $\lim_{x \rightarrow 0} \frac{\int_0^x x e^{t^2} dt}{1 + x - e^x}$ is equal to -2
 (C) If $\vec{a}$ and $\vec{b}$ are any two unit vectors, then least value of $\frac{1}{|\vec{a} + \vec{b}|^2} + \frac{1}{|\vec{a} - \vec{b}|^2}$ is equal to 2.
 (D) The angle between pair of tangents drawn to the curve $7x^2 - 12y^2 = 84$ from $M(1, 2)$ is $\frac{\pi}{4}$
90. If a is an integer and m_1, m_2, m_3 are the slopes of all three straight lines represented by equation $y^3 + (2a + 5)xy^2 - 6x^2y - 2ax^3 = 0$ which are also integers, then which of the following can holds good?
 (A) $a + \sum_{i=1}^3 m_i = -1$ (B) $a + \sum_{i=1}^3 m_i = -5$
 (C) $a + \prod_{i=1}^3 m_i = 0$ (D) $a + \prod_{i=1}^3 m_i = 32$
91. If $\text{fog}(x) = 1 - x$, $\forall x \in \mathbb{R}$ and $\text{fof}(x) = 1 + \sin(f(x))$, $\forall x \in \mathbb{R}$, then
 (A) $f(0) = 1$ (B) $f(\pi) = 1$ (C) $f(x)$ is periodic (D) $f(x) \geq 0$, $\forall x \in \mathbb{R}$

SECTION - 3 (True & False Statement Type)

This Section Contains 2 questions. Each question contains 4 statements S_1, S_2, S_3 & S_4 . Each statement is either true (T) or false (F). Each question has 4 choices (A), (B), (C) and (D) each of which contains whether S_1, S_2, S_3 & S_4 are true or false. Exactly one choice contains the correct order of truthness or falseness of S_1, S_2, S_3 & S_4 respectively and is the correct choice.

92. Consider the following statements :
- S_1 : Let $ax^2 + bx + c = 0$ be a quadratic equation, if $b^2 - 4ac > 0$ then roots are real and distinct.
- S_2 : Let $a, b, c, d \in I$. Then the possible rational roots of the equation $ax^3 + bx^2 + cx + d = 0$ are of the form $\frac{p}{q}$ where p is an integral divisor of d and q is an integral divisor of a .
- S_3 : Let $z = a + ib$ be a complex number where $a, b \in C - R$. Then $\bar{z} \neq a - ib$.
- S_4 : If $(a + b\sqrt{5})^2 = \frac{3 - \sqrt{5}}{2}$, then $a, b \in Q$.
- State, in order, whether S_1, S_2, S_3, S_4 are true or false
 (A) FTTT (B) TFTT (C) FTTF (D) FFTF
93. Consider the following statements :
- S_1 : Let $a, b, c \in C$ and $ax^2 + bx + c = 0$ be a quadratic equation. Then $b^2 - 4ac = 0 \Rightarrow$ roots are real and equal.
- S_2 : Let $b, c \in I$ and $b^2 - 4c$ be a perfect square. Then roots of the equation $x^2 + bx + c = 0$ may not be integers.
- S_3 : If the quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ have common root, then $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.
- S_4 : $f(x) = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$, $g(x) = \frac{\ell x + m}{ax + b}$ where a_1, a_2, ℓ, a are non zero real and other coefficients are also real. Then range of $f(x) \neq$ range of $g(x)$.
- State, in order, whether S_1, S_2, S_3, S_4 are true or false
 (A) TTFT (B) TTTF (C) FFTT (D) FFFF

SECTION - 4 : (Paragraph Type)

This section contains 10 paragraphs each describing theory, experiment, data etc. **twenty questions** relate to two paragraphs with **one or two or three questions** on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 94

Let $\frac{x^2}{36} + \frac{y^2}{27} = 1$ be an ellipse and P be any point on it and S and S' be its foci.

94. The point of intersection of straight lines joining each focus to the foot of the perpendicular from the other focus upon the tangent at a point $P\left(\frac{\pi}{3}\right)$, is
- (A) $\left(\frac{21}{8}, \frac{9}{4}\right)$ (B) $\left(\frac{15}{8}, \frac{9}{4}\right)$ (C) $\left(\frac{15}{4}, \frac{9}{8}\right)$ (D) $\left(\frac{21}{4}, \frac{9}{8}\right)$

Paragraph for Question Nos. 95 to 96

Consider $f(x) = \log_{[x]}[x]$ where $[.]$ is the greatest integer function and $\{x\}$ is fractional part function.

95. $f(x)$ is
 (A) a one-one function (B) a many-one function
 (C) a odd function (D) a periodic function
96. Consider the inequality $f(x) < 2$. Number of solutions of this inequality in $x \in (-1, 2)$ is/are
 (A) 0 (B) 1 (C) 3 (D) infinitely many

Paragraph for Question Nos. 97 to 98

Consider a polynomial $y = P(x)$ of the least degree passing through $A(-1, 1)$ and whose graph has two points of inflection $B(1, 1)$ and C with abscissa 0 at which the curve is inclined to the positive direction of x -axis at an angle of $\sec^{-1}(\sqrt{2})$.

97. The value of $P(0)$ is
 (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 0 (D) 2
98. The equation $P'(x) = 0$ has
 (A) three distinct real roots
 (B) one real root
 (C) three real roots such that one root is repeated.
 (D) none of these

Paragraph for Question Nos. 99 to 100

If $\cos^{-1} \frac{x}{2} + \cos^{-1} \frac{y}{3} = \theta$ then the value of $9x^2 - 12xy \cos \theta + 4y^2$ is equal to $N \sin^2 \theta$ (where N is a positive integer) and complete set of values of x for which $(\cos^{-1} x)^2 - (\sin^{-1} x)^2 > 0$ is satisfied, is $[p, q]$.

99. Which of the following values lie in $[p, q]$?
 (A) $\sqrt{N} - 3$ (B) $\sqrt{N} - 4$ (C) $\sqrt{N} - 5$ (D) $\sqrt{N} - 6$
100. Which of the following functions cannot be defined at $x = \sqrt{N} - 6$?
 (A) $\sin^{-1} x$ (B) $\cos^{-1} x$ (C) $\tan^{-1} x$ (D) $\sec^{-1} x$

Paragraph for Question Nos. 101 to 102

Base of a pyramid is rectangular, three of its vertices of the base are $A(2, 2, -1)$, $B(3, 1, 2)$ and $C(1, 1, 1)$ and its vertex at the top is $P\left(4, \frac{-26}{3}, \frac{-10}{3}\right)$. Fourth vertex of the base is D . Then answer the following questions.

101. If the coordinates of foot of normal drawn from P on the base of the pyramid are (ℓ, m, n) , then
 (A) $\ell + m + n = 4$ (B) $\ell = 3m$ (C) $n = 3m$ (D) $\ell < m < n$
102. Volume (in cubic unit) of the pyramid is
 (A) 20 (B) 30 (C) $\sqrt{20}$ (D) $\sqrt{30}$

Paragraph for Question Nos. 103 to 104

Consider $f(x) = \tan^{-1} \left(\frac{(\sqrt{12} - 2)x^2}{x^4 + 2x^2 + 3} \right)$ and m and M are respectively minimum and maximum values of $f(x)$ and $x = a$ ($a > 0$) is the point in the domain of $f(x)$, where $f(x)$ attains its maximum value.

103. If $\cos^{-1}x + \cos^{-1}y = 3 \left(\tan^{-1} \left(\tan \frac{7M}{2} \right) + \tan^{-1} \left(m + \tan \frac{3\pi}{8} \right) \right)$, then $(x + y)$ is equal to
 (A) 2 (B) -2 (C) 0 (D) $\frac{3}{2}$
104. If α and β are roots of the equation $x^2 - (\tan(3\sin^{-1}(\sin M)))x + a^4 = 0$, then $\alpha\beta - (\alpha + \beta)$ equals to
 (A) 1 (B) 4 (C) 3 (D) 2

Paragraph for Question Nos. 105 to 106

Consider the quadratic equation $ax^2 + (2 - a)x - 2 = 0$, where $a \in \mathbb{R}$.

105. If exactly one root is negative, then the range of $a^2 + 2a + 5$ is
 (A) $[4, \infty)$ (B) $[-2, \infty)$ (C) $(-\infty, 4]$ (D) $(5, \infty)$
106. If α, β are roots of the quadratic equation and if there are at least four negative integers between α and β , then the complete set of values of a is
 (A) $\left(-\frac{7}{2}, -3\right)$ (B) $\left(0, \frac{1}{2}\right)$ (C) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$ (D) $\left(\frac{7}{2}, 3\right)$

Paragraph for Question Nos. 107 to 108

Let M be the maximum value of c^2 for which $O(0, 0)$ and $A(1, 1)$ does not lie on opposite side of the straight line $(a + b)^2x - (ab + bc + ca + 1)y + 2 = 0$ for all $a, b \in \mathbb{R}$. Also $lx + my + n = 0$ be a variable line, where l, m, n are 1st, 3rd and 7th terms of an arithmetic progression and the variable straight line always passes through a fixed point $P(\alpha, \beta)$.

107. If the circles $x^2 + y^2 = M + 1$ and $x^2 + y^2 - 24x - 10y + \lambda^2 = 0$ have exactly two common tangents, then the number of possible integral values of λ , is
 (A) 11 (B) 13 (C) 12 (D) 10
108. The tangent of the circle $x^2 + y^2 = M + 2$ at the point $(\alpha - 1, \beta + 1)$ also touches the circle $x^2 + y^2 - 4\alpha x - 2\beta y + 20 = 0$ then its point of contact is
 (A) $(3, -1)$ (B) $(-3, 0)$ (C) $(-1, -1)$ (D) $(-2, 1)$

Paragraph for Question Nos. 109 to 110

Let $f(x) = x^2 + b_1x + c_1$, $g(x) = x^2 + b_2x + c_2$, α, β be real roots of $f(x) = 0$ and $\alpha + \delta, \beta + \delta$ be real roots of $g(x) = 0$, minimum value of $y = f(x)$ be $-\frac{1}{4}$ and minimum value of $y = g(x)$ occurs at $x = \frac{7}{2}$.

109. Minimum value of $y = g(x)$ is
 (A) -1 (B) $-\frac{1}{2}$ (C) $-\frac{1}{4}$ (D) $-\frac{1}{3}$
110. The value of b_2 is
 (A) 6 (B) -7 (C) 8 (D) 7

Paragraph for Question Nos. 111 to 113

There exists a matrix Q such that $PQP^T = N$, where $P = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Given N is a diagonal matrix of form $N = \text{diag.}(n_1, n_2, n_3)$ where n_1, n_2, n_3 are three values of n satisfying the equation $\det.(P - nI) = 0$, $n_1 < n_2 < n_3$.

[Note : I is an identity matrix of order 3×3]

111. The value of $\det.(\text{adj}N)$ is equal to

[Note : $\text{adj} M$ denotes the adjoint of a square matrix M.]

- (A) 4 (B) $\frac{1}{4}$ (C) $\frac{1}{9}$ (D) 9

112. If $Q^T = Q + \alpha I$, then the value of α is equal to

- (A) -1 (B) 0 (C) 1 (D) $-\frac{1}{3}$

113. The trace of matrix P^{2012} is equal to

- (A) $3^{2011} + 2$ (B) 3^{2012} (C) $3^{2012} + 2$ (D) 3^{2011}

SECTION - 5
Matrix - Match Type

This section contains 4 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column-I** are labelled A, B, C and D, while the statements in **Column-II** are labelled p, q, r, s and t. Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example. If the correct matches are A-p, s and t; B-q and r; C-p and q; and D-s and t; then the correct darkening of bubbles will look like the following :

	p	q	r	s	t
A	☒	☐	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

114. Match the inequality in column-I with their complete solution set in column-II.

Column - I

(A) $\log_{\sin x} \log_3 \log_{0.2} x < 0$

(B) $\frac{(e^x - 1)(2x - 3)(x^2 + x + 2)}{(\sin x - 2)x(x + 1)} \leq 0$

(C) $|2 - [x] - 1| \leq 2$,

[.] represents greatest integer function.

(D) $|\sin^{-1}(3x - 4x^3)| \leq \frac{\pi}{2}$

Column - II

(p) $[-1, 1]$

(q) $[-3, 6]$

(r) $\left(0, \frac{1}{125}\right)$

(s) $(-\infty, -1) \cup \left[\frac{3}{2}, \infty\right)$

(t) ϕ

115. If $f(x) = \frac{2x-1}{x-2}$ and $g(x) = \frac{x+1}{x}$.

Column – I

- (A) Domain of $f(g(x))$ is
 (B) Domain of $g(f(x))$ is
 (C) Range of $f(f(x))$ is
 (D) Range of $g(g(x))$ is

Column – II

- (p) $(-\infty, 1) \cup (1, 2) \cup (2, \infty)$
 (q) $(-\infty, 2) \cup (2, \infty)$
 (r) $(-\infty, 0) \cup (0, 1) \cup (1, \infty)$
 (s) $(-\infty, 1) \cup (2, \infty)$
 (t) $(-\infty, 1/2) \cup (1/2, 2) \cup (2, \infty)$

116. Column-I

(A) If $f : (1/2, 1) \rightarrow [0, \infty)$ and $f(x) = \sqrt{\log_e \left(\frac{x}{1-x} \right)}$, then $f(x)$ is

(B) If $f : [0, 1] \rightarrow [\cos(\sin 1) + \sin(\cos 1), 1 + \sin 1]$ and

$f(x) = \cos(\sin x) + \sin(\cos x)$, then $f(x)$ is

(C) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = \begin{cases} x - 2m + 2, & x \leq -1 \\ mx - 4, & x > -1 \end{cases}$, then for all values of $m \in (6, \infty)$. The function f is

(D) Let $f(x) = \frac{(x-\alpha)(x-\beta)}{(x-\gamma)(x-\delta)}$, where $\alpha < \beta < \gamma < \delta$, then $f(x)$ is

Column-II

- (p) one-one function
 (q) many-one function
 (r) into function
 (s) surjective function
 (t) invertible function

117. Column-I

(A) $a_n = \sqrt{1 + \left(1 + \frac{1}{n}\right)^2} + \sqrt{1 + \left(1 - \frac{1}{n}\right)^2}$ Then, the value of $\sum_{n=1}^{20} \frac{1}{a_n}$ is not divisible by

(C) Number of solutions satisfying the equation $\frac{1}{\sin x} - \frac{1}{\sin 2x} = \frac{2}{\sin 4x}$ in $[0, 4\pi]$ equals

(D) Integral multiple of solution of the equation $\cot \left(\sum_{r=1}^4 \tan^{-1} \frac{1}{2r^2} \right) = \frac{3x+4}{3x+2}$ is

Column-II

- (p) 2
 (q) 5
 (r) 7
 (s) 4
 (t) 6

SECTION – 6 : (Matching list Type)

This section contains 1 multiple choice questions. Each question has matching lists. The codes for the lists have choice (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

118. Match List I with List II and select the correct answer using the code given below the lists :

List - I

List - II

- P. $\sin \frac{\pi}{21} \sin \frac{2\pi}{21} \sin \frac{3\pi}{21} \dots \sin \frac{9\pi}{21} \sin \frac{10\pi}{21} =$
- Q. $\sin \frac{\pi}{22} \sin \frac{2\pi}{22} \sin \frac{3\pi}{22} \dots \sin \frac{9\pi}{22} \sin \frac{10\pi}{22} =$
- R. $\cos \frac{\pi}{21} \cos \frac{2\pi}{21} \cos \frac{3\pi}{21} \dots \cos \frac{9\pi}{21} \cos \frac{10\pi}{21} =$
- S. $\cos \frac{\pi}{22} \cos \frac{2\pi}{22} \cos \frac{3\pi}{22} \dots \cos \frac{9\pi}{22} \cos \frac{10\pi}{22} =$

1. $\frac{\sqrt{11}}{2^{10}}$
2. $\frac{1}{2^{10}}$
3. $\frac{\sqrt{11}}{2^5}$
4. $\frac{\sqrt{21}}{2^{10}}$

Codes:

	P	Q	R	S
(A)	3	4	4	2
(B)	4	1	2	1
(C)	2	3	1	4
(D)	4	3	1	3

SECTION – 7 : (Integer value correct Type)

This section contains 43 questions. The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

119. If $f(x) = \begin{cases} \frac{x \log(x + \sqrt{x^2 + 1})}{\left[\frac{x+e}{e} \right] - \frac{1}{2}} & x \neq ne, n \in \mathbb{I} \\ 0 & x = 0 \end{cases}$ (where $[.]$ denotes the greatest integer function),

then find the value of $f(0) + f(1947) + f(-1947)$

120. The equation of common tangent to the curve $xy = 4$ and $x^2 + y^2 = 8$, whose x and y intercepts are positive is $x + y = \lambda$. Find the value of λ .
121. If function $y = f(x)$ has only two points of discontinuities say x_1, x_2 , where $x_1, x_2 < 0$, then find the number of discontinuities of $y = f(|x|)$.
122. Given equation $|x^2 - 5x + 4 + \sin x| = |x^2 - 5x + 4| + |\sin x|$, where $0 \leq x \leq 2\pi$ then find sum of all integral values present in complete solution set of given equation.
123. If $f(x) + f\left(1 - \frac{1}{x}\right) = 1 + x \quad \forall x \in \mathbb{R} - \{0, 1\}$. Then find the value of $4f(2)$.
124. If $|z_1| = 2$, $(1 - i)z_2 + (1 + i)\bar{z}_2 = 8\sqrt{2}$, (z_1, z_2 are complex variables) then the minimum value of $|z_1 - z_2| =$
125. If $\left(\frac{x}{x^2 - 5x + 9}\right)^2 + \left(\frac{x}{x^2 - 5x + 9}\right) + k < 0$ for all real x , then find sum of modulus of all integral values of k in $[-4, 7]$.
126. Find the number of integral values of 'a' for which the equation $(3\sin x - 4\cos x)^2 - (a^2 + a + 5)|3\sin x - 4\cos x| + a^3 + 3a^2 + 2a + 6 = 0$ has a real solution.

127. In a sequence of circles $c_1, c_2, c_3, \dots, c_n$, the centres lie along positive x-axis with abscissa forming an A.P. of first term unity and common difference 3. The radius of these circles are in G.P. with first term unity and common ratio 2. If the tangents with slope m_1 and m_2 of c_3 intersect each other at the centre of c_5 , then find the value of $10|m_1 m_2|$.
128. In a meeting there are six ministers all speak exactly two languages. M_1 speaks only L_1 and L_2 , M_2 speaks only L_2 and L_3 , M_3 speaks only L_3 and L_4 , M_4 speaks only L_4 and L_2 , M_5 speaks only L_4 and L_1 , M_6 speaks only L_1 and L_3 . If two ministers are chosen at random, the probability that they speak common language is λ , then find the value of 5λ .
129. If a, b and c are distinct real numbers, such that the quadratic expressions $Q_1(x) = ax^2 + bx + c$, $Q_2(x) = bx^2 + cx + a$ and $Q_3(x) = cx^2 + ax + b$ are always non-negative, then find the number of possible integers in the range of the expression $y = \frac{a^2 + b^2 + c^2}{ab + bc + ca}$.
130. Find sum of all real solutions of the equation $x \log 5 + \log 2 = x + \log(2^x - 1)$.
131. Given $f: \mathbb{R} \rightarrow \mathbb{R}$; $f(x) = 2x^3 - 3(k+2)x^2 + 12kx - 7$, $-4 \leq k \leq 6$, $k \in \mathbb{I}$, then find the number of values of k for $f(x)$ to be invertible.
132. The real value of a for which the integral $\int_{a-2}^{a+2} e^{-(x-2)^2} dx$ attains its maximum value is λ . Find the value of $|\lambda|$.
133. If $f(x) = \frac{x^3}{3} + 3x^2 + k^2x - 10$ is a many one function, find sum of all positive integral values of k .
134. Find the minimum value of $e^{(2x^2 - 2x + 1)\sin^2 x}$.
135. If a is the number of onto functions on $A = \{a, b, c, d\}$ and b is the numbers of one one functions on A , find $|a - b|$.
136. Find number of solutions of the equation $\sin 3x = \cos 4x$ in the interval $\left[0, \frac{\pi}{4}\right]$.
137. If sum of the series $\cot^{-1}(2.1^2) + \cot^{-1}(2.2^2) + \cot^{-1}(2.3^2) + \dots \infty$ is equal to $k\pi$, then find the value of $[k]$, where $[.]$ denotes greatest integer function.
138. The function $f(x) = x^2 e^{-2x}$, $x > 0$. If ℓ is maximum value of $f(x)$ then find $[\ell^{-1}]$.
(where $[.]$ represents greater integer function)
139. Let A be any variable point on the x-axis and B be the point $(2, 3)$. The perpendicular at A to the line AB meets the y-axis at C . Locus of mid-point of the segment AC as A moves is $ax^2 - 2x + by = 0$. Find the value of $a + b$.
140. $x^2 + (a - b)x + (1 - a - b) = 0$, $a, b \in \mathbb{R}$. Find the least integral value of ' a ' for which both roots of the equation are real and unequal $\forall b \in \mathbb{R}$.
141. If $a_1, a_2, a_3, 5, 4, a_6, a_7, a_8, a_9$ are in H.P. and

$$\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$$
 then find the value of $[\Delta]$ is, where $[.]$ denotes G.I.F.
142. If $f(x) = \begin{vmatrix} \sin(x + \alpha) & \sin(x + \beta) & \sin(x + \gamma) \\ \cos(x + \alpha) & \cos(x + \beta) & \cos(x + \gamma) \\ \cos(\beta - \gamma) & \cos(\gamma - \alpha) & \cos(\alpha - \beta) \end{vmatrix}$ and $f(9) = \lambda$ ($\lambda \neq 0$), then find the value of $\frac{\sum_{k=1}^9 f(k)}{\lambda}$.

143. Find number of solutions of the equation $\left[\log_{10}^2 x + \log_{10} x \right] = 2 \log_{10} x + 1$ (where $[.]$ denotes greatest integer function)
144. If the value of the definite integral $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx = \frac{\pi^2}{\sqrt{n}}$ ($n \in \mathbb{N}$), then value of $\left(\frac{n}{27} \right)$ is
145. If $K\vec{r} + \vec{r} \times \vec{a} = \vec{b}$, where K is a non zero scalar and $\vec{a}, \vec{b}$ are two given vectors. Then $\vec{r}$ will be $\frac{1}{K^2 + |\vec{a}|^2} \left(\ell K \vec{b} + \frac{\vec{a} \cdot \vec{b}}{K^m} \vec{a} + n(\vec{a} \times \vec{b}) \right)$, find the value of $\ell + m + n$.
146. Sum of all elements in range of $f(x) = {}^{16-x}C_{2x-1} + {}^{20-3x}C_{4x-5}$ is λ then find the value of $\frac{\lambda}{469}$.
147. If $z = \frac{1}{2} (i\sqrt{3} - 1)$, then find the value of $(z - z^2 + 2z^3)(2 - z + z^2)$.
148. If $\begin{vmatrix} \sin 3\theta & -1 & 1 \\ \cos 2\theta & 4 & 3 \\ 2 & 7 & 7 \end{vmatrix} = 0$, then find the number of values of θ in $[0, 2\pi]$.
149. If 1 lies between the roots of the quadratic equation $3x^2 - 3(\sin \theta)x - 2\cos^2 \theta = 0$ then find sum of all integral values of θ in $[0, 2\pi]$, where θ is in radians.
150. The volume of the parallelepiped whose coterminous edges are represented by the vectors $3(\vec{b} \times \vec{c})$, $2(\vec{a} \times \vec{b})$ and $4(\vec{c} \times \vec{a})$ where
- $$\vec{a} = (1 + \sin \theta) \hat{i} + (\cos \theta) \hat{j} + (\sin 2\theta) \hat{k}$$
- $$\vec{b} = \sin \left(\theta + \frac{2\pi}{3} \right) \hat{i} + \cos \left(\theta + \frac{2\pi}{3} \right) \hat{j} + \sin \left(2\theta + \frac{4\pi}{3} \right) \hat{k}$$
- $$\vec{c} = \sin \left(\theta - \frac{2\pi}{3} \right) \hat{i} + \cos \left(\theta - \frac{2\pi}{3} \right) \hat{j} + \sin \left(2\theta - \frac{4\pi}{3} \right) \hat{k}$$
- is 18 cubic units then find number of values of θ in the interval $\left(0, \frac{\pi}{2} \right)$.
151. Let $f(x) = \int x^{\sin x} (1 + x \cos x \cdot \ell n x + \sin x) dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$ then find the value $|\cos(f(\pi))|$, where $x > 0$.
152. Find $\frac{T}{2}$ if T is the fundamental period of $f(x) = \sin\left(\frac{2[x]\pi}{3}\right) + \cos\left(\frac{3[x]\pi}{2}\right)$, where $[.]$ denotes greatest integer function.
153. $f(x) = x^3 + 4x^2 + \lambda x + 2$ is monotonically decreasing in the largest possible interval $\left(-2, \frac{-2}{3}\right)$. Then find greatest value of λ .
154. If range of $f(x) = \ell n \left(2 \sin x + \tan x - \frac{3x}{\pi} + 1 \right) \forall x \in \left[\frac{\pi}{6}, \frac{\pi}{3} \right]$ is $[a, b]$, then find the value of $[a + b]$, where $[.]$ denotes greatest integer function.
155. Find numbers of values of k if $-x^2 + kx + 2 < 0$ for exactly two integers.

156. Let z_1, z_2 and z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = |z_1 + z_2 + z_3| = 2$. If $|z_1 - z_2| = |z_1 - z_3|$ and $z_2 \neq z_3$. Then find the value of $|z_1 + z_2| |z_1 + z_3|$.
157. Following usual notations in a triangle ABC, if $R(a+b) = c\sqrt{ab}$ then $r = \frac{a}{\lambda + \sqrt{\mu}}$. Find the value of $\lambda + \mu$.
158. Let a function $f(x)$ be such that $f''(x) = f'(x) + e^x$ and $f(0) = 0, f'(0) = 1$, then find the value of $\ln\left(\frac{(f(2))^2}{4}\right)$.
159. If the area bounded by $|y| = \sin^{-1}|x|$ and $x = 1$ is $a(\pi + b)$, then find the value of $(a - b)$.
160. If $z_1, z_2, z_3 \in \mathbb{C}$ satisfy the system of equations given by $|z_1| = |z_2| = |z_3| = 1, z_1 + z_2 + z_3 = 1$ and $z_1 z_2 z_3 = 1$ such that $\text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$, then find the value of $[|z_1 + z_2^2 + z_3^3|]$, where $[.]$ denotes the greatest integer function
161. If $\left(\frac{x}{x^2 - 5x + 9}\right)^2 + \left(\frac{x}{x^2 - 5x + 9}\right) + k < 0$ for all real x , then find sum of modulus of all integral values of k in $[-4, 7]$.
162. Line L is a tangent to a unit circle S at a point P . Point A and the circle S are on the same side of L and the distance from A to L is 3 unit. Two tangents from point A intersect line L at the point B and C . Find the value of $(PB)(PC)$

SECTION – 8 : (Integer value correct Type)

This section contains 17 **questions**. The answer to each question is a **Two digit integer**, ranging from 00 to 99 (both inclusive).

163. If x, y, z are non-negative integers such that $2(x^3 + y^3 + z^3) = 3(x + y + z)^2$ then maximum value of $x + y + z$ is
164. If $[.]$ denotes the greatest integer function and let f be a function defined on the set of all non-negative integers and taking values in the same set. Given that -
 (A) $x - f(x) = 20\left[\frac{x}{20}\right] - 10\left[\frac{f(x)}{10}\right]$ for all non negative integers
 (B) $2008 < f(2010) < 2012$
 then find the number of possible values of $f(2010)$.
165. Eccentricity of an ellipse of minimum area, circumscribing two circles, of equal radius, touching externally, is $\sqrt{\frac{22}{\mu}}$. Find the value of μ .
166. Tangent at a point P (other than origin) on the curve $y=x^3$ meets the curve again at P_1 . The tangent at P_1 meets the curve again at P_2 and so on. then find the ratio $\frac{\text{area of } \Delta P_1 P_2 P_3}{\text{area of } \Delta P P_1 P_2}$
167. Let $f : \mathbb{N} \rightarrow [0, \infty)$ be a function satisfying the following conditions :
 (a) $f(100) = 10$,
 (b) $\frac{1}{f(0)+f(1)} + \frac{1}{f(1)+f(2)} + \frac{1}{f(2)+f(3)} + \dots + \frac{1}{f(n)+f(n+1)} = f(n+1)$, for all non-negative integers n . Find the value of $f(9801)$.

168. Complex number z satisfies $\arg(z - (3 + 3i)) = \alpha$ and $\arg\left(\frac{z-2}{z-2i}\right) = \frac{\pi}{11}$ then range of α such that no z exists is $[a, b]$, then find the value of $[10|a+b|]$ where $[.]$ represents the greatest integer function and $\arg$ means principal value of argument.
169. Range of $f(x) = \ell n\left(2\sin x + \tan x - \frac{3x}{\pi} + 1\right)$, $\forall x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ is $[a, b]$, then find the value of $11[a + b]$. $[.]$ represents greatest integer function.
170. A function $g(x)$ is defined as $g(x) = \frac{1}{4}f(2x^2 - 1) + \frac{1}{2}f(1 - x^2)$ and $f'(x)$ is an strictly increasing function then $g(x)$ is strictly increasing on the interval $\left(-\sqrt{\frac{p}{q}}, 0\right) \cup \left(\sqrt{\frac{p}{q}}, \infty\right)$ then find the value of $p^2 + q^2$. (where p and q are coprime to each other)
171. If $\int \frac{\sin 2x}{(3 + 4\cos x)^3} dx = \frac{a + b\cos x}{c(3 + 4\cos x)^d} + e$ where a, b, c, d are positive integers and e is arbitrary constant then minimum value of $a + b + c + d$ equal to
172. If $f(x) = x^3 - x^2 + 100x + 1001$, then find the number of correct statements among the following
(i) $f(2000) > f(2001)$, (ii) $f\left(\frac{1}{1999}\right) > f\left(\frac{1}{2000}\right)$, (iii) $f(x + 1) > f(x - 1)$, (iv) $f(3x - 5) > f(3x)$
173. The numbers 2,3,4,5,6,7,8 are to be placed, one per square, in the diagram shown so that the sum of the four numbers in the horizontal row equals 21 and the sum of the four numbers in the vertical column also equals 21. In how many different ways can this be done ?
-
174. If the function $f(x) = \sin(\ell n x) - \cos(\ell n x)$, where $-\pi \leq \ell n x \leq \pi$, strictly increases in interval $[e^\lambda, e^\mu]$ and the length of $[e^\lambda, e^\mu]$ is greatest possible, then find the value of $-5 \cos(\mu - \lambda)$.
175. If $\int \frac{\cos 5x - \cos 4x}{2\cos 3x + 1} dx = \frac{\sin 2x}{a} - \sin x + C$, find a .
176. If largest subset of $(0, p)$ at each point of which the function $f(x) = 3 \cos^4 x + 10 \cos^3 x + 6 \cos^2 x - 3$, is decreasing is $\left(0, \frac{\pi}{p}\right) \cup \left(\frac{2\pi}{r}, \pi\right)$, then find the value of $(p + r)$.
177. Consider seven digit number $x_1 x_2 \dots x_7$, where $x_1, x_2, \dots, x_7, \neq 0$ having the property that x_4 is the greatest digit and all digits towards the left and right of x_4 are in decreasing order. Then total number of such number in which all digits are distinct is a , then find the sum of digits of a .
178. Let $a = \tan^{-1} \tan 30, b = \cos^{-1} \cos 40, c = \sin^{-1} \sin(b - a)$ then product of all integral values of x satisfying $\cos^{-1} \cos x \geq |x + a - b - c|$ is
179. If $S = \sum_{n=2}^{\infty} \frac{3n^2 + 1}{(n^2 - 1)^3}$ then $\frac{9}{S} =$
180. If $\lim_{n \rightarrow \infty} \left(\frac{{}^{3n}C_n}{{}^{2n}C_n}\right)^{\frac{1}{n}} = \frac{A}{B}$, where A & B are relatively prime numbers, then $A+B$ is equal to-

ANSWER KEY

1.	(D)	2.	(D)	3.	(D)	4.	(B)	5.	(A)	6.	(A)	7.	(D)
8.	(B)	9.	(D)	10.	(C)	11.	(A)	12.	(B)	13.	(B)	14.	(C)
15.	(A)	16.	(B)	17.	(D)	18.	(B)	19.	(C)	20.	(D)	21.	(B)
22.	(A)	23.	(C)	24.	(B)	25.	(B)	26.	(D)	27.	(C)	28.	(D)
29.	(C)	30.	(B)	31.	(B)	32.	(C)	33.	(B)	34.	(A)	35.	(A)
36.	(C)	37.	(B)	38.	(D)	39.	(B)	40.	(B)	41.	(D)	42.	(D)
43.	(C)	44.	(B)	45.	(D)	46.	(C)	47.	(A)	48.	(B)	49.	(B)
50.	(B)	51.	(C)	52.	(B)	53.	(A)	54.	(BD)	55.	(ABCD)	56.	(BD)
57.	(BD)	58.	(ABC)	59.	(AB)	60.	(ABC)	61.	(ABCD)	62.	(AC)	63.	(BC)
64.	(AC)	65.	(BC)	66.	(ABCD)	67.	(ABCD)	68.	(ABCD)	69.	(ACD)	70.	(AB)
71.	(CD)	72.	(AC)	73.	(BC)	74.	(AC)	75.	(AD)	76.	(C)	77.	(BD)
78.	(ABC)	79.	(ABD)	80.	(ABD)	81.	(ABCD)	82.	(AD)	83.	(ACD)	84.	(CD)
85.	(BD)	86.	(ABCD)	87.	(BD)	88.	(BC)	89.	(AB)	90.	(ABC)	91.	(ABCD)
92.	(C)	93.	(D)	94.	(B)	95.	(B)	96.	(D)	97.	(B)	98.	(C)
99.	(D)	100.	(D)	101.	(A)	102.	(A)	103.	(B)	104.	(D)	105.	(D)
106.	(B)	107.	(B)	108.	(A)	109.	(C)	110.	(B)	111.	(D)	112.	(B)
113.	(C)	114.	(A) - (r), (B) - s, (C) - q, (D) - p					115.	(A) - (r), (B) - (t), (C) - (q), (D) - (p)				
116.	(A) - (p,r), (B) - (p,s,t), (C) - (p,r), (D) - (q)						117.	(A) -p,q,s,t ; (B) - p;		(C) - s; (D) - p,s,t			
118.	(B)	119.	0	120.	4	121.	0	122.	5	123.	3	124.	2
125.	7	126.	6	127.	8	128.	4	129.	3	130.	1	131.	1
132.	2	133.	3	134.	1	135.	0	136.	1	137.	0	138.	7
139.	5	140.	2	141.	2	142.	9	143.	0	144.	4	145.	3
146.	5	147.	7	148.	5	149.	3	150.	1	151.	1	152.	6
153.	4	154.	1	155.	0	156.	8	157.	4	158.	4	159.	3
160.	2	161.	7	162.	3	163.	12	164.	00	165.	33	166.	16
167.	99	168.	47	169.	11	170.	13	171.	29	172.	02	173.	72
174.	05	175.	02	176.	05	177.	09	178.	24	179.	16	180.	43



SOLUTIONS

1. Method - I

$$f'(x) = 2\sin(x + \alpha) \cos(x + \alpha) + 2\sin(x + \beta) \cos(x + \beta) - 2\cos(\alpha - \beta) \cos(x + \alpha) \sin(x + \beta) - 2\cos(\alpha - \beta) \sin(x + \alpha) \cos(x + \beta)$$

$$= \sin(2x + 2\alpha) + \sin(2x + 2\beta) - 2\cos(\alpha - \beta) \sin(2x + \alpha + \beta)$$

$$= 2\sin(2x + \alpha + \beta) \cos(\alpha - \beta) - 2\cos(\alpha - \beta) \sin(2x + \alpha + \beta)$$

$$= 0$$

$\therefore f(x)$ is a constant function

Method - II

$$f(x) = 1 - \cos^2(x + \alpha) + \sin^2(x + \beta) - 2\cos(\alpha - \beta) \sin(x + \alpha) \sin(x + \beta)$$

$$= 1 - \cos(2x + \alpha + \beta) \cos(\alpha - \beta) - 2\cos(\alpha - \beta) \sin(x + \alpha) \sin(x + \beta)$$

$$= 1 - \cos(\alpha - \beta) [\cos(2x + \alpha + \beta) - \cos(2x + \alpha + \beta)] + \cos(\alpha - \beta)$$

$$= \sin^2(\alpha - \beta)$$

a constant function

2. Product of roots

$$\Rightarrow \frac{c(a-b)}{a(b-c)} = 1 \quad \Rightarrow \quad b = \frac{2ac}{(a+c)}$$

$$\Rightarrow \frac{b}{a} = \frac{2c}{a+c} = \frac{2}{\left(\frac{a}{c}\right)+1} \quad \Rightarrow \quad (\alpha + \beta) = \frac{2}{\frac{1}{\alpha\beta} + 1}$$

$$\Rightarrow \frac{2\alpha\beta}{\alpha + \beta} = -1 - \alpha\beta$$

3. $AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+2c & b+2d \\ 3a+4c & 3b+4d \end{bmatrix}$

$$BA = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} a+3b & 2a+4b \\ c+3d & 2c+4d \end{bmatrix}$$

if $AB = BA$, then $a + 2c = a + 3b$
 $\Rightarrow 2c = 3b \quad \Rightarrow \quad b \neq 0$
 $b + 2d = 2a + 4b \quad \Rightarrow \quad 2a - 2d = -3b$

$$\frac{3a-3d}{3b+c} = \frac{-\frac{9}{2}b}{3b+\frac{3}{2}b} = -1$$

4. We know $\lim_{x \rightarrow 0^+} x^x = 1$

$$\lim_{x \rightarrow 0^+} (x^x - 1) = 0$$

$$\lim_{x \rightarrow 0^+} \frac{e^{(x^x-1)} - x^x}{((x^x)^x - 1)^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{\left(1 + (x^x - 1) + \frac{(x^x - 1)^2}{2!} + \frac{(x^x - 1)^3}{3!} + \dots\right) - 1 - (x^x - 1)}{((x^x)^x - 1)^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{(x^x - 1)^2}{2!} + \frac{(x^x - 1)^3}{3!} + \dots}{((x^x)^x - 1)^2} = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$$

5. Let $4m^3 - 3am^2 - 8a^2m + 8 = 0$ has three roots m_1, m_2, m_3

$$\Rightarrow m_1 m_2 m_3 = -2$$

$$\text{Given } m_1 m_2 = -1$$

$$\Rightarrow m_3 = 2 \quad \Rightarrow \quad 32 - 12a - 16a^2 + 8 = 0$$

$$\Rightarrow 16a^2 + 12a - 24 = 0 \Rightarrow \text{Sum of values of } a = -\frac{12}{16} = -\frac{3}{4}$$

6. $P = (a - d) + pd$

$$Q = (a - d) + qd$$

$$R = (a - d) + rd$$

option A is correct

For option B

$$T_{p+q+r} = (a - d) + (p + q + r)d$$

$$P + Q + R = 3(a - d) + (p + q + r)d$$

$\therefore a - d$ need not to be zero

$$\Rightarrow T_{p+q+r} \neq P + Q + R$$

For option C $p < q < r$

If Common diff. is $-ve$, then

$$pd > qd > rd \quad \Rightarrow \quad P > Q > R$$

For option D not necessarily true. if first term is not an integer.

7. Let $y = x^2 - 4x + 5$

$$y = (x - 2)^2 + 1$$

$\therefore \sin^{-1} y, \cos^{-1} y$ defined only at $y = 1$
at which $x = 2$

So given equation

$$4 + 2a + \frac{\pi}{2} = 0$$

$$a = -\frac{\pi + 8}{4}$$

8. $\alpha = \log_{10} 15 = 1 + \log_{10} 3 - \log_{10} 2$

$$\beta = 4 \log_{10} 2$$

$$4 \log_{10} 3 = 4\alpha + \beta - 4$$

we required those elements of A which can be written in terms

of $\log_{10} 2$ & $\log_{10} 3$.

i.e. number of integers which are divisible 2 or 3, or 5 only.

required numbers = 24

9. Roots are $2k, -3$

$$\therefore -6 < 2k < 10 \quad k \in (-3, 5)$$

$\therefore$ largest value of k is not defined.

10. $\frac{(2x-1)}{(2x^3+3x^2+x)} \geq 0$

$$\Rightarrow x \in (-\infty, -1) \cup \left(\frac{-1}{2}, 0\right) \cup \left[\frac{1}{2}, \infty\right) \dots \dots (1)$$

for $\log_2 x, x > 0$

for $\sqrt{\sin^{-1}(\log_2 x)}, 0 \leq \log_2 x \leq 1$

$\therefore x \in [1, 2] \dots \dots (2)$

So by (1) and (2) we get
 $x \in [1, 2]$

$$11. R^2 - SG^2 = \frac{1}{9} (a^2 + b^2 + c^2) \geq \frac{3}{9} (abc)^{2/3} \dots\dots\dots(1)$$

$$abc = 4R\Delta = 4Rrs = 4Rr \left(\frac{a+b+c}{2} \right) \geq 2Rr(abc)^{1/3} \cdot 3$$

$$(abc)^{2/3} \geq 6Rr$$

$$R^2 - SG^2 \geq \frac{1}{3} 6Rr$$

$$SG^2 \leq R^2 - 2Rr$$

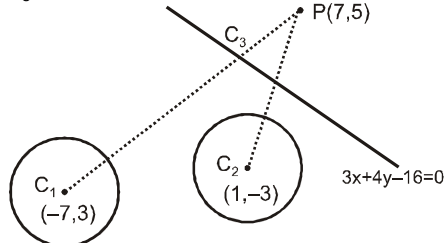
$$SG^2 \leq SI^2$$

$$SG \leq SI$$

12. Image of the centre C_2 (1, -3) in the line $3x + 4y - 16 = 0$ is P (7, 5).

Now for $C_1C_2 + C_2C_3 + C_3C_1$ to be minimum C_1, C_3 and P should be on same line so $C_3 = (0, 4)$

$$C_3 = (0, 4)$$



Distance between C_3 and C_1

$$= \sqrt{50} = 5\sqrt{2}$$

$$\text{radius of } C_1 = 3\sqrt{2}$$

$$\text{so radius of } C_3 = 2\sqrt{2}$$

$$\text{Equation of } C_3 (x-0)^2 + (y-4)^2 = 8$$

$$x^2 + y^2 - 8y + 8 = 0, a = 0, b = -8, c = 8$$

13. For $0 < n < 1$, $\sin x < \sin^n x$ and for $n > 1$, $\sin x > \sin^n x$
 Now, for $0 < n < 1$,

$$f(x) = \frac{2(\sin x - \sin^n x) - (\sin x - \sin^n x)}{2(\sin x - \sin^n x) + (\sin x - \sin^n x)} = \frac{1}{3}$$

and for ,

$$f(x) = \frac{2(\sin x - \sin^n x) + (\sin x - \sin^n x)}{2(\sin x - \sin^n x) - (\sin x - \sin^n x)} = 3$$

$$\text{For } n > 1, g(x) = 3, x \in (0, \pi)$$

$\therefore g(x)$ is continuous and differentiable at $x = \frac{\pi}{2}$, for

$$0 < n < 1,$$

$$g(x) = \begin{cases} \left[\frac{1}{3} \right] = 0, & x \in \left(0, \frac{\pi}{2} \right) \cup \left(\frac{\pi}{2}, \pi \right) \\ 3, & x = \frac{\pi}{2} \end{cases}$$

$\therefore g(x)$ is not continuous at $x = \frac{\pi}{2}$, Hence, $g(x)$ is also not

differentiable at $x = \frac{\pi}{2}$

$$14. \int \frac{\sec^2 x - 2010}{\sin^{2010} x} dx = \int \sec^2 x \cdot (\sin x)^{-2010} dx - 2010$$

$$\int \frac{1}{(\sin x)^{2010}} dx = I_1 - I_2$$

Applying by parts on I_1 , we get

$$\frac{\tan x}{(\sin x)^{2010}} + 2010 \int \frac{\tan x \cos x}{(\sin x)^{2011}} dx$$

$$= \frac{\tan x}{(\sin x)^{2010}} + 2010 \int \frac{dx}{(\sin x)^{2010}}$$

$$\Rightarrow I_1 - I_2 = \frac{\tan x}{(\sin x)^{2010}} = \frac{P(x)}{(\sin x)^{2010}} + c \Rightarrow P(x) = \tan x$$

$$\therefore P(x) = \tan \frac{\pi}{3} = \sqrt{3}$$

$$15. \vec{c} = x(\hat{i} + \hat{j}) + y(\hat{j} + \hat{k})$$

$$\frac{\vec{c} \cdot (\hat{i} + \hat{j})}{|\vec{c}| |\sqrt{2}|} = \frac{(\hat{i} + \hat{j}) \cdot (\hat{j} + \hat{k})}{\sqrt{2} \cdot \sqrt{2}}$$

$$\frac{2x + y}{\sqrt{x^2 + (x+y)^2 + y^2}} = \frac{1}{\sqrt{2}}$$

$$2(4x^2 + y^2 + 4xy) = 2x^2 + 2y^2 + 2xy$$

$$6x^2 + 6xy = 0$$

$$x = 0 \text{ or } x = -y$$

$$\vec{c} = x(\hat{i} - \hat{k})$$

$$\hat{c} = \frac{1}{\sqrt{2}}(\hat{i} - \hat{k})$$

$$\hat{c} \cdot \hat{j} = 0$$

$$16. (A + B)^2 = (A + B) \cdot (A + B) = A^2 + AB + BA + B^2 \dots\dots\dots(1)$$

$$\text{Since } B = -A^{-1}BA$$

$$\therefore AB = A(-A^{-1}BA)$$

$$= -(AA^{-1})BA$$

$$\therefore AB = -BA$$

put in (1), we get

$$(A + B)^2 = A^2 - BA + BA + B^2$$

$$(A + B)^2 = A^2 + B^2$$

$$17. \theta = \sin^{-1} \left(\sqrt{\frac{(\sqrt{3}-1)^2}{8}} \right) + \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) + \sec^{-1} (\sqrt{2})$$

$$\theta = \sin^{-1} \left(\frac{\sqrt{3}-1}{2\sqrt{2}} \right) + \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) + \sec^{-1} (\sqrt{2})$$

$$\theta = \frac{\pi}{12} + \frac{\pi}{6} + \frac{\pi}{4}$$

$$\theta = \frac{\pi + 2\pi + 3\pi}{12} = \frac{\pi}{2}$$

18. f is continuity iff $x = \cos x$
 which is possible only for one 'x'

19. Since $f'(x) = 0 - \frac{1}{\sqrt{1+x^3}}$

$$\therefore \int_0^2 xf(x) dx = \left(f(x) \cdot \frac{x^2}{2} \right)_0^2 - \int_0^2 f'(x) \frac{x^2}{2} dx$$

$$= 2f(2) + \frac{1}{2} \int_0^2 \frac{x^2}{\sqrt{1+x^3}} dx \quad \therefore f(2) = 0$$

$$= 0 + \frac{1}{6} \int_0^2 \frac{3x^2}{\sqrt{1+x^3}} dx = \frac{1}{6} \times \left((1+x^3)^{1/2} \right)_0^2$$

$$= \frac{1}{3} [3-1] = \frac{2}{3}$$

20. Clearly $P \equiv (1, 2)$

Equation of tangent at P is $4x(1) + y(2) = 8$

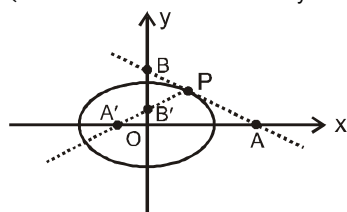
P पर स्पर्श रेखा का समीकरण $4x(1) + y(2) = 8$

$\Rightarrow A \equiv (2, 0)$ and $B \equiv (0, 4)$

$\Rightarrow A \equiv (2, 0)$ और $B \equiv (0, 4)$

Similarly normal at P is $2x - 4y + 6 = 0$

इसी प्रकार P पर अभिलम्ब $2x - 4y + 6 = 0$



$$\Rightarrow A'(-3, 0) \text{ and } B'\left(0, \frac{3}{2}\right) \Rightarrow A'(-3, 0) \text{ तथा}$$

$$B'\left(0, \frac{3}{2}\right)$$

$$\therefore \text{Area } \triangle BPB' = \frac{1}{2} \begin{vmatrix} 0 & 4 & 1 \\ 1 & 2 & 1 \\ 0 & 3/2 & 1 \end{vmatrix} = \frac{5}{4} \text{ and area } \triangle APA' = 5$$

$\therefore \text{Ratio} = 4$

21. $I = \int \frac{\left(\cos \frac{x}{2} - \sin \frac{x}{2} \right) e^{-x/2} dx}{2 \cos^2 \frac{x}{2}}$

Put $-x/2 = t$

$$I = - \int e^t (\sec t + \sec t \tan t) dt$$

$$= -e^t \sec t + C$$

$$= -e^{-x/2} \sec \frac{x}{2} + C$$

22. $L = \lim_{x \rightarrow 0^-} \frac{e^{-1-x} - 2}{-1-x} = \frac{e^{-1} - 2}{-1} = 2 - \frac{1}{e}$

23. $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi$

$$\Rightarrow \tan^{-1}2 = \frac{\pi}{4} + \cot^{-1}3 \dots \dots \dots (1)$$

$$\tan^{-1}(1) + \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{4} - \cot^{-1}3 \dots \dots \dots (2)$$

From 1

$$\alpha = \frac{13\pi}{24} + \frac{1}{6} \cot^{-1}3$$

From 2

$$\beta = \frac{3\pi}{4} + \cot^{-1}3$$

$$\Rightarrow |\beta - \alpha| = \frac{3\pi}{4} - \frac{13\pi}{24} + \left(1 - \frac{1}{6}\right) \cot^{-1}3$$

$$= \frac{5\pi}{24} + \frac{5}{6} \cot^{-1}3$$

$$\Rightarrow p = 5, q = 24, r = 5, s = 6$$

24. $\therefore \left| y \cdot \frac{dy}{dx} \right| = 1$

$$\Rightarrow \int y dy = \left| \int dx \right|$$

$$\Rightarrow \frac{y^2}{2} = \pm x + c$$

$\therefore Q(0, 1)$ lies on curve

$$\frac{y^2}{2} = \pm x + \frac{1}{2}$$

$$\Rightarrow y^2 = 2x + 1 \text{ and } y^2 = -2x + 1$$

$$\Rightarrow y^2 = 2\left(x + \frac{1}{2}\right) \text{ and } y^2 = -2\left(x - \frac{1}{2}\right) \therefore \text{Required area} = 4$$

$$= 4 \times \frac{2}{3} \times \frac{-1}{2} \left((1-2x)^{3/2} \right)_0^1$$

$$= \frac{-4}{3} (0-1) = \frac{4}{3}$$

25. $f(f(x)) = f(x) + f^2(x) + f^4(x) + f^8(x) + \dots$

Coefficient of x^{10} in $f(x) = 0$

Coefficient of x^{10} in $f^2(x) = 2$

Coefficient of x^{10} in $f^4(x) = {}^4C_2 + {}^4C_1 = 10$

Coefficient of x^{10} in $f^8(x) = {}^8C_2 = 28$

26. Let $S = \sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$

$$S = 2009(\alpha_1 + \alpha_2 + \dots + \alpha_{2008})$$

$$S = -2009$$

27. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{(\cos x)^\alpha}{\left(x - \frac{\pi}{2}\right)^2}$ must be ∞

iff $\alpha > 2$

28. Let variable line be $\ell x + my + n = 0$

$$\Rightarrow P_1 = \frac{\frac{3a\ell x_1}{a+b+c} + \frac{3amy_1}{a+b+c} + n}{\sqrt{\ell^2 + m^2}}$$

$$\Rightarrow P_2 = \frac{\frac{3b\ell x_2}{a+b+c} + \frac{3bmy_2}{a+b+c} + n}{\sqrt{\ell^2 + m^2}}$$

$$\Rightarrow P_3 = \frac{\frac{3c\ell x_3}{a+b+c} + \frac{3cmy_3}{a+b+c} + n}{\sqrt{\ell^2 + m^2}}$$

$$\text{Given } P_1 + P_2 + P_3 = 0$$

$$\Rightarrow \frac{3\ell(ax_1 + bx_2 + cx_3)}{(a+b+c)} + \frac{3m(ay_1 + by_2 + cy_3)}{(a+b+c)} + 3n = 0$$

$$\Rightarrow \frac{\ell(ax_1 + bx_2 + cx_3)}{(a+b+c)} + \frac{m(ay_1 + by_2 + cy_3)}{(a+b+c)} + n = 0$$

Hence line $\ell x + my + n = 0$ passes through incentre

$$\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c} \right) \text{ of triangle ABC.}$$

29. According to LMVT

$$\left| \frac{f'(x) - f'(0)}{x} \right| = |f''(C_1)| \leq 1$$

$$\Rightarrow |f'(x)| \leq |x|$$

$$\text{again by LMVT } \left| \frac{f(x) - f(0)}{x} \right| = |f'(C_2)| \leq |x|$$

$$\Rightarrow f(x) \leq x^2$$

30. $f(x) = \cos 8\pi \{x\} = \cos (8\pi x - 8\pi [x]) = \cos 8\pi x$. Its period is $\frac{1}{4}$.

Period of $\sin 2\pi x \operatorname{cosec} 2\pi x$ is $\frac{1}{2}$. $\therefore$ period of $f(x)$ is $\frac{1}{2}$

32. $f(0) = k$ and $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x^2 \{e^{1/x}\} = 0$

$\therefore$ for $f(x)$ to be continuous $f(0) = \lim_{x \rightarrow 0} f(x) \Rightarrow k = 0$

$$\therefore f'(0^+) = \lim_{h \rightarrow 0^+} \frac{h^2 \{e^{1/h}\}}{h} = 0$$

$$\therefore f(x) \text{ is derivable at } x = 0 \text{ and } f'(0^-) = \lim_{h \rightarrow 0^+} \frac{h^2 \{e^{-1/h}\}}{-h} = 0$$

due to $\{e^{1/x}\}$, will be discontinuous at all value of x , where $e^{1/x}$ becomes an integer.

i.e. at $x = \log_2 e, \log_3 e, \dots$ etc.

33. f is even and g is odd

$\therefore$ gogog is odd gogog
 $\Rightarrow$ fogogog is even fogogog
 $\Rightarrow$ gogogogogogog is even. gogogogogogog

34. n^{th} term of $1 + 3 + 6 + 10 + \dots$ is $n = \frac{n(n+1)}{2}$

$$\therefore T_n = \cot^{-1} \left[2a^{-1} + \frac{n(n+1)a}{2} \right] = \cot^{-1} \left(\frac{4 + n(n+1)a^2}{2a} \right)$$

$$= \tan^{-1} \left[\frac{2a}{4 + n(n+1)a^2} \right] = \tan^{-1} \left(\frac{\frac{a}{2}}{1 + \frac{na(n+1)a}{2}} \right)$$

$$= \tan^{-1} \left[\frac{\frac{(n+1)a}{2} - \frac{na}{2}}{1 + \left(\frac{(n-1)a}{2} \right) \left(\frac{na}{2} \right)} \right]$$

$$= \tan^{-1} \left((n+1) \frac{a}{2} \right) - \tan^{-1} \left(\frac{na}{2} \right)$$

Put $n = 1, 2, 3, \dots, n$ we have

$$S_n = \tan^{-1} \frac{(n+1)a}{2} - \tan^{-1} \frac{a}{2}; S_\infty = \frac{\pi}{2} - \tan^{-1} \frac{a}{2} = \cot^{-1} \left(\frac{a}{2} \right)$$

$$\therefore a = 2 \quad S_\infty = \frac{\pi}{4}$$

35. $5x^5 - 10x^3 + x + 2y + 6 = 0 \dots\dots\dots (i)$
 Differentiating w.r.t. x

$$25x^4 - 30x^2 + 1 + 2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{1}{2} (25x^4 - 30x^2 + 1)$$

$$\left(\frac{dy}{dx} \right)_{(0,-3)} = -\frac{1}{2} (0 - 0 + 1) = -\frac{1}{2}$$

$\therefore$ equation of the normal at $P(0,-3)$ is $P(0,-3)$
 $y + 3 = 2(x - 0)$

$$\Rightarrow y = 2x - 3 \dots\dots\dots (ii)$$

solving equation (i) and (ii)

$$5x^5 - 10x^3 + x + 4x - 6 + 6 = 0$$

$$5x(x^4 - 2x^2 + 1) = 0$$

$$5x(x^2 - 1)^2 = 0$$

$$\therefore x = 0, 1, -1$$

$$y = -3, -1, -5$$

$\therefore$ Normal at point $P(0,-3)$ meets the curve again at two points $(1,-1)$ and $(-1,-5)$.

Now equation of tangent at point $(1,-1)$ is

$$y + 1 = \left(\frac{dy}{dx} \right)_{(1,-1)} (x - 1)$$

$$\Rightarrow y + 1 = 2(x - 1)$$

$$\Rightarrow 2x - y - 3 = 0 \dots\dots\dots (iii)$$

Equation of tangent at point $(-1,-5)$ is

$$y + 5 = \left(\frac{dy}{dx} \right)_{(-1,-5)} (x + 1)$$

$$\Rightarrow y + 5 = 2(x + 1) \Rightarrow 2x - y - 3 = 0 \dots\dots\dots (iv)$$

36. $f(0) = 1$

$$\text{and } f(x + y + 1) = \left(\sqrt{f(x)} + \sqrt{f(y)} \right)^2$$

substituting $x = 0, y = 0$, we get $f(1) = (1 + 1)^2 = 4$

$$\text{substituting } y = 0, \text{ we get } f(x + 1) = \left(\sqrt{f(x)} + 1 \right)^2$$

$f(0) = 1, f(1) = 2^2, f(2) = 3^2, f(3) = 4^2$ and so on
 hence $f(x) = (1 + x)^2$

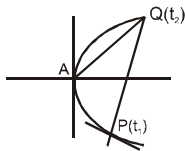
37. Composition of one - one function is one - one

$$38. \therefore AQ = \sqrt{a^2 t_2^4 + 4a^2 t_2^2} = at_2 \sqrt{t_2^2 + 4} = t_2 \sqrt{t_2^2 + 4}$$

($\therefore a = 1$)

$$\text{Since } t_2 = -t_1 - \frac{2}{t_1} = (-t_1) + \left(\frac{-2}{t_1} \right)$$

$\therefore t_1 < 0$ (as shown is diagram
now apply A.M. $\geq$ G.M



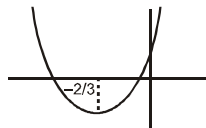
$$\Rightarrow \frac{(-t_1) + \left(\frac{-2}{t_1}\right)}{2} \geq \sqrt{2}$$

$$\Rightarrow (-t_1) + \left(\frac{-2}{t_1}\right) \geq 2\sqrt{2} \quad \therefore t_2 \geq 2\sqrt{2}$$

$$\therefore (AQ)_{\min} = 2\sqrt{2} \sqrt{8+4} = 4\sqrt{6}$$

39. $p' = \lambda(x-2)(x-4) = \lambda(x^2 - 6x + 8)$
 $p(x) = \lambda(x^3/3 - 3x^2 + 8x) + \mu$

40. $y = f(x) = (3x+2)^2 - 1, -\infty < x \leq -\frac{2}{3}$



It is clear from the graph that if $x \in (-\infty, -2/3]$, then $y \in [-1, \infty)$

$$y+1 = (3x+2)^2 \Rightarrow 3x+2 = \pm \sqrt{y+1}$$

($\therefore x \leq -2/3$, $\therefore$ negative sign)

$$3x+2 = -\sqrt{y+1}$$

$$\Rightarrow x = \frac{-2-\sqrt{y+1}}{3} = g(y) = f^{-1}(y)$$

$$\therefore g(x) = \frac{-2-\sqrt{x+1}}{3}$$

41. $\cot^{-1} \frac{9}{2} = \tan^{-1} \frac{2}{9} = \tan^{-1} \frac{4-2}{1+4 \cdot 2} = \tan^{-1} 4 - \tan^{-1} 2$

$$\cot^{-1} \frac{33}{4} = \tan^{-1} \frac{4}{33} = \tan^{-1} \frac{8-4}{1+4 \cdot 8} = \tan^{-1} 8 - \tan^{-1} 4$$

$$\cot^{-1} \frac{129}{8} = \tan^{-1} \left(\frac{8}{129} \right) = \tan^{-1} \frac{16-8}{1+16 \cdot 8} = \tan^{-1} 16 - \tan^{-1} 8$$

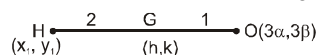
Similarly $T_n = \tan^{-1} 2^{n+1} - \tan^{-1} 2^n$ on adding we get

$$\text{sum} = \tan^{-1} 2^{n+1} - \tan^{-1} 2$$

$$\text{as } n \rightarrow \infty \text{ sum} = \frac{\pi}{2} - \tan^{-1} 2 = \cot^{-1} 2.$$

42. Let $S = {}^2C_0 + {}^3C_1 + {}^4C_2 + {}^5C_3 + \dots + {}^{99}C_{97} + {}^{100}C_{98}$
 $\therefore S = {}^3C_0 + {}^3C_1 + {}^4C_2 + {}^5C_3 + \dots + {}^{99}C_{97} + {}^{100}C_{98}$
 $S = {}^4C_1 + {}^4C_2 + {}^5C_3 + \dots + {}^{99}C_{97} + {}^{100}C_{98}$
 $S = {}^5C_2 + {}^5C_3 + \dots + {}^{99}C_{97} + {}^{100}C_{98}$
 $S = {}^{101}C_{98}$

43. Orthocentre lies on the rectangular hyperbola and



$$\therefore h = \frac{2(3\alpha) + x_1}{3} \text{ and } k = \frac{2(3\beta) + y_1}{3}$$

$$x_1 = 3(h-2\alpha) \text{ and } y_1 = 3(k-2\beta)$$

orthocentre (x_1, y_1) lies on $x^2 - y^2 = 36$

$$\therefore 9(h-2\alpha)^2 - 9(k-2\beta)^2 = 36$$

$$(h-2\alpha)^2 - (k-2\beta)^2 = 4$$

$\Rightarrow$ locus of centroid $G(h, k)$ is $(x-2\alpha)^2 - (y-2\beta)^2 = 4$

$$\Rightarrow \lambda = 4$$

44. Here $f'(x) = \begin{cases} 4-x^2 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$

$\Rightarrow f(x)$ is decreasing in $[2, 3)$ and increasing in $[3, \infty)$.
Now, $f(x)$ will have least value at $x = 3$ if

$$= \lim_{x \rightarrow 3^-} f(x) \geq f(3) \text{ i.e. } -15 + \ln(b^2 - 3b + 3) \geq -15$$

$$\text{i.e. } \ln(b^2 - 3b + 3) \geq 0$$

$$\Rightarrow b^2 - 3b + 3 \geq 1 \quad \text{i.e.} \quad b^2 - 3b + 2 \geq 0$$

$$\text{i.e. } b \in (-\infty, 1] \cup [2, \infty).$$

45. Let $\operatorname{cosec}^{-1} x = t$

$$\therefore t^2 - 2t \geq \frac{\pi}{6} (t-2)$$

$$\Rightarrow (t-2)(t-\pi/6) \geq 0$$

$$\Rightarrow t \leq \pi/6 \text{ or } t \geq 2 \text{ (not possible)}$$

$$\Rightarrow t \leq \pi/6$$

$$\Rightarrow 0^+ < t \leq \pi/6 \text{ or } -\pi/2 \leq t < 0^-$$

$$\Rightarrow 0^+ < \operatorname{cosec}^{-1} x \leq \pi/6 \text{ or } -\pi/2 \leq \operatorname{cosec}^{-1} x < 0^-$$

$$\Rightarrow 2 \leq x < \infty \text{ or } -\infty < x \leq -1$$

$$\Rightarrow x \in [2, \infty) \cup (-\infty, -1]$$

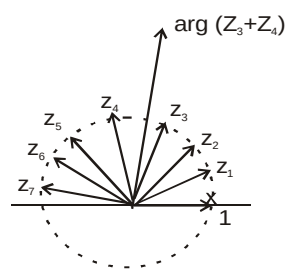
$$\Rightarrow a = -1, b = 2 \quad \therefore a + b = 1$$

46. $\arg \left(\frac{1 + Z_1 + Z_2 + \dots + Z_7}{1 + Z_8 + Z_9 + Z_{10} + \dots + Z_{14}} \right)$

$$= \arg \left(\frac{1 + \overline{Z_1} + \overline{Z_2} + \dots + \overline{Z_7}}{1 + \overline{Z_8} + \overline{Z_9} + \dots + \overline{Z_{14}}} \right)$$

$$= 2 \arg (1 + Z_1 + Z_2 + \dots + Z_7)$$

$$\left(\because \arg \frac{Z}{\overline{Z}} = 2 \arg (Z) \right)$$



$$= 2 \arg (Z_3 + Z_4)$$

$$= 2 \times \frac{7\pi}{15} = \frac{14\pi}{15}$$

Also $\arg (Z_1 Z_2 Z_3 \dots Z_7)$

$$= \frac{2\pi}{15} + \frac{4\pi}{15} + \frac{6\pi}{15} + \dots + \frac{14\pi}{15}$$

$$= \frac{2\pi}{15} (1 + 2 + 3 + \dots + 7) = \frac{56\pi}{15}$$

47. $x + \sqrt{x^2 + 1} = \frac{1}{y + \sqrt{y^2 + 1}}$

take $\log_e \ell n(x + \sqrt{x^2 + 1}) = \ell n\left(\frac{1}{y + \sqrt{y^2 + 1}}\right)$

$= \ell n\left(\frac{\sqrt{y^2 + 1} - y}{y^2 + 1 - y^2}\right)$

$\sqrt{x^2 + 1} + x = \sqrt{y^2 + 1} - y$

$x + y = \sqrt{x^2 + 1} + \sqrt{y^2 + 1} \dots\dots(1)$

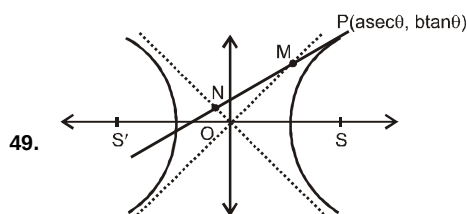
similarly $y + \sqrt{y^2 + 1} = \frac{1}{x + \sqrt{x^2 + 1}}$

$\Rightarrow \sqrt{y^2 + 1} + y = \sqrt{x^2 + 1} - x$

$x + y = -\sqrt{x^2 + 1} - \sqrt{y^2 + 1} \dots\dots(2)$

adding (1) & (2) $x + y = 0$
Image of $xy = 1$ in $xy = 1$ is itself.

48. $|1 + Z_1 + Z_2 + Z_3 + \dots\dots\dots Z_7|$
 $= |1 + Z_1| + |Z_1 + Z_2| + |Z_2 + Z_3| + |Z_3 + Z_4|$
 $= 2\cos \frac{7\pi}{15} + 2\cos \frac{5\pi}{15} + 2\cos \frac{3\pi}{15} + 2\cos \frac{\pi}{15}$
 $= 2 \left[2\cos \frac{4\pi}{15} \cos \frac{3\pi}{15} + 2\cos \frac{4\pi}{15} \cos \frac{\pi}{15} \right]$
 $= 4 \cos \frac{4\pi}{15} \left[\cos \frac{3\pi}{15} + \cos \frac{\pi}{15} \right]$
 $= 8 \cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} = \frac{8\sin(8\pi/15)}{8\sin(\pi/15)}$
 $= \frac{1}{2} \sec \frac{7\pi}{15}$



49.

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Tangent at $P(a \sec \theta, b \tan \theta)$

$\frac{x}{a} \sec \theta + \frac{y}{b} \tan \theta = 1 \dots\dots\dots(1)$

Asymptotes are

$y = \frac{b}{a} x \dots\dots\dots(2)$

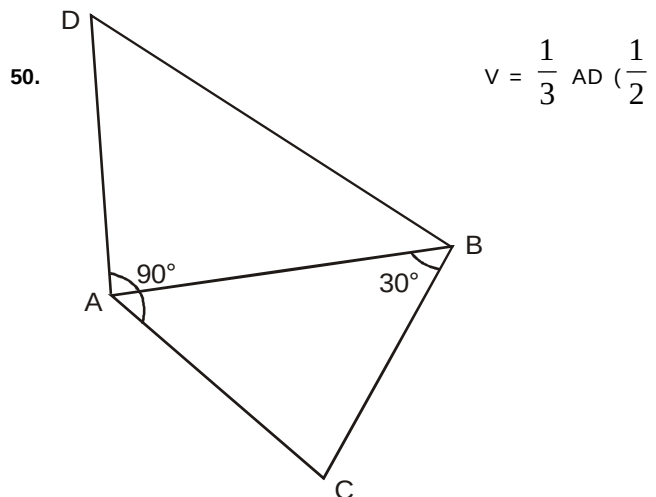
$y = -\frac{b}{a} x \dots\dots\dots(3)$

$M \equiv [a(\sec \theta - \tan \theta), b(\sec \theta - \tan \theta)]$
and $N \equiv [a(\sec \theta + \tan \theta), b(\sec \theta + \tan \theta)]$

$\Rightarrow ON = \sqrt{a^2 + b^2} (\sec \theta + \tan \theta) = ae(\sec \theta + \tan \theta)$

$\Rightarrow OM + ON = ae(2 \sec \theta) = 2ae(\sec \theta)$

$SP + S'P = e(a \sec \theta - \frac{a}{e}) + e(a \sec \theta + \frac{a}{e}) = 2ae \sec \theta$



50.

AB. BC $\sin 30^\circ$

$\Rightarrow AD \cdot AB \cdot BC = 216$

Now, $AB + BC + AD \geq 3(AD \cdot AB \cdot BC)^{1/3}$

$\Rightarrow AB + BC + AD \geq 18$

and minimum value occurs when $AB = BC = AD = 6$

Hence $AC = \sqrt{AB^2 + BC^2 - 2AB \cdot BC \cos 30^\circ}$

$= 3(\sqrt{6} - \sqrt{2})$

52. Since $f''(x) > 0$

$\Rightarrow f'(x)$ is always increasing.

$g'(x) = 2f'(2x^3 - 3x^2) \times (6x^2 - 6x) + f'(6x^2 - 4x^3 - 3)(12x - 12x^2)$
 $= 12(x^2 - x) \cdot (f'(2x^3 - 3x^2) - f'(6x^2 - 4x^3 - 3))$
 $= 12x(x - 1)[f'(2x^3 - 3x^2) - f'(6x^2 - 4x^3 - 3)]$

For increasing $g'(x) > 0$

Case-I $x < 0$ or $x > 1 \Rightarrow f'(2x^3 - 3x^2) > f'(6x^2 - 4x^3 - 3)$

$\Rightarrow 2x^3 - 3x^2 > 6x^2 - 4x^3 - 3 \{ \because f'(x) \text{ is increasing} \}$

$\Rightarrow (x - 1)^2 \left(x + \frac{1}{2} \right) > 0 \Rightarrow x > -\frac{1}{2} \therefore x \in \left(-\frac{1}{2}, 0 \right) \cup (1, \infty)$

Case-II If $0 < x < 1$

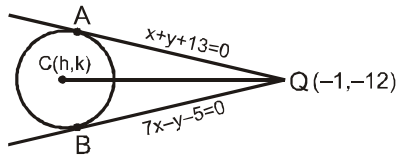
$f'(2x^3 - 3x^2) < f'(6x^2 - 4x^3 - 3)$

$(x - 1)^2 \left(x + \frac{1}{2} \right) < 0 \Rightarrow x < -\frac{1}{2}$, so there

is no solution

$\Rightarrow$ Hence the values are $x \in \left(-\frac{1}{2}, 0 \right) \cup (1, \infty)$

53. Clearly, the point lies on $7x - y = 5$



Also, centre of the circle must lie on the bisectors of the lines $x + y + 13 = 0$ and $7x - y - 5 = 0$ given by

$$\frac{x + y + 13}{\sqrt{2}} = \pm \frac{7x - y - 5}{\sqrt{50}} \Rightarrow x - 3y = 35 \text{ and } 3x + y + 15 = 0$$

Let (h, k) be the centre of the circle, then $h - 3k = 35$ (1)

and $3h + k + 15 = 0$

..... (2)

Clearly CA is perpendicular AQ

$$\Rightarrow \frac{k-2}{h-1} \times 7 = -1 \Rightarrow h + 7k - 15 = 0 \text{ (3)}$$

On solving, we get centres as $C_1 = (29, -2)$ and $C_2 = (-6, 3)$

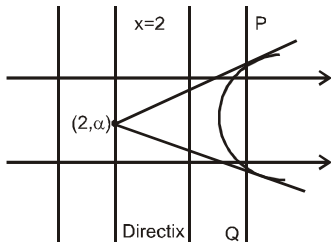
$$\Rightarrow r_1^2 = 800 \text{ and } r_2^2 = 50 \Rightarrow \text{Smaller circle has radius} = \sqrt{50}$$

Therefore area of quadrilateral ACBQ = $\sqrt{50} \times \sqrt{200}$ sq. units = 100 sq. units. Hence (A)

54. $z = x + iy$

$$(x-2)^2 = (x-7)^2 + (y+2)^2$$

$$(x-2)^2 - (x-7)^2 = (y+2)^2$$



$$(y+2)^2 = 5(2x-9)$$

$$(y+2)^2 = 10 \left(x - \frac{9}{2} \right)$$

$$Y^2 = 4aX \Rightarrow a = \frac{5}{2}$$

$$PQ|_{\min} = L(LR) = 10$$

55. $(x^2 + y) = (f(x))^2 + f(y)$

$$f(0) = (f(0))^2 + f(0) \Rightarrow f(0) = 0$$

$$\text{for } y = 0 : f(x^2) = (f(x))^2 \text{ (i)}$$

$$\text{for } y = -x^2 : 0 = f(0) = (f(x))^2 + f(-x^2)$$

$$\Rightarrow (f(x))^2 = -f(-x^2) \text{ (ii)}$$

from (i) and (ii)

$$f(-x^2) = -f(x^2)$$

$$\Rightarrow f(-x) = -f(x) \text{ (iii)}$$

thus $f(x)$ is an odd function

$$\text{if } f(x) \text{ is even also, then } f(-x) = f(x) \text{ (iv)}$$

$$\therefore f(x) = 0 \text{ for all } x \text{ {by (iii) and (iv)}}$$

since $f(x)$ is continuous at $x = 0$,

$$\therefore \lim_{h \rightarrow 0} f(h) = 0$$

$$\Rightarrow \lim_{h^2 \rightarrow 0} f(x + h^2) = \lim_{h^2 \rightarrow 0} \{(f(h))^2 + f(x)\} = f(x)$$

$\Rightarrow$ it is continuous everywhere

$$\text{since } \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h} \text{ exists}$$

$$\therefore \lim_{h \rightarrow 0} \frac{f(x+h^2) - f(x)}{h^2} = \lim_{h \rightarrow 0} \frac{(f(h))^2 + f(x) - f(x)}{h^2}$$

$$= (f'(0))^2$$

$$\therefore f'(x) = (f'(0))^2 \Rightarrow f'(0) = (f'(0))^2$$

$$\Rightarrow f'(0) = 0 \text{ or } f'(0) = 1$$

$$\therefore f(x) = (f'(0))^2 x$$

Now (i) if $f'(0) = 0$, then $f(x) = 0 = x f'(0)$

(ii) if $f'(0) = 1$, then $f(x) = x = x f'(0)$

56. $f(0) = r$ is odd. Let $r = 2n + 1$, $n \in \mathbb{I}$.

$$f(-1) = -1 + p - q + 2n + 1 = p - q + 2n \text{ is odd}$$

$\Rightarrow$ exactly one of p, q is odd

$$f(1) = 1 + p + q + 2n + 1 = p + q + 2n + 2 \text{ is odd}$$

If possible suppose $\alpha, \beta, \gamma \in \mathbb{I}$ be zeros of $f(x)$.

$$\Rightarrow x^3 + px^2 + qx + r = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma$$

$$\Rightarrow r = -\alpha\beta\gamma \Rightarrow \alpha, \beta, \gamma \text{ are odd integers}$$

$$\Rightarrow p = -(\alpha + \beta + \gamma) \text{ is odd}$$

and $q = \alpha\beta + \beta\gamma + \gamma\alpha$ is also odd.

It is a contradiction. Hence $f(x) = 0$

cannot have three integer roots.

$$57. f(x) = x^2 + \frac{b}{a}x + \frac{c}{a}$$

$$f(-1) < 0 \text{ \& } f(1) < 0$$

$$58. 1 - \log_3(x+1)^2 = \frac{1}{2} \log_{3^{1/2}} \left(\frac{x+5}{x+3} \right)$$

$$\Rightarrow 1 - \frac{2}{2} \log_3 |x+1| = \frac{2}{2} \log_3 \left(\frac{x+5}{x+3} \right)$$

$$\Rightarrow \log_3 \left(\frac{3}{|x+1|} \right) = \log_3 \left(\frac{x+5}{x+3} \right)$$

$$\Rightarrow \frac{3}{|x+1|} = \frac{x+5}{x+3}$$

Case - I $x+1 > 0 \Rightarrow x > -1$

$$\Rightarrow 3(x+3) = (x+1)(x+5)$$

$$\Rightarrow x^2 + 3x - 4 = 0$$

$$\Rightarrow x = -4 \text{ or } x = 1$$

$x = -4$ rejected

($\because x > -1$)

$$\therefore x = 1$$

Case - II $x+1 < 0 \Rightarrow x < -1$

$$\Rightarrow 3(x+3) = -(x+1)(x+5)$$

$$\Rightarrow x^2 + 9x + 14 = 0$$

$$\Rightarrow x = -2 \text{ or } x = -7$$

$$\therefore \text{Set of value of } x = \{-7, -2, 1\}$$

$$59. \lim_{x \rightarrow 1^+} g(x) = \lim_{x \rightarrow 1^+} \left\{ \lim_{m \rightarrow \infty} \frac{x^m f(x) + h(x) + 3}{2x^m + 4x + 1} \right\} = \lim_{m \rightarrow \infty}$$

$$\lim_{x \rightarrow 1^+} \left\{ \frac{x^m f(x) + h(x) + 3}{2x^m + 4x + 1} \right\}$$

$$= \lim_{m \rightarrow \infty} \lim_{x \rightarrow 1^+} \left\{ \frac{f(x) + \frac{h(x) + 3}{x^m}}{2 + \frac{4x + 1}{x^m}} \right\} = \frac{f(1)}{2}$$

$$\Rightarrow f(1) = 2e^3$$

$$\text{Similarly } \lim_{x \rightarrow 1^-} g(x) = \frac{h(1) + 3}{5}$$

$$\Rightarrow h(1) = 5e^3 - 3$$

60. The tangent $3x + 4y - 25 = 0$ is tangent at vertex and axis is $4x - 3y = 0$
So, PS = a = 5
Latus rectum = AB = 20

61. Let $\sqrt{768} = 32 \cos \theta$

$$16\sqrt{3} = 32 \cos \theta$$

$$\cos \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$$

$$\text{So } \sqrt{4 + \sqrt{8 - \sqrt{32 + 32 \cos \frac{\pi}{6}}}}$$

$$= \sqrt{4 + \sqrt{8 - 8 \cos \frac{\pi}{12}}}$$

$$= \sqrt{4 + 4 \sin \frac{\pi}{24}} = \sqrt{4 + 4 \cos \frac{11\pi}{24}}$$

$$= \sqrt{4 \left(1 + \cos \frac{11\pi}{24} \right)} = \sqrt{4 \cdot 2 \cos^2 \frac{11\pi}{48}} = 2\sqrt{2} \cos \frac{11\pi}{48}$$

$$\text{So } a = 2, b = 48$$

$$\therefore \frac{b}{a} = 24$$

62. $f(x) = e^x (x^2 + bx + c) > 0$ iff $D = b^2 - 4c < 0$
 $g(x) = e^x (x^2 + (b+2)x + (b+c)) > 0$ iff $D' = b^2 - 4c + 4 = D + 4 < 0$

63. $\left(2^{\frac{\pi}{\cos^{-1}x}} \right)^2 - \left(a + \frac{1}{2} \right) 2^{\frac{\pi}{\cos^{-1}x}} - a^2 = 0$

$$\text{let } 2^{\frac{\pi}{\cos^{-1}x}} = t \Rightarrow t \geq 2$$

$$\text{Now equation } t^2 - \left(a + \frac{1}{2} \right) t - a^2 = 0 \text{ has one root } 2 \text{ or}$$

greater than 2 and other root less than 2

$$\Rightarrow f(2) \leq 0$$

$$\Rightarrow 4 - \left(a + \frac{1}{2} \right) 2 - a^2 \leq 0$$

$$4 - 2a - 1 - a^2 \leq 0$$

$$a^2 + 2a - 3 \geq 0$$

$$(a+3)(a-1) \geq 0 \Rightarrow a \leq -3 \text{ or } a \geq 1$$

64. f is discontinuous at points outside its domain.

65. $\therefore T_{r+1} {}^{15}C_r (x^4)^{(15-r)} (x^{-3})^r = {}^{15}C_r x^{60-7r}$

(A) for the term independent of x, $60 - 7r = 0 \Rightarrow r$ is not an integer

$\therefore$ there is no term independent of x.

- (B) $\therefore n = 15$ is odd

$$\therefore {}^nC_r \text{ will be maximum if } r = \frac{n-1}{2} \text{ or } r = \frac{n+1}{2}$$

i.e. $r = 7$ or $r = 8$

$\therefore$ binomial coefficient of 8th and 9th terms will be greatest

- (C) for the coefficient of x^{32} ; $60 - 7r = 32 \Rightarrow r = 4$

$\therefore$ coefficient of x^{32} is ${}^{15}C_4$

for the coeff. of x^{-17} ; $60 - 7r = -17$; $r = 11$

$\therefore$ coeff. of x^{-17} is ${}^{15}C_{11} = {}^{15}C_4$

$\therefore$ (C) is correct

- (D) If $x = \sqrt{2}$, $T_{r+1} = {}^{15}C_r 2^{\frac{60-7r}{2}}$

$\therefore$ for rational terms $r = 0, 2, 4, 6, \dots, 14$

66. Solution set = ϕ

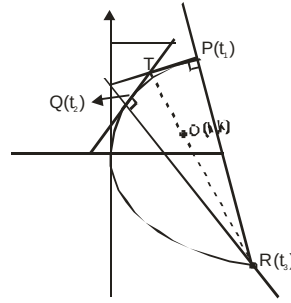
67. A & B are obvious

$$\left(f\left(\frac{1}{x}\right) - 1 \right) = \frac{1}{(f(x)-1)} \text{ i.e., } \left(f\left(\frac{1}{x}\right) - 1 \right) \text{ is reciprocal of } (f(x)-1)$$

$$\text{Now, for } f(x) = \frac{\pi/2}{\tan^{-1}|x|}, f(x)-1 = \frac{\cot^{-1}|x|}{\tan^{-1}|x|}, f\left(\frac{1}{x}\right) - 1 = \frac{\tan^{-1}|x|}{\cot^{-1}|x|}$$

$$\text{Also for } f(x) = \frac{2}{1+k \ln|x|}, f(x)-1 = \frac{1-k \ln|x|}{1+k \ln|x|}, f\left(\frac{1}{x}\right) - 1 = \frac{1+k \ln|x|}{1-k \ln|x|}$$

68. $t_1 t_2 = 2$ and $t_1 + t_2 + t_3 = 0$ and $a = 2$



$\therefore$ Let circumcentre of ΔTPQ be $O(h, k)$

$O(h, k)$ will be mid point of RT

$$\therefore h = \frac{at_1 t_2 + at_3^2}{2} \Rightarrow h = 2 + t_3^2 \quad \dots\dots (1)$$

$$\text{and } k = \frac{a(t_1 + t_2) + 2at_3}{2} \Rightarrow k = t_3 \quad \dots\dots$$

(2)

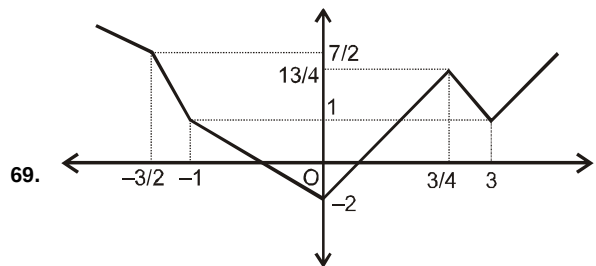
$$\therefore h = 2 + k^2$$

$\Rightarrow$ locus of $O(h, k)$ is

$$\Rightarrow O(h, k)$$

$$y^2 = x - 2$$

$$y^2 = 1(x - 2)$$



- 69.

70. $a_0 z^4 + a_1 z^3 + a_2 z^2 + a_3 z + a_4 = 0$

Taking conjugate on both sides.

$$a_0 (\bar{z})^4 + a_1 (\bar{z})^3 + a_2 (\bar{z})^2 + a_3 \bar{z} + a_4 = 0$$

$\therefore \bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4$ are the roots of the equation if z_1 is real,

then $\bar{z}_1$ is also real and if z_1 is non real, then $\bar{z}_1$ is also root because imaginary roots occur in conjugate pair.

71. $\lim_{x \rightarrow 3} \frac{\sin(e^{x-3} - 1)}{\ln(x - 2)}$

put $x = 3 + h$

$$\lim_{h \rightarrow 0} \frac{\sin(e^h - 1)}{\ln(1+h)} = \lim_{h \rightarrow 0} \frac{\sin(e^h - 1) \cdot (e^h - 1)}{(e^h - 1)h} \times \frac{h}{\ln(1+h)}$$

$$= 1 \times 1 \times 1 = 1$$

$$\begin{aligned} (\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) &= (\vec{b} \cdot \vec{c})(\vec{a} \cdot \vec{b}) - \vec{a} \cdot \vec{c} \\ (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) &= (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{c}) - \vec{a} \cdot \vec{b} \\ (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) &= (\vec{a} \cdot \vec{b})(\vec{a} \cdot \vec{c}) - \vec{b} \cdot \vec{c} \end{aligned}$$

Given that

$$\begin{aligned} |\vec{a} + \vec{b} + \vec{c}| &= \sqrt{3} \Rightarrow (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = 3 \Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0 \\ \lambda &= (\vec{a} \cdot \vec{b}) \cdot (\vec{b} \cdot \vec{c}) + (\vec{b} \cdot \vec{c})(\vec{c} \cdot \vec{a}) + (\vec{c} \cdot \vec{a})(\vec{a} \cdot \vec{b}) - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} - \vec{c} \cdot \vec{a} \\ \Rightarrow \lambda &= (\vec{a} \cdot \vec{b}) \cdot (\vec{b} \cdot \vec{c}) + (\vec{b} \cdot \vec{c})(\vec{c} \cdot \vec{a}) + (\vec{c} \cdot \vec{a})(\vec{a} \cdot \vec{b}) \\ \Rightarrow \lambda &\leq 0 \text{ [since, } x + y + z = 0, xy + yz + zx \leq 0] \\ \lambda_{\max} &= 0 \text{ only when } \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0 \\ \Rightarrow \vec{a} &\perp \vec{b}, \vec{b} \perp \vec{c} \text{ and } \vec{c} \perp \vec{a} \\ (2\vec{a} + 3\vec{b} + 4\vec{c}) \cdot (\vec{a} \times \vec{b} + 5\vec{b} \times \vec{c} + 6\vec{c} \times \vec{a}) &= 10\vec{a} \cdot (\vec{b} \times \vec{c}) + 18\vec{b} \cdot (\vec{c} \times \vec{a}) + 4\vec{c} \cdot (\vec{a} \times \vec{b}) = 32 \end{aligned}$$

73. As the point (α, β) , lies on both $f(x)$ and $g(x)$, the point (β, α) will also lie on both the curves and as the functions are continuous they must cross (meet on) the line $y = x$ in between. f must be on decreasing path, for all these to happen.

$$74. f'(x) = ab \cos x - b \sqrt{1-a^2} \sin x = b \cos(x + \phi)$$

$$\text{Where } \phi = \cos^{-1}a. \text{ So } f'(x) = 0 \Rightarrow x = (2m+1)\frac{\pi}{2} - \phi$$

$$\text{Also } f''(x) = -b \sin(x + \phi). \text{ Thus } f''\left(\frac{\pi}{2} - \phi\right) < 0 \text{ and } f''\left(\frac{3\pi}{2} - \phi\right) > 0$$

$$\text{Hence } f \text{ has maximum at } \frac{\pi}{2} - \phi \text{ and minimum at } \frac{3\pi}{2} - \phi.$$

$$\text{More over, } \max f = ab \sin(\pi/2 - \phi) + b \sqrt{1-a^2} \cos(\pi/2 - \phi) +$$

$$c = a^2b + b(1-a^2) + c = b + c$$

$$\text{and } \min f = -b + c.$$

$$75. \lim_{x \rightarrow 2} (\log_3(ax^2 + 3x + 1)) \text{ must be equal to } 1$$

$$\text{So } 4a + 7 = 3 \Rightarrow a = -1$$

$$\text{Also } l = e^{\lim_{x \rightarrow 2} \left(\frac{\log_3(-x^2 + 3x + 1) - 1}{\log_3(x-1)} \right)} = e^{\lim_{x \rightarrow 2} \left(\frac{(x-1)(3-2x)}{(-x^2 + 3x + 1)} \right)} = e^{-1/3}$$

$$76. f(x) = \frac{2^x + 1}{4^x - 1} = \frac{1}{2^x - 1} \text{ is one-one function.}$$

$$77. \text{ Let } S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\therefore \text{ it cuts } x^2 + y^2 = 4 \text{ orthogonally}$$

$$\Rightarrow c = 4$$

$$\text{Moreover } -2g + 2f + 9 = 0$$

$$(\therefore (-g, -f) \text{ satisfy the given equation})$$

$$\therefore S \equiv x^2 + y^2 + 2gx + 2fy + 4 = 0$$

$$\Rightarrow x^2 + y^2 + (2f + 9)x + 2fy + 4 = 0$$

$$\Rightarrow (x^2 + y^2 + 9x + 4) + 2f(x + y) = 0$$

It is of the form $S + \lambda P = 0$ and hence passes through the intersection of $S = 0$ and $P = 0$ which when solved give $(-1/2, 1/2), (-4, 4)$.

$$78. \text{ Replace } x \text{ by } 2, 2f(2) + 2f\left(\frac{1}{2}\right) - 2f(1) = 4$$

$$\Rightarrow f(2) + f\left(\frac{1}{2}\right) = 2 + f(1) \text{ -----(1)}$$

$$\text{Replace } x \text{ by } 1, f(1) = -1 \text{ -----(2)}$$

$$\text{Replace } x \text{ by } \frac{1}{2}, 2f\left(\frac{1}{2}\right) + \frac{1}{2}f(2) + 2 = \frac{5}{2} \text{ -----(3)}$$

$$\text{Solve (1) and (3)} \Rightarrow f\left(\frac{1}{2}\right) = 0; f(2) = 1$$

$$79. \text{ Let } \sqrt{768} = 32 \cos \theta$$

$$16\sqrt{3} = 32 \cos \theta$$

$$\cos \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$$

$$\text{So } \sqrt{4 + \sqrt{8 - \sqrt{32 + 32 \cos \frac{\pi}{6}}}} = \sqrt{4 + \sqrt{8 - 8 \cos \frac{\pi}{12}}}$$

$$= \sqrt{4 + 4 \sin \frac{\pi}{24}} = \sqrt{4 + 4 \cos \frac{11\pi}{24}}$$

$$= \sqrt{4 \left(1 + \cos \frac{11\pi}{24} \right)} = \sqrt{4 \cdot 2 \cos^2 \frac{11\pi}{48}} = 2\sqrt{2} \cos \frac{11\pi}{48}$$

$$\text{So } a = 2, b = 48$$

$$\therefore (b - a) = 46$$

80. Equation of the required plane is

$$(\vec{r} \cdot \vec{n}_1 - q_1) + \lambda(\vec{r} \cdot \vec{n}_2 - q_2) = 0 \Rightarrow \vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = (q_1 + \lambda q_2)$$

$$\text{Now, perpendicular to } \vec{n}_3 \times \vec{n}_4$$

$$(\vec{n}_1 + \lambda \vec{n}_2) \cdot (\vec{n}_3 \times \vec{n}_4) = 0$$

$$\lambda = - \frac{[\vec{n}_1 \vec{n}_3 \vec{n}_4]}{[\vec{n}_2 \vec{n}_3 \vec{n}_4]}$$

So independent on q_3 and q_4 only.

81.

Let $C = (\lambda_1 + 1, 2\lambda_1 + 2, 3\lambda_1 + 3)$ and

$D = (\lambda_2 + 1, 2\lambda_2 + 2, 3\lambda_2 + 3)$.

$$CD = \sqrt{14}$$

$$\Rightarrow (\lambda_1 - \lambda_2)^2 + 4(\lambda_1 - \lambda_2)^2 + 9(\lambda_1 - \lambda_2)^2 = 14$$

$$\Rightarrow \lambda_2 - \lambda_1 = 1 \Rightarrow \lambda_2 = 1 + \lambda_1$$

Let $G = (\alpha, \beta, \gamma)$

$$4\alpha = 5 + 2\lambda_1$$

$$4\beta = 10 + 4\lambda_1$$

$$4\gamma = 9 + 6\lambda_1$$

$$\text{Locus of } G \text{ is } \frac{4x-5}{2} = \frac{4y-10}{4} = \frac{4z-9}{6}$$

$$\frac{2x - \frac{5}{2}}{1} = \frac{y - \frac{5}{2}}{1} = \frac{z - \frac{9}{4}}{\frac{3}{2}}$$

$$82. \frac{3x+6y}{k} = 1$$

..... (1)

$$2x^2 + 2xy + 3y^2 - 1 = 0$$

..... (2)

Homogenizing (2) with the help of (1), we get

$$2x^2 + 2xy + 3y^2 - \left(\frac{3x+6y}{k}\right)^2 = 0$$

$$\Rightarrow k^2(2x^2 + 2xy + 3y^2) - (3x + 6y)^2 = 0$$

..... (3)

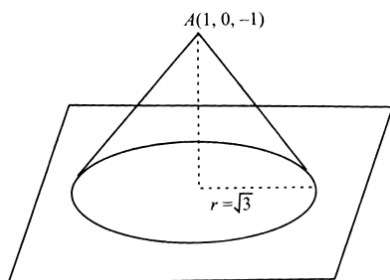
$\therefore$ circle described on AB as Diameter passes through (0, 0)

$\therefore$ AB will subtend right angle at (0, 0)

$\therefore$ coefficient of x^2 + coefficient of y^2 = 0

$$\Rightarrow (2k^2 - 9) + (3k^2 - 36) = 0 \Rightarrow k^2 = 9 \Rightarrow k = \pm 3$$

83.



The rod sweeps out the figure which is a cone.
The distance of point $A(1, 0, -1)$ from the plane

$$\text{is } \frac{|1+1-2+4|}{\sqrt{9}} = 1 \text{ unit.}$$

The slant height l of the cone is 2 units.

Then the radius of the base of the cone is

$$\sqrt{l^2 - 1} = \sqrt{4 - 1} = \sqrt{3}$$

Hence, the volume of the cone is $\frac{\pi}{3}(\sqrt{3})^2 \cdot 1 = \pi$

cubic units.

Area of the circle on the plane which the rod traces is 3π .

Also, the centre of the circle is $Q(x, y, z)$. Then

$$\frac{x-1}{1} = \frac{y-0}{-2} = \frac{z+1}{2} = \frac{-(1-0-2+4)}{1^2 + (-2)^2 + 2^2}, \quad \text{or}$$

$$Q(x, y, z) \equiv \left(\frac{2}{3}, \frac{2}{3}, \frac{-5}{3}\right).$$

84. (A) Consider $g(x) = \sin^{-1} f(x) - x$

$$\text{since } g(0) = 0, g\left(\frac{\pi}{2}\right) = 0$$

$\therefore$ there is at least one value of $\alpha \in \left(0, \frac{\pi}{2}\right)$ such that $g'(\alpha)$

$$= \frac{f'(\alpha)}{\sqrt{1 - (f(\alpha))^2}} - 1 = 0$$

i.e. $f'(\alpha) = \sqrt{1 - (f(\alpha))^2}$ for atleast one value of α but may

not be for all $\alpha \in \left(0, \frac{\pi}{2}\right) \therefore$ false

(B) Consider $g(x) = f(x) - \frac{2x}{\pi}$

$$\text{since } g(0) = 0, g\left(\frac{\pi}{2}\right) = 0$$

$\therefore$ there is at least one value of $\alpha \in \left(0, \frac{\pi}{2}\right)$ such that $g'(\alpha)$

$$= f'(\alpha) - \frac{2}{\pi} = 0$$

i.e. $f'(\alpha) = \frac{2}{\pi}$ for atleast one value of α but may not be for all

$$\alpha \in \left(0, \frac{\pi}{2}\right)$$

$\therefore$ false

(C) Consider $g(x) = (f(x))^2 - \frac{2x}{\pi}$

$$\text{since } g(0) = 0, g\left(\frac{\pi}{2}\right) = 0$$

$\therefore$ there is at least one value of $\alpha \in \left(0, \frac{\pi}{2}\right)$ such that $g'(\alpha)$

$$= 2f(\alpha) f'(\alpha) - \frac{2}{\pi} = 0$$

$$\therefore f(\alpha) f'(\alpha) = \frac{1}{\pi} \quad \therefore \text{true}$$

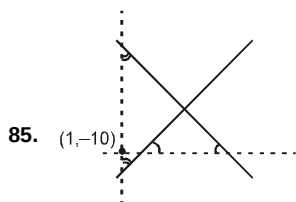
(D) Consider $g(x) = f(x) - \frac{4x^2}{\pi^2}$

$$\text{since } g(0) = 0, g\left(\frac{\pi}{2}\right) = 0$$

∴ there is at least one value of $\alpha \in \left(0, \frac{\pi}{2}\right)$ such

$$\text{that } g'(\alpha) = f'(\alpha) - \frac{8\alpha}{\pi^2} = 0$$

$$\therefore f'(\alpha) = \frac{8\alpha}{\pi^2} \quad \therefore \text{true}$$

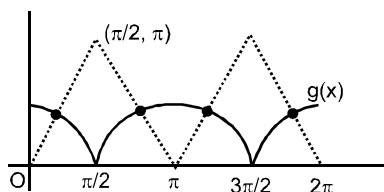


∴ third side will be parallel to bisectors of two given line

$$\therefore \text{Bisectors are } \frac{7x - y + 3}{5\sqrt{2}} = \pm \frac{(-x - y + 3)}{\sqrt{2}}$$

∴ Bisectors are $3x + y - 1 = 0$ and $x - 3y + 9 = 0$
Now required third side will be parallel to these bisectors
∴ $3x + y + 7 = 0$ or $x - 3y - 31 = 0$

86. $f(x) = \cos^{-1}(\cos 2x)$, $g(x) = |\cos x|$
 $f(x)$, and $g(x)$ both are even and periodic so $\max\{f(x), g(x)\}$ and $\min\{f(x), g(x)\}$ will also be periodic and even.



but $\max\{f(x), g(x)\}$ will be non-differentiable when $f(x) = g(x)$
no of points where $f(x) = g(x)$ are four in $[0, 2\pi]$

87. $f'(x) = 0 \Rightarrow 2\cos x + 2\cos 2x = 0$

$$\Rightarrow x = \frac{\pi}{3}, \pi \text{ are critical points}$$

$f(x)$ is increasing in $\left[0, \frac{\pi}{3}\right]$ and decreasing in $\left[\frac{\pi}{3}, \frac{3\pi}{2}\right]$

$$\text{Hence } f(0) = 0, f\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{2}, f(\pi) = 0, f\left(\frac{3\pi}{2}\right) = -2$$

$$\text{Least value} = -2, \text{Greatest value} = \frac{3\sqrt{3}}{2}$$

88. (A) If function is one - one & onto then $a = b$ since every element of set B should have exactly one pre-image in A.
(B) For one - one, into function, every element of set B should have either one pre-image or no pre-image in set A. $\Rightarrow$ no. of elements in set B $>$ no. of elements in A. $\therefore b > a$
(C) For many - one, onto function, every element of set B should have one or more than one pre-images in A. $\Rightarrow n$
(B) $< n$ (A) $\Rightarrow b < a$ or $a > b$
(D) For many - one, into function $a \in \mathbb{R}$ & $b \in \mathbb{R}$.

89. (A) $\therefore f(x) = 1 - x^{6/5} \Rightarrow f'(x) = -\frac{6}{5} x^{-1/5}$ exist $x \in (-1, 1)$
and $\therefore f(1) = 0 = f(-1)$ $\therefore$ Rolle's theorem is applicable

$$\begin{aligned} \text{(B)} \quad \therefore \lim_{x \rightarrow 0} \frac{\int_0^x x e^{t^2} dt}{1 + x - e^x} &= \lim_{x \rightarrow 0} \frac{x \left(\int_0^x e^{t^2} dt \right)}{1 + x - e^x} \\ &= \lim_{x \rightarrow 0} \frac{x(e^{x^2}) + \int_0^x e^{t^2} dt}{1 - e^x} \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{x(e^{x^2} \cdot 2x) + e^{x^2} + e^{x^2}}{-e^x} = \frac{0 + 1 + 1}{-1} = -2 \end{aligned}$$

$$\text{(C)} \quad \therefore |\vec{a} + \vec{b}| = 1 + 1 + 2\cos\theta = 2(1 + \cos\theta) = 4\cos^2\theta/2$$

$$\text{and } |\vec{a} - \vec{b}|^2 = 1 + 1 - 2\cos\theta = 2(1 - \cos\theta) = 4\sin^2\theta/2$$

$$\begin{aligned} \therefore \frac{1}{|\vec{a} + \vec{b}|^2} + \frac{1}{|\vec{a} - \vec{b}|^2} &= \frac{1}{4} \left(\sec^2 \frac{\theta}{2} + \operatorname{cosec}^2 \frac{\theta}{2} \right) \\ &= \frac{1}{4} \left(1 + \tan^2 \frac{\theta}{2} + 1 + \cot^2 \frac{\theta}{2} \right) \\ &= \frac{1}{4} \left(2 + \tan^2 \frac{\theta}{2} + \cot^2 \frac{\theta}{2} \right) \quad \therefore \text{A.M.} \geq \text{G.M} \\ \therefore \tan^2 \frac{\theta}{2} \cot^2 \frac{\theta}{2} &\geq 2 \geq 1 \end{aligned}$$

$$\text{(D)} \quad \therefore M(1, 2) \text{ lies on the director circle of } 7x^2 - 12y^2 = 84$$

$$90. m^3 + (2a + 5)m^2 - 6m - 2a = 0 \quad \begin{matrix} m_1 \\ m_2 \\ m_3 \end{matrix}$$

$$\begin{aligned} m_1 + m_2 + m_3 &= -(2a + 5) \\ m_1 m_2 + m_2 m_3 + m_3 m_1 &= -6 \\ m_1 m_2 m_3 &= 2a \end{aligned}$$

For (A)

$$a + \sum_{i=1}^3 m_i = -1$$

$$\Rightarrow a - 2a - 5 = -1 \Rightarrow a = -4 \Rightarrow m_1 m_2 m_3 = -8$$

$$\Rightarrow m_1 = 1, m_2 = -2, m_3 = 4 \Rightarrow m_1 m_2 m_3 = -8$$

for (B)

$$a - 2a - 5 = -5 \Rightarrow a = 0 \Rightarrow m_1 m_2 m_3 = 0$$

$$\Rightarrow m_1 = 1, m_2 = 0, m_3 = -6$$

$$\Rightarrow a + \sum_{i=1}^3 m_i = 0 + 0 = 0 \quad (\text{C is correct})$$

for (D)

$$a + 2a = 32 \text{ not possible}$$

Hence (D) is incorrect.

91. $\text{fof}(g(x)) = 1 + \sin(\text{fog}(x)) \Rightarrow f(1 - x) = 1 + \sin(1 - x)$
 $\Rightarrow f(x) = 1 + \sin x$

99. $\therefore \sqrt{N} - 6 = \sqrt{36} - 6 = 0 \in \left[-1, \frac{1}{\sqrt{2}}\right]$

100. $\sec^{-1}x$ is not defined at $x = 0$

101. to 102.

$P\left(4, \frac{-26}{3}, \frac{-10}{3}\right)$

$\therefore \overrightarrow{AB} = \hat{i} - \hat{j} + 3\hat{k}, \overrightarrow{AC} = -\hat{i} - \hat{j} + 2\hat{k} \text{ and } \overrightarrow{BC} = -2\hat{i} - \hat{k}$

$\overrightarrow{AC} \perp \overrightarrow{BC}$

$\therefore$ Fourth vertex D is (4, 2, 0)

$\overrightarrow{AB} \times \overrightarrow{AC} = \hat{i} - 5\hat{j} - 2\hat{k}$

$\therefore$ Equation of base is $x - 5y - 2z + 6 = 0$

Let E(x, y, z) be the foot of the perpendicular drawn from P to the base

$\therefore \frac{x-4}{1} = \frac{y+26/3}{-5} = \frac{z+10/3}{-2} = r \text{ (let)}$

$\therefore \left(4+r, -5r - \frac{26}{3}, -2r - \frac{10}{3}\right)$

It lies on the base

$\therefore 4+r-5\left(-5r-\frac{26}{3}\right)-2\left(-\frac{10}{3}-2r\right)+6=0$

$\therefore r = -2$

$\therefore E\left(2, \frac{4}{3}, \frac{2}{3}\right) \therefore \ell + m + n = 4$

$\overrightarrow{EP} = 2\hat{i} - 10\hat{j} - 4\hat{k}$

height $= \overrightarrow{EP} = \sqrt{120}$

$\therefore \text{Volume} = \frac{1}{3} |\overrightarrow{AC} \times \overrightarrow{BC}| |\overrightarrow{EP}| = \frac{1}{3} \sqrt{25+1+4} \sqrt{120}$

$= \frac{60}{3} = 20 \text{ cubic unit}$

103. to 104

$f(x) = \tan^{-1}\left(\frac{2(\sqrt{3}-1)}{x^2 + \frac{3}{x^2} + 2}\right) \geq 2\sqrt{3} \therefore x^2 + \frac{3}{x^2} + 2$

$\geq 2(\sqrt{3}+1)$

$\therefore f_{\max} = \tan^{-1}\left(\frac{2(\sqrt{3}-1)}{2(\sqrt{3}+1)}\right) = \frac{\pi}{12} = M \text{ which occurs at } x^2 =$

$\frac{3}{x^2} \text{ i.e. at } x = 3^{1/4} = a$

and $f_{\min} = 0 = m \text{ at } x = 0$
 $\cos^{-1}x + \cos^{-1}y =$

$3\left(\tan^{-1}\left(\tan\frac{7\pi}{24}\right) + \tan^{-1}\left(\tan\frac{3\pi}{8}\right)\right) = 3\left(\frac{7\pi}{24} + \frac{3\pi}{8}\right) = 2\pi$

$\Rightarrow x = y = -1 \Rightarrow x + y = -2$

$3\sin^{-1}(\sin M) = 3\sin^{-1}\left(\sin\frac{\pi}{12}\right) = \frac{\pi}{4}$

$\therefore x^2 - (\tan(3\sin^{-1}(\sin M)))x + a^4 = 0$

$\Rightarrow x^2 - (\tan\pi/4)x + 3 = 0$

$x^2 - x + 3 = 0$

$\therefore \alpha + \beta = 1, \alpha\beta = 3$

$\therefore (\alpha\beta) - (\alpha + \beta) = 3 - 1 = 2$

105. to 106.

Roots are 1, $-\frac{2}{a}$

$-\frac{2}{a} < 0 \Rightarrow a > 0$

$\Rightarrow a^2 + 2a + 5 \in (5, \infty)$

$-\frac{2}{a} < -4 \Rightarrow a \in \left(0, \frac{1}{2}\right)$

107. to 108

Given, $(a+b)^2x - (ab+bc+ca+1)y + 2 = 0$

Now, $(a+b)^2 - (ab+bc+ca) + 1 \geq 0$ for all $a, b \in \mathbb{R}$

$\Rightarrow a^2 + b^2 + ab - bc - ca + 1 \geq 0$

$\Rightarrow a^2 + (b-c)a + (b^2 - bc + 1) \geq 0 \quad \forall a \in \mathbb{R}$

$\Rightarrow (b-c)^2 - 4(b^2 - bc + 1) \leq 0 \Rightarrow 3b^2 - 2bc + (4 - c^2) \geq 0 \quad \forall b \in \mathbb{R}$

$\Rightarrow 4c^2 - 12(4 - c^2) \leq 0 \Rightarrow c^2 - 12 + 3c^2 \leq 0 \Rightarrow c^2 \leq 3$

$\therefore M = 3$

Also, $lx + my + n = 0$ (l, m, n are 1st, 3rd and 7th terms of an arithmetic progression)

$\therefore lx + (l+2d)y + (l+6d) = 0$

('d' be their common difference) 'd'

$\Rightarrow l(x+y+1) + d(2y+6) = 0 \Rightarrow x+y+1 = 0, 2y+6 = 0$

$\Rightarrow x = 2, y = -3$

Hence $\alpha = 2, \beta = -3$

$x^2 + y^2 = 4$; $x^2 + y^2 - 24x - 10y + \lambda^2 = 0$

$c_1(0, 0)$; $c_2 = (12, 5)$

$r_1 = 2$; $r_2 = \sqrt{169 - \lambda^2}$

$169 - \lambda^2 > 0 \Rightarrow \lambda \in (-\sqrt{48}, \sqrt{48})$

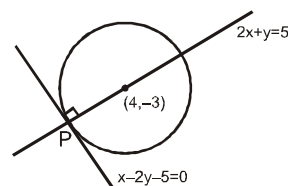
Since λ is integer

$\lambda = 0, \pm 1, \pm 2, \dots, 2 \pm 6$

$\therefore$ Number of possible integral values of λ is **13 Ans.**

$x^2 + y^2 = M + 2 = 3 + 2 = 5$

tangent at (1, -2)



$x - 2y - 5 = 0 \dots\dots(1)$

This line touches the circle $x^2 + y^2 - 8x + 6y + 20 = 0$

Solving $\begin{cases} 2x + y = 5 \\ x - 2y = 5 \end{cases} \Rightarrow x = 3, -1$

Hence point of contact (3, -1) **Ans.**

109 to 110

$(\beta - \alpha) = ((\beta + \delta) - (\alpha + \delta))$

$(\beta + \alpha)^2 - 4\alpha\beta = [(\beta + \delta) + (\alpha + \delta)]^2 - 4(\beta + \delta)(\alpha + \delta)$

$(-b_1)^2 - 4c_1 = (-b_2)^2 - 4c_2$

$D_1 = D_2 \dots\dots\dots(i)$

minimum of $y = f(x)$ is $-\frac{D_1}{4} = -\frac{1}{4}$

$\Rightarrow D_1 = 1 \Rightarrow D_2 = 1$

$$\therefore \text{minimum of } y = g(x) \text{ is } -\frac{D_2}{4} = -\frac{1}{4}$$

$$\text{minimum of } y = g(x) \text{ occur at } x = -\frac{b_2}{2} = \frac{7}{2}$$

$$\Rightarrow b_2 = -7$$

$$b_2^2 - 4c_2 = D_2$$

$$49 - 4c_2 = 1 \Rightarrow \frac{48}{4} = c_2$$

$$\Rightarrow c_2 = 12$$

111 to 113

$$\text{As } \det. (P - nI) = 0$$

$$\Rightarrow \begin{vmatrix} 1-n & 2 & 0 \\ 2 & 1-n & 0 \\ 0 & 0 & 1-n \end{vmatrix} = 0$$

$$\Rightarrow (1-n)(n-3)(n+1) = 0 \Rightarrow n = -1, 1, 3$$

$$\text{So } N = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\therefore \det. (N) = -3 \Rightarrow \det. (\text{adj } N) = (\det N)^2 = (-3)^2 = 9 \text{ Ans.}$$

$$112. \text{ Given } P Q P^T = N \Rightarrow \underline{P^{-1}P} Q \underline{P^T(P^T)^{-1}} = P^{-1}N(P^T)^{-1}$$

$$\Rightarrow Q = P^{-1}N(P^T)^{-1} \quad [\text{As } (P^T)^{-1} = (P^{-1})^T]$$

$$\therefore Q$$

$$= \begin{bmatrix} -1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\therefore Q^T = Q + \alpha I$$

$$\Rightarrow \alpha = 0 \text{ Ans.}$$

$$113. \text{ Given } P = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{So } P^2 = PP = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Now } P^3 = P^2P = \begin{bmatrix} 5 & 4 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 13 & 14 & 0 \\ 14 & 13 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P^4 = P^3P = \begin{bmatrix} 13 & 14 & 0 \\ 14 & 13 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 41 & 40 & 0 \\ 40 & 41 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Also } P^5 = P^4P = \begin{bmatrix} 41 & 40 & 0 \\ 40 & 41 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 121 & 122 & 0 \\ 122 & 121 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{Tr.}(P) = 3, \text{Tr.}(P^3) = 27 = 3^3$$

$$\text{Tr.}(P^5) = 243 = 3^5 \Rightarrow \text{Tr.}(P^{2011}) = 3^{2011}$$

$$\text{Also } \text{Tr.}(P^2) = 11 = 3^2 + 2$$

$$\text{Tr.}(P^4) = 3^4 + 2$$

$$\text{Tr.}(P^{2012}) = 3^{2012} + 2 \text{ Ans.}$$

$$114. (A) \log_{\sin x}(\log_3(\log_{0.2} x)) < 0 = \log_{\sin x} 1$$

$$\Rightarrow \log_3(\log_{0.2} x) > 1 \Rightarrow \log_{0.2} x > 3 = \log_{0.2}(0.2)^3$$

$$\Rightarrow 0 < x < (0.2)^3 \Rightarrow 0 < x < \frac{1}{125}$$

$$(B) \frac{(e^x - 1)(2x - 3)(x^2 + x + 2)}{(\sin x - 2)x(x + 1)} \leq 0$$

$$\begin{array}{c} + \quad - \quad - \quad + \\ -1 \quad 0_e \quad 3/2 \end{array}$$

$$\Rightarrow \frac{(e^x - 1)(x - 3/2)}{x(x + 1)} \geq 0 \Rightarrow x < -1 \text{ or } x \geq \frac{3}{2}$$

$$\Rightarrow x \in (-\infty, -1) \cup \left[\frac{3}{2}, \infty\right)$$

$$(C) |2 - [x] - 1| \leq 2$$

$$\Rightarrow ||[x] - 1| - 2| \leq 2 \Rightarrow 0 \leq |[x] - 1| \leq 4$$

$$\Rightarrow -3 \leq [x] \leq 5 \Rightarrow x \in [-3, 6)$$

$$(D) |\sin^{-1}(3x - 4x^3)| \leq \frac{\pi}{2} \Rightarrow -\frac{\pi}{2} \leq \sin^{-1}(3x - 4x^3) \leq \frac{\pi}{2}$$

$$\Rightarrow -1 \leq 3x - 4x^3 \leq 1 \Rightarrow -1 \leq x \leq 1$$

$$115. (A) x \neq 0 \text{ and } g(x) \neq 2 \Rightarrow x \neq 1$$

$$\text{Domain is } x \in (-\infty, 0) \cup (0, 1) \cup (1, \infty)$$

$$(B) x \neq 2 \text{ and } f(x) \neq 0 \Rightarrow x \neq 1/2$$

$$\text{Domain is } x \in (-\infty, 1/2) \cup (1/2, 2) \cup (2, \infty)$$

$$(C) f(f(x)) = x \text{ and } x \neq 2$$

$$\text{Range is } (-\infty, 2) \cup (2, \infty)$$

$$(D) g(g(x)) = \frac{2x + 1}{x + 1} \text{ and } x \neq 0$$

$$\text{Range is } (-\infty, 1) \cup (1, \infty)$$

$$116. (A) \text{ Range of the function is } (0, \infty)$$

$$(B) \text{ Function is decreasing}$$

$$(C) \text{ Range of function is not } \mathbb{R} \text{ and function is one-one.}$$

$$(D) \text{ The function is clearly many-one because the value of function is zero at } x = \alpha \text{ and } \beta.$$

$$\text{And range cannot be } \mathbb{R}.$$

$$117. (A) a_n = \frac{\sqrt{2n^2 + 2n + 1} - \sqrt{2n^2 - 2n + 1}}{n}$$

$$a_n = \frac{4}{\sqrt{2n^2 + 2n + 1} - \sqrt{2n^2 - 2n + 1}}$$

$$\frac{1}{a_n} = \frac{1}{4} \left(\sqrt{2n^2 + 2n + 1} - \sqrt{2n^2 - 2n + 1} \right)$$

$$\sum_{n=1}^{20} = \frac{1}{4} (\sqrt{5} - 1)$$

$$+ \frac{1}{4} (\sqrt{13} - \sqrt{5}) \dots \dots \dots \frac{1}{4} (\sqrt{841} - \sqrt{761})$$

$$= \frac{1}{4} (29 - 1) = 7$$

$$(C) \frac{1}{\sin x} - \frac{1}{\sin 2x} = \frac{2}{\sin 4x}$$

$$\frac{\sin 2x - \sin x}{\sin x \sin 2x} = \frac{2}{2 \sin 2x \cos 2x}$$

$$\frac{2 \cos 3x / 2 \sin x / 2}{2 \sin x / 2 \cos x / 2} = \frac{1}{\cos 2x}$$

$$\cos \frac{3x}{2} \cos 2x = \cos x / 2$$

$$\cos \frac{7x}{2} + \cos \frac{x}{2} = 2 \cos \frac{x}{2}$$

$$\cos \frac{7x}{2} = \cos \frac{x}{2} \Rightarrow \sin 2x \sin \frac{3x}{2} = 0$$

$$\sin 2x \neq 0 \sin \frac{3x}{2} = 0 \Rightarrow \frac{3x}{2} = n\pi$$

$$x = \frac{2n\pi}{3} \quad x \in$$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}$$

$$(D) \cot \left(\sum_{r=1}^4 \tan^{-1} \frac{1}{2r^2} \right)$$

$$\cot \left(\sum_{r=1}^4 \tan^{-1} \left(\frac{2}{4r^2} \right) \right)$$

$$\cot \left(\sum_{r=1}^4 \tan^{-1} \left(\frac{(2r+1) - (2r-1)}{1 + (2r+1)(2r-1)} \right) \right)$$

$$\cot \left(\sum_{r=1}^4 \tan^{-1} (2r+1) - \tan^{-1} (2r-1) \right)$$

$$\cot (\tan^{-1} 9 - \tan^{-1} (1))$$

$$\cot (\tan^{-1} 4/5)$$

$$51x = \frac{3x+4}{3x+2}$$

$$15x + 10 = 12x + 16$$

$$3x = 6$$

$$x = 2$$

$$118. \text{ We have } z^n - 1 = (z-1) \left(z - e^{\frac{i2\pi}{n}} \right) \dots \dots \left(z - e^{\frac{i2\pi(n-1)}{n}} \right)$$

$$1 + z + \dots + z^{n-1} = \left(z - e^{\frac{i2\pi}{n}} \right) \dots \dots \left(z - e^{\frac{i2\pi(n-1)}{n}} \right) \dots \dots (1)$$

$$\text{Put } z = 1 \text{ and take modulus } n = 2^{n-1} \sin \frac{\pi}{n} \sin \frac{2\pi}{n} \dots \dots \sin (n-1)$$

$$\frac{\pi}{n}$$

$$(P) \text{ If } n = 21$$

$$21 = 2^{20} \sin \frac{\pi}{21} \sin \frac{2\pi}{21} \dots \dots \sin \frac{9\pi}{21} \cdot \sin \frac{10\pi}{21} \dots \dots \sin \frac{20\pi}{21}$$

$$21 = \left(\sin \frac{\pi}{21} \dots \dots \sin \frac{10\pi}{21} \right)^2 \cdot 2^{20}$$

$$\therefore \sin \frac{\pi}{21} \dots \dots \sin \frac{10\pi}{21} = \frac{\sqrt{21}}{2^{10}}$$

$$(Q) \text{ If } n = 22$$

$$22 = 2^{21} \left(\sin \frac{\pi}{22} \sin \frac{2\pi}{22} \dots \dots \sin \frac{10\pi}{22} \right)^2$$

$$\therefore \left(\sin \frac{\pi}{22} \dots \dots \sin \frac{10\pi}{22} \right)^2 = \frac{22}{2^{21}} = \frac{11}{2^{20}}$$

$$\therefore \sin \frac{\pi}{22} \sin \frac{2\pi}{22} \dots \dots \sin \frac{10\pi}{22} = \frac{\sqrt{11}}{2^{10}}$$

$$(R) \text{ Again put } z = -1 \text{ is } \dots \dots (1) \text{ and take modulus}$$

$$1 = \left| \cos \frac{\pi}{n} \cos \frac{2\pi}{n} \dots \dots \cos (n-1) \frac{\pi}{n} \right| 2^{n-1}$$

$$n = 21$$

$$1 = \left| \cos \frac{\pi}{21} \cos \frac{2\pi}{21} \dots \dots \cos \frac{20\pi}{21} \right| 2^{20}$$

$$\therefore \cos \frac{\pi}{21} \dots \dots \cos \frac{10\pi}{21} = \frac{1}{2^{10}}$$

$$(S) \left(\sin \frac{\pi}{22} \cdot \sin \frac{2\pi}{22} \dots \sin \frac{10\pi}{22} \right) = \frac{\sqrt{11}}{2^{10}}$$

$$\left(\cos \frac{10\pi}{22} \cos \frac{9\pi}{22} \dots \cos \frac{\pi}{22} \right) = \frac{\sqrt{11}}{2^{10}}$$

$$\therefore \sin \theta = \cos(90 - \theta)$$

119. Let $f(x) = \frac{p(x) \cdot g(x)}{h(x)}$

where $p(x) = x \Rightarrow$ odd function

$$g(x) = \log(x + \sqrt{x^2 + 1})$$

$$g(x) + g(-x) = \log(x + \sqrt{x^2 + 1}) + \log(-x + \sqrt{x^2 + 1})$$

$$= \log(-x^2 + x^2 + 1)$$

$$= \log(1)$$

$$= 0 \Rightarrow g(x) \text{ is an odd function.}$$

$$h(x) = \left[\frac{x+e}{e} \right] - \frac{1}{2} = \left[\frac{x}{e} \right] + \frac{1}{2}$$

$$h(x) + h(-x) = \left[\frac{x}{e} \right] + \frac{1}{2} + \left[\frac{-x}{e} \right] + \frac{1}{2}$$

$$= \left[\frac{x}{e} \right] + \left[\frac{-x}{e} \right] + 1$$

$$= -1 + 1 \quad \therefore [x] + [-x] = -1 \text{ if } x \in \mathbb{R} - \mathbb{I}$$

$$= 0$$

$h(x)$ is an odd function

$\therefore f(x)$ is an odd function

$$\Rightarrow f(1947) + f(-1947) + f(0) = 0.$$

120. Any point on the curve $xy = 4$ is $\left(2t, \frac{2}{t} \right)$

$$\text{slope of the tangent at } \left(2t, \frac{2}{t} \right) \text{ is } -\frac{1}{t^2}$$

$\therefore$ equation of tangent is $x + t^2y - 4t = 0$
since it is tangent to $x^2 + y^2 = 8$ also

$$\therefore \left| \frac{4t}{\sqrt{1+t^4}} \right| = 2\sqrt{2}$$

$$\text{i.e. } t^2 = 1$$

$\therefore$ equation of the tangent is $x + y = \pm 4$

Since the intercepts are positive

$\therefore$ the tangent is $x + y = 4$

121. $y = f(|x|)$

$$\Rightarrow y = \begin{cases} f(x) & , x \geq 0 \\ f(-x) & , x < 0 \end{cases}$$

$$= \begin{cases} f(x) & , x \geq 0 \\ f(t) & , t > 0 \end{cases} \text{ where } t = -x \text{ and } y = f(x) \text{ has no point}$$

of discontinuity when $x \geq 0$

$y = f(|x|)$ has no discontinuous points

122. $|a+b| = |a| + |b|$

$$\Rightarrow ab \geq 0 \Rightarrow (x^2 - 5x + 4) \sin x \geq 0$$

$$\Rightarrow x \in [0, 1] \cup [\pi, 4] \cup \{2\pi\}$$

sum of all integral values is $0 + 1 + 4 = 5$

123. Putting $x = 2, \frac{1}{2}$ and -1 successively

$$f(2) + f\left(\frac{1}{2}\right) = 3 \quad \dots (1)$$

$$f\left(\frac{1}{2}\right) + f(-1) = \frac{3}{2} \quad \dots (2)$$

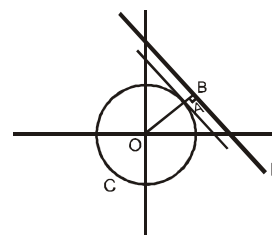
$$\text{and } f(-1) + f(2) = 0 \quad \dots (3)$$

$$\text{Solving, we get } f(2) = \frac{3}{4}$$

$$\therefore 4f(2) = 3$$

124. $|z_1| = 2$ and $(1-i)z_2 + (1+i)\bar{z}_2 = 8\sqrt{2}$

$$C: x^2 + y^2 = 4 \quad L: x + y = 4\sqrt{2}$$



$$AB = OB - r$$

$$AB = 2.$$

125. Let $f = \frac{x}{x^2 - 5x + 9} \quad f \in \left[\frac{-1}{11}, 1 \right]$

$$\therefore f^2 + f + k < 0 \quad \forall f \in \left[\frac{-1}{11}, 1 \right] \Rightarrow f \left[\frac{-1}{11}, 1 \right] < 0 \text{ \& } f(1) < 0$$

126. Let $|3\sin x - 4\cos x| = t$

$$\therefore t \in [0, 5] \quad x \in \mathbb{R}$$

Given equation can be written as

$$t^2 - (a^2 + a + 5)t + a^2(a + 3) + 2(a + 3) = 0$$

$$t^2 - ((a^2 + 2) + (a + 3))t + (a + 3)(a^2 + 2) > 0$$

$$\Rightarrow t = a + 3 \quad \text{or } t = a^2 + 2$$

$$\Rightarrow 0 \leq a + 3 \leq 5 \quad \text{or } 0 \leq a^2 + 2 \leq 5$$

$$\Rightarrow -3 \leq a \leq 2 \quad \text{or } -2 \leq a^2 \leq 3$$

$$\Rightarrow -3 \leq a \leq 2 \quad \text{or } -\sqrt{3} \leq a \leq \sqrt{3}$$

$\therefore$ acceptable integer values of a are $-3, -2, -1, 0, 1, 2$

$\therefore 6$ is the answer

127. Centre of circles c_1, c_2, c_3 are in A.P.

General term for abscissa of centres $= 1 + (n-1) \cdot 3 = 3n - 2$

$\therefore$ centre of c_5 is $(13, 0)$

Radius of circles are in G.P.

$$\therefore R_n = 1 \cdot 2^{n-1} = 2^{n-1}$$

$$\therefore R_3 = 4 \text{ and centre of } c_3 \text{ is } (7, 0)$$

tangents of circle c_3 intersect each other at $(13, 0)$

equation of any line passing through $(13, 0)$ is $y - 0 = m(x - 13)$

$\Rightarrow mx - y - 13m = 0$ now it will be required tangents if

$$\left| \frac{7m - 0 - 13m}{\sqrt{m^2 + 1}} \right| = 4$$

$$\Rightarrow 36m^2 = 16m^2 + 16 \Rightarrow 20m^2 = 16 \Rightarrow m = \pm \frac{2}{\sqrt{5}}$$

$$\text{Let } m_1 = \frac{2}{\sqrt{5}}, m_2 = -\frac{2}{\sqrt{5}}$$

$$\therefore 10|m_1 m_2| = 8$$

128.

	M ₁	M ₂	M ₃	M ₄	M ₅	M ₆
L ₁	✓				✓	✓
L ₂	✓	✓		✓		
L ₃		✓	✓			✓
L ₄			✓	✓	✓	

$$\text{Required probability} = \frac{{}^4C_1 \times {}^3C_2}{{}^6C_2} = \frac{4}{5}$$

Alter : Required probability = 1 - (no languages is common)

$$= 1 - 1 \times \frac{1}{5} = \frac{4}{5}$$

$$129. Q_1(x) \geq 0 \Rightarrow a > 0 \text{ and } b^2 - 4ac \leq 0$$

$$Q_2(x) \geq 0 \Rightarrow b > 0 \text{ and } c^2 - 4ab \leq 0$$

$$Q_3(x) \geq 0 \Rightarrow c > 0 \text{ and } a^2 - 4bc \leq 0$$

$$\Rightarrow \sum a^2 - 4(\sum ab) \leq 0$$

$$\Rightarrow \sum a^2 \leq 4(\sum ab) \quad \because a > 0, b > 0 \text{ and } c > 0$$

$$\therefore \sum ab > 0$$

$$\Rightarrow \frac{\sum a^2}{\sum ab} \leq 4 \quad \dots\dots\dots(1)$$

$$\text{we know that } (a-b)^2 + (b-c)^2 + (c-a)^2 > 0$$

$$\Rightarrow \sum a^2 - \sum ab > 0$$

$$\Rightarrow \frac{\sum a^2}{\sum ab} > 1 \quad \dots\dots\dots(2)$$

from (1) and (2), we can say that

$$1 < \frac{\sum a^2}{\sum ab} \leq 4$$

$\therefore$ possible of integers in the range are 2,3,4

$\therefore$ Number of integers in the range is 3.

$$130. \log_{10} \left(\frac{5^x \cdot 2}{2^x - 1} \right) = x \quad \Rightarrow x = 1 \text{ only}$$

$$131. f(x) = 2x^3 - 3(k+2)x^2 + 12kx - 7, -4 \leq k \leq 6, k \in \mathbb{I}$$

$$f'(x) = 6(x^2 - (k+2)x + 2k) = 6(x-k)(x-2)$$

For f(x) to be invertible, k must be 2 only.

$$132. \text{ Let } f(a) = \int_{a-2}^{a+2} e^{-(x-2)^2} dx$$

$$\Rightarrow f'(a) = e^{-(a)^2} - e^{-(a-4)^2} = 0$$

$$\Rightarrow e^{-a^2} - e^{-(a-4)^2} \Rightarrow -a^2 = -(a-4)^2$$

$$\Rightarrow a = 2$$

$$f''(a) = e^{-a^2}(-2a) - e^{-(a-4)^2}(-2(a-4))$$

$$= -2ae^{-a^2} + 2(a-4)e^{-(a-4)^2}$$

$$\Rightarrow f''(2) = -4e^{-4} - 4e^{-4}$$

$$\Rightarrow f''(2) \text{ is negative hence at } a = 2$$

$$\Rightarrow f(a) \text{ is maximum}$$

$$133. f'(x) \text{ has } D > 0 \quad \Rightarrow 36 - 4k^2 > 0$$

$$\Rightarrow k \in (-3, 3)$$

$$\text{Ans. } 3$$

$$134. 2x^2 - 2x + 1 = 2 \left(x - \frac{1}{2} \right)^2 + \frac{1}{2}$$

$$\therefore \text{ minimum value of } 2x^2 - 2x + 1 \text{ is } \frac{1}{2} \text{ and minimum value of } \sin^2 x \text{ is } 0$$

$$\therefore \text{ minimum value of } e^{(2x^2 - 2x + 1)\sin^2 x} \text{ is } 1$$

$$135. a = b$$

$$136. \sin 3x = \cos 4x$$

$$\Rightarrow \cos 4x = \cos \left(\frac{\pi}{2} - 3x \right) \Rightarrow \cos 4x = \sin x \pm \left(\frac{\pi}{2} - 3x \right)$$

$$137. T_r = \cot^{-1}(2r^2)$$

$$S = \sum_{r=1}^{\infty} \cot^{-1}(2r^2) = \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{1}{2r^2} \right)$$

$$= \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{(2r+1) - (2r-1)}{1 + (2r+1)(2r-1)} \right)$$

$$= \sum_{r=1}^{\infty} \tan^{-1}(2r+1) - \tan^{-1}(2r-1)$$

$$= (\tan^{-1}3 - \tan^{-1}1) + (\tan^{-1}5 - \tan^{-1}3) + (\tan^{-1}7 - \tan^{-1}5) + \dots + \tan^{-1}\infty$$

$$S = -\frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4} \quad \text{Given } S = \frac{\pi}{4} = k\pi \Rightarrow k = \frac{1}{4}$$

$$[k] = 0$$

$$138. f'(x) = -2x^2 \cdot e^{-2x} + 2x \cdot e^{-2x}$$

$$= 2(-x^2 + x) e^{-2x}$$

$$f'(x) = 0 \text{ at } x = 1 \text{ and } f'(x) > 0 \text{ in } (0, 1) \text{ and } f'(x) < 0 \text{ in } (1, \infty)$$

$$f(x) \text{ is maximum at } x = 1$$

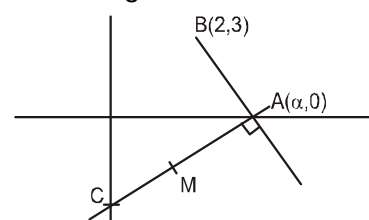
$$f(1) = \frac{1}{e^2} = \ell \quad \Rightarrow \quad \frac{1}{\ell} = e^2$$

$$[\ell^{-1}] = 7$$

$$139. \text{ Slope of AB} = \frac{3}{2-\alpha}$$

Equation of AC

$$y - 0 = \frac{(2-\alpha)}{3}(x-\alpha)$$



$$\therefore C \left(0, \frac{\alpha(2-\alpha)}{3} \right)$$

Let $M(h,k) \Rightarrow a = 2h, 2k = \frac{\alpha(2-\alpha)}{3}$

$\Rightarrow 6k = 2h(2-2h)$
 $3y = x(2-2x)$
 $2x^2 - 2x + 3y = 0$
 $a = 2, b = 3$
 $a + b = 5$

140. $x^2 + (a-b)x + (1-a-b) = 0$

$\therefore D > 0$
 $\Rightarrow (a-b)^2 - 4 \times 1 \times (1-a-b) > 0$
 $a^2 + b^2 - 2ab - 4 + 4a + 4b > 0$
 $\therefore b^2 + 2b(2-a) + (a^2 + 4a - 4) > 0$
 $\therefore 4(2-a)^2 - 4 \times 1 \times (a^2 + 4a - 4) < 0$
 $4 + a^2 - 4a - a^2 - 4a + 4 < 0$
 $8a - 8 > 0$
 $a > 1$

141. $\therefore a_n = \frac{20}{n}$ and $d = \frac{1}{20}$

$\therefore \Delta = \begin{vmatrix} 20 & \frac{20}{2} & \frac{20}{3} \\ \frac{20}{4} & \frac{20}{5} & \frac{20}{6} \\ \frac{20}{7} & \frac{20}{8} & \frac{20}{9} \end{vmatrix} = \frac{(20)^3}{4 \times 7} \begin{vmatrix} 1 & 1/2 & 1/3 \\ 1 & 4/5 & 2/3 \\ 1 & 7/8 & 7/9 \end{vmatrix}$

applying $R_1 \rightarrow R_1 - R_2$ and $R_2 \rightarrow R_2 - R_3$

$\therefore \Delta = \frac{(20)^3}{4 \times 7} \begin{vmatrix} 0 & -3/10 & -1/3 \\ 0 & -3/40 & -1/9 \\ 1 & 7/8 & 7/9 \end{vmatrix}$

$\therefore \Delta = \frac{50}{21} \quad \therefore [\Delta] = 2 \text{ Ans.}$

142. Using the idea of the differentiation of determinant, we get

$f'(x) = \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \cos(\beta-\gamma) & \cos(\gamma-\alpha) & \cos(\alpha-\beta) \end{vmatrix} +$

$\begin{vmatrix} \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ -\sin(x+\alpha) & -\sin(x+\beta) & -\sin(x+\gamma) \\ \cos(\beta-\gamma) & \cos(\gamma-\alpha) & \cos(\alpha-\beta) \end{vmatrix}$

$+ \begin{vmatrix} \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ 0 & 0 & 0 \end{vmatrix}$

$= 0 + 0 + 0$

$f'(x) = 0$ for all x

$f(x) = a$ constant

But $f(9) = \lambda$

$\therefore f(x) = \lambda$ for all x .

$\sum_{k=1}^9 f(k) = 9\lambda$

143. Put $\log_{10} x = t$

$[t^2 + t] = 2t + 1$

$2t + 1 \leq t^2 + t < 2t + 2$

and $2t \in I$

$t \in \phi$

144. Let $I = \int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx$ (1)

$\Rightarrow I = \int_0^1 \frac{\sin^{-1} \sqrt{1-x}}{(1-x)^2 - (1-x) + 1} dx = \int_0^1 \frac{\sin^{-1} \sqrt{1-x}}{x^2 - x + 1} dx$

$I = \int_0^1 \frac{\cos^{-1} \sqrt{x}}{x^2 - x + 1} dx$ (2)

$\therefore \sin^{-1} x = \cos^{-1} \sqrt{1-x^2}$

(1) + (2) we get

$\therefore \sin^{-1} \sqrt{1-x} = \cos^{-1} \sqrt{x}$

$2I = \int_0^1 \frac{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}}{x^2 - x + 1} dx$

$I = \frac{\pi}{4} \int_0^1 \frac{1}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx$

$\Rightarrow I = \frac{\pi}{4} \times \frac{1}{\sqrt{3}} \left(\tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) \right)_0^1$

$\Rightarrow I = \frac{\pi}{2\sqrt{3}} \left[\frac{\pi}{6} + \frac{\pi}{6} \right] = \frac{\pi}{2\sqrt{3}} \times \frac{\pi}{3} = \frac{n^2}{\sqrt{108}}$

$\therefore n = 108$

$\frac{n}{27} = 4 \text{ Ans.}$

145. $K\vec{r} + \vec{r} \times \vec{a} = \vec{b}$ (i)

pre cross (i) with $\vec{a}$ both side

$\Rightarrow K(\vec{a} \times \vec{r}) + \vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}$

$\Rightarrow K(\vec{a} \times \vec{r}) + |\vec{a}|^2 \vec{r} - (\vec{a} \cdot \vec{r}) \vec{a} = \vec{a} \times \vec{b}$

$\Rightarrow K(K\vec{r} - \vec{b}) + |\vec{a}|^2 \vec{r} - (\vec{a} \cdot \vec{r}) \vec{a} = \vec{a} \times \vec{b}$

(using equation (i))

$\Rightarrow \vec{r}(K^2 + |\vec{a}|^2) = \vec{a} \times \vec{b} + K\vec{b} + (\vec{a} \cdot \vec{r}) \vec{a}$ (ii)

Dot (i) with $\vec{a}$ both side

$K(\vec{a} \cdot \vec{r}) = \vec{a} \cdot \vec{b}$

$\Rightarrow \vec{a} \cdot \vec{r} = \frac{\vec{a} \cdot \vec{b}}{K}$ (iii)

by (ii) & (iii)

$\vec{r} = \frac{1}{K^2 + |\vec{a}|^2} \left(K\vec{b} + \frac{\vec{a} \cdot \vec{b}}{K} \vec{a} + \vec{a} \times \vec{b} \right)$

$\Rightarrow \ell + m + n = 3$

146. For defining $f(x)$

$$(16-x), (20-3x) \in \mathbb{N}; (2x-1), (4x-5) \in \mathbb{W}$$

$$(16-x) \geq (2x-1); (20-3x) \geq (4x-5)$$

$$\text{Hence } x = 2, 3 \text{ } x = 2, 3$$

$$\text{when } x = 2, f(x) = {}^{14}C_3 + {}^{14}C_3 = 728$$

$$\text{when } x = 3, f(x) = {}^{13}C_5 + {}^{11}C_4 = 1617$$

$$\text{Hence sum } 728 + 1617 = 2345$$

$$147. z = \omega \Rightarrow (z - z^2 + 2z^3)(2 - z + z^2) = (\omega - \omega^2 + 2)(2 - \omega + \omega^2)$$

$$= |2 - \omega + \omega^2|^2 = |2 - i\sqrt{3}|^2 = 7$$

148. On expanding the determinant, we get

$$7\sin 3\theta + 14\cos 2\theta - 14 = 0$$

$$\Rightarrow \sin 3\theta - 2(1 - \cos 2\theta) = 0 \Rightarrow 3\sin \theta - 4\sin^3 \theta - 4\sin^2 \theta = 0$$

$$\Rightarrow \sin \theta (2\sin \theta - 1)(2\sin \theta + 3) = 0 \Rightarrow \sin \theta = 0 \text{ or } \sin \theta = \frac{1}{2}$$

$$\Rightarrow \theta = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6} \quad \therefore 5 \text{ solutions}$$

149. $\therefore 1$ lies between the roots

$$\text{hence } f(1) < 0$$

$$\Rightarrow 3 - 3\sin \theta - 2\cos^2 \theta < 0 \Rightarrow 2\sin^2 \theta - 3\sin \theta + 1 < 0$$

$$\Rightarrow \frac{1}{2} < \sin \theta < 1 \Rightarrow 2n\pi + \frac{\pi}{6} < \theta < \frac{5\pi}{6} + 2n\pi, n \in \mathbb{I}$$

$$\text{In } [0, 2\pi], \theta \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right) - \left\{\frac{\pi}{2}\right\}$$

$$\theta = 1, 2$$

$$\text{Required sum } 1 + 2 = 3$$

$$150. \text{Volume} = |[3(\vec{b} \times \vec{c}) 2(\vec{a} \times \vec{b}) 4(\vec{c} \times \vec{a})]| = 18$$

$$[\vec{a} \vec{b} \vec{c}]^2 = \frac{18}{24}$$

$$[\vec{a} \vec{b} \vec{c}] = \pm \frac{\sqrt{3}}{2} \quad \dots(1)$$

$$[\vec{a} \vec{b} \vec{c}]$$

$$= \begin{vmatrix} 1 + \sin \theta & \cos \theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix}$$

$$(R_1 \rightarrow R_1 + R_2 + R_3)$$

$$= -\frac{\sqrt{3}}{2} \cos 3\theta$$

$$\cos 3\theta = \pm 1$$

$$3\theta = \pi$$

$$1 \text{ values of } \theta$$

$$151. \therefore f(x) = \int x^{\sin x} (1 + x \cos x \cdot \ln x + \sin x) dx$$

$$= \int \left(x^{\sin x} + x \cdot x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right) \right) dx$$

$$\text{Let } F(x) = x^{\sin x}$$

$$\therefore F'(x) = \left(\sin x \cdot \frac{1}{x} + \cos x \cdot \ln x \right) x^{\sin x}$$

$$\therefore f(x) = \int (F(x) + xF'(x)) dx$$

$$\Rightarrow f(x) = x F(x) + c$$

$$\therefore f\left(\frac{\pi^2}{4}\right) = \left(\frac{\pi}{2}\right) \therefore f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} F\left(\frac{\pi}{2}\right) + c$$

$$\frac{\pi^2}{4} = \frac{\pi}{2} \left(\frac{\pi}{2}\right)^1 + c \Rightarrow c = 0$$

$$\therefore f(x) = xF(x)$$

$$\therefore f(\pi) = x F(\pi) = \pi(\pi)^0$$

$$\therefore |\cos(f(\pi))| = |\cos \pi| = 1$$

$$152. [x + a] = [x] + a; \text{ if } a \in \mathbb{I}$$

$$\text{Minimum if 'a' for } \sin\left(\frac{2[x]\pi}{3}\right) \text{ is } 3$$

$$\text{and for } \cos\left(\frac{3[x]\pi}{2}\right) \text{ is } 4$$

$$T = 12$$

$$153. f'(x) = 3 \left(x^2 + \frac{8x}{3} + \frac{16}{9} \right) + \lambda - \frac{16}{3} = 3 \left(x + \frac{4}{3} \right)^2 + \lambda - \frac{16}{3}$$

$$\text{now } -2 < x < -\frac{2}{3} \Leftrightarrow -2 + \frac{4}{3} < x + \frac{4}{3} < \frac{2}{3}$$

$$\Leftrightarrow -\frac{2}{3} < \left(x + \frac{4}{3} \right) < \frac{2}{3} \Leftrightarrow \left(x + \frac{4}{3} \right)^2 < \frac{4}{9}$$

$$\therefore f'(x) < 3 \cdot \frac{4}{9} + \lambda - \frac{16}{3}$$

$$\therefore \lambda \leq 4$$

$$\therefore \text{greatest value of } \lambda \text{ is } 4$$

$$154. f(x) = \ell n \left(2 \sin x + \tan x - \frac{3x}{\pi} + 1 \right), x \in \left[\frac{\pi}{6}, \frac{\pi}{3} \right]$$

$$\text{Let } g(x) = 2 \sin x + \tan x - \frac{3x}{\pi} + 1, x \in \left[\frac{\pi}{6}, \frac{\pi}{3} \right]$$

$$g'(x) = 2 \cos x + \sec^2 x - \frac{3}{\pi} > 0, \forall x \in \left[\frac{\pi}{6}, \frac{\pi}{3} \right]$$

$$\therefore g(x) \text{ is strictly increasing}$$

$$\text{range of } g(x) \text{ is } \left[g\left(\frac{\pi}{6}\right), g\left(\frac{\pi}{3}\right) \right]$$

$$\therefore g(x) \left[g\left(\frac{\pi}{6}\right), g\left(\frac{\pi}{3}\right) \right]$$

$$\text{i.e. } \left[\frac{1}{\sqrt{3}} + \frac{3}{2}, 2\sqrt{3} \right]$$

$$\text{range of } f(x) \text{ is } \left[\ell n \left(\frac{3}{2} + \frac{1}{\sqrt{3}} \right), \ell n(2\sqrt{3}) \right]$$

$$\therefore f(x) = \left[\ln\left(\frac{3}{2} + \frac{1}{\sqrt{3}}\right), \ln(2\sqrt{3}) \right]$$

$$\therefore a+b = \ln\left(\frac{3}{2} + \frac{1}{\sqrt{3}}\right), \ln(2\sqrt{3})$$

$$= \ln(3\sqrt{3} + 2)$$

$$[a+b] = 1$$

155. $+k, -x^2 + kx + 2$ is $-ve$ for infinitely many integers.

156. The triangle has circumcentre at origin and its orthocentre lying on the circumcircle.

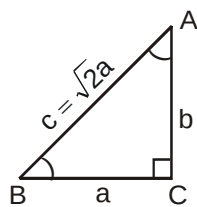
$$157. R(a+b) = c\sqrt{ab}$$

$$R(a+b) = 2R \sin C \sqrt{ab}$$

$$\frac{a+b}{2\sqrt{ab}} = \sin C$$

$$\text{we know } \frac{a+b}{2\sqrt{ab}} \geq 1 \text{ and } \sin C \leq 1$$

$$\Rightarrow a+b = 2\sqrt{ab} \quad \text{and } \sin C = 1$$



$$\Rightarrow a = b \text{ and } c = 90^\circ$$

$$r = \frac{\Delta}{s} = \frac{\frac{1}{2}aa}{\frac{a+a+a\sqrt{2}}{2}} = \frac{a}{2+\sqrt{2}}$$

$$158. f''(x) - f'(x) = e^x$$

$$\text{Put } f'(x) = v$$

$$\frac{dv}{dx} + v(-1) = e^x \Rightarrow ve^{-x} = \int e^x \cdot e^{-x} dx$$

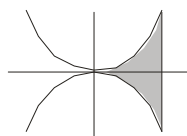
$$\Rightarrow ve^{-x} = x + C_1, f'(0) = 1 \Rightarrow C_1 = 1$$

$$f'(x) = xe^x + e^x$$

$$f(x) = xe^x + C_2 \Rightarrow f(0) = 0 \Rightarrow C_2 = 0$$

$$\Rightarrow f(x) = xe^x \Rightarrow f(2) = 2e^2$$

$$\ln\left(\frac{(f(2))^2}{4}\right) = 4$$



159.

$$\text{Required area} = 2 \int_0^1 \sin^{-1} x dx$$

$$= 2 \left[\left(x \sin^{-1} x \right)_0^1 - \int_0^1 \frac{1 \cdot x}{\sqrt{1-x^2}} dx \right]$$

$$= 2 \left[\frac{\pi}{2} - \int_0^1 (1-x^2)^{-\frac{1}{2}} x dx \right]$$

$$= \pi + \int_0^1 (1-x^2)^{-\frac{1}{2}} (-2x) dx$$

$$= \pi + 2 \left(\sqrt{1-x^2} \right)_0^1$$

$$= \pi + 2(0-1) = \pi - 2$$

$$\therefore a = 1, b = -2 \quad \therefore a - b = 3$$

$$160. z_1 z_2 + z_2 z_3 + z_3 z_1 = z_1 z_2 z_3 \left(\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right) = 1$$

$$\left(\frac{|z_1|^2}{z_1} + \frac{|z_2|^2}{z_2} + \frac{|z_3|^2}{z_3} \right) = \bar{z}_1 + \bar{z}_2 + \bar{z}_3$$

$$= (\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = 1$$

$\therefore$ The cubic equation with roots z_1, z_2 and z_3 will be

$$z^3 - (\Sigma z_1)z^2 + (\Sigma z_1 z_2)z - z_1 z_2 z_3 = 0$$

$$\Rightarrow z^3 - (1)z^2 + (1)z - 1 = 0$$

$$\Rightarrow (z-1)(z^2+1) = 0$$

$$\Rightarrow z = 1, i, -i$$

$$\therefore \text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$$

$$\therefore z_1 = -i, z_2 = 1, z_3 = i$$

$$\therefore |z_1 + z_2^2 + z_3^3| = |-i + 1 - i| = |1 - 2i| = \sqrt{5} \quad \therefore [\sqrt{5}] = 2$$

$$161. \text{ Let } f = \frac{x}{x^2 - 5x + 9}$$

$$f \in \left[\frac{-1}{11}, 1 \right]$$

$$\therefore f^2 + f + k < 0 \quad \forall f \in \left[\frac{-1}{11}, 1 \right]$$

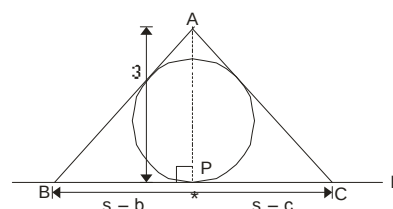
$$\Rightarrow t \left[\frac{-1}{11} \right] < 0 \text{ \& } f(1) < 0$$

$$162. \therefore r = \frac{\Delta}{s} = 1 \Rightarrow \Delta = s$$

$$(PB)(PC) = (s-b)(s-c)$$

$$= \frac{s(s-a)(s-b)(s-c)}{s(s-a)}$$

$$= \frac{\Delta^2}{s(s-a)} = \frac{\Delta}{s} \cdot \frac{\Delta}{s-a}$$



$$= r \cdot \frac{\Delta}{\Delta - a} \quad \because r = 1 \text{ and } s = \Delta$$

$$(PB)(PC) = \frac{\Delta}{\Delta - a} \quad \dots\dots (1)$$

$$\Delta = \frac{1}{2} (BC) (\text{height})$$

$$\Delta = \frac{1}{2} (s - b + s - c) (3) = \frac{3a}{2}$$

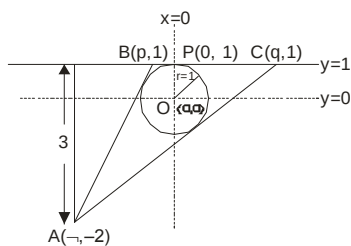
$$\therefore \text{from equation (1)} (PB)(PC) = \frac{3a/2}{\frac{3a}{2} - a} = 3$$

Alter

Let circle be $x^2 + y^2 = 1$

Pair of tangents from $A(\alpha, -2)$ is

$$(x^2 + y^2 - 1)(\alpha^2 + 3) = (\alpha x - 2y - 1)^2$$



$$x^2(\alpha^2 + 3) = (\alpha x - 3)^2 \quad [\text{put } y = 1 \text{ for points B and C}]$$

$$\Rightarrow 3x^2 + 6\alpha x - 9 = 0$$

$$\Rightarrow pq = -3 \quad \Rightarrow (PB)(PC) = 3$$

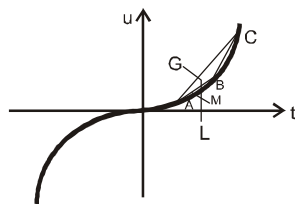
163. Let $f(t) = t^3$, $u = f(t)$

$A(x, x^3)$, $B(y, y^3)$, $C(z, z^3)$

Centroid G lies inside $\triangle ABC$

$GL \geq ML$

$$\frac{x^3 + y^3 + z^3}{3} \geq f\left(\frac{x + y + z}{3}\right)$$



$$\frac{x^3 + y^3 + z^3}{3} \geq \frac{(x + y + z)^3}{27} \quad \dots\dots (1)$$

$$\text{Given } \frac{(x + y + z)^2}{2} = \frac{x^3 + y^3 + z^3}{3} \quad \dots\dots (2)$$

$$\Rightarrow \frac{(x + y + z)^2}{2} \geq \frac{(x + y + z)^3}{27} \Rightarrow (x + y + z) \leq \frac{27}{2}$$

If $x + y + z = 13$, x, y, z will not satisfy (2)

If $x + y + z = 12$, $x = 3, y = 4, z = 5$ will not satisfy (2)

Hence maximum value is 12.

Alternate :

$$\frac{a\alpha + b\beta + c\delta}{3} \geq \left(\frac{a + b + c}{3}\right) \left(\frac{\alpha + \beta + \delta}{3}\right)$$

$$\frac{x^3 + y^3 + z^3}{3} \geq \left(\frac{x + y + z}{3}\right) \left(\frac{x^2 + y^2 + z^2}{3}\right) \quad \dots\dots (1)$$

$$\frac{x^2 + y^2 + z^2}{3} \geq \left(\frac{x + y + z}{3}\right) \left(\frac{x + y + z}{3}\right) \quad \dots\dots (2)$$

$$\Rightarrow \frac{x^3 + y^3 + z^3}{3} \geq \left(\frac{x + y + z}{3}\right)^3 \quad \dots\dots (3)$$

$$\text{Given } \frac{(x + y + z)^2}{2} = \frac{x^3 + y^3 + z^3}{3} \quad \dots\dots (4)$$

$$(3), (4) \Rightarrow x + y + z \leq \frac{27}{2}$$

If $x + y + z = 13$, x, y, z will not satisfy (4)

If $x + y + z = 12$, then $x = 3, y = 4, z = 5$ will not satisfy (4)

Hence maximum value is 12.

164. given that $f(2010) \in (2008, 2012)$

$$\Rightarrow \frac{f(2010)}{10} \in (200.8, 201.2)$$

$$\Rightarrow \left[\frac{f(2010)}{10}\right] \in \{200, 201\}$$

$$\text{Now } x - f(x) = 20 \left[\frac{x}{20}\right] - 10 \left[\frac{f(x)}{10}\right]$$

Put $x = 2010$

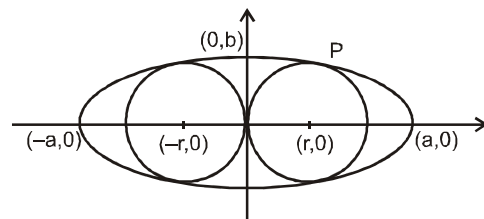
$$\begin{aligned} 2010 - f(2010) &= 20 \times \left[\frac{2010}{20}\right] - 10 \times (200 \text{ or } 201) \\ &= 20 \times 100 - 10 \times (200 \text{ or } 201) \\ &= 2000 - [2000 \text{ or } 2010] \\ &= 0 \text{ or } -10 \end{aligned}$$

$\Rightarrow$ not possible

165. $(x - r)^2 + y^2 = r^2$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$P(\cos\theta, \sin\theta) \Rightarrow (\cos\theta - r)^2 + (\sin\theta)^2 = r^2 \quad \dots\dots (1)$$



Normal passes through $(r, 0)$

$$\frac{a \cdot r}{\cos\theta} - \frac{b \cdot 0}{\sin\theta} = a^2 - b^2 \Rightarrow \cos\theta = \frac{r}{ae^2}$$

$$\text{put in (1)} b^2 = \frac{r^2}{e^2} \Rightarrow b = \frac{r}{e}$$

$$\text{Area} = \pi ab = \frac{\pi b^2}{\sqrt{1 - e^2}} = \frac{\pi r^2}{e^2 \sqrt{1 - e^2}} \quad \text{differentiable w.r.t. } e$$

$$\Rightarrow e = \sqrt{\frac{2}{3}}$$

166. $y = x^3$ (1)

$$\text{let } P(h, k) \Rightarrow k = h^3 \Rightarrow P(h, h^3)$$

$$y = x^3 \Rightarrow \frac{dy}{dx} = 3x^2$$

$$\Rightarrow \text{Slope of tangent at } P = 3h^2$$

equation of tangent at $P(h, k)$

$$y - h^3 = 3h^2(x - h)$$

$$\Rightarrow y = 3h^2x - 2h^3 \text{(2)}$$

Solving 1 & 2

$$x^3 - 3h^2x + 2h^3 = 0$$

$$(x - h)^2(x + 2h) = 0$$

$$\Rightarrow x = -2h \text{ \& } y = -8h^3 \Rightarrow \text{point}$$

$$P_1(-2h, -8h^3)$$

Now tangent at $P_1((-2h), (-2h)^3)$ will meet the curve at

$$P_2(-2(-2h), -8(-2h)^3)$$

$$\text{i.e. } P_2(4h, 64h^3)$$

$$\text{similarly } P_3(-8h, -512h^3)$$

Area

$$\Delta P_1P_2P_3 = \frac{1}{2} \begin{vmatrix} -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \\ -8h & -512h^3 & 1 \end{vmatrix} = 8h^4 \begin{vmatrix} 1 & 1 & 1 \\ -2 & -8 & 1 \\ 4 & 64 & 1 \end{vmatrix}$$

$$\text{Area of } \Delta PP_1P_2 = \frac{1}{2} \begin{vmatrix} h & h^3 & 1 \\ -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \end{vmatrix} = \frac{h^4}{2} \begin{vmatrix} 1 & 1 & 1 \\ -2 & -8 & 1 \\ 4 & 64 & 1 \end{vmatrix}$$

$$\Rightarrow \frac{\text{area of } \Delta P_1P_2P_3}{\text{area of } \Delta PP_1P_2} = 16$$

167. Replace n by $n-1$

$$\frac{1}{f(0)+f(1)} + \frac{1}{f(1)+f(2)} + \dots + \frac{1}{f(n-1)+f(n)} = f(n)$$

put in (b)

$$f(n) + \frac{1}{f(n)+f(n+1)} = f(n+1)$$

$$\Rightarrow 1 = f^2(n+1) - f^2(n)$$

$$\Rightarrow 1 = f^2(1) - f^2(0)$$

$$1 = f^2(2) - f^2(1)$$

$\vdots$

$$1 = f^2(100) - f^2(99)$$

Adding

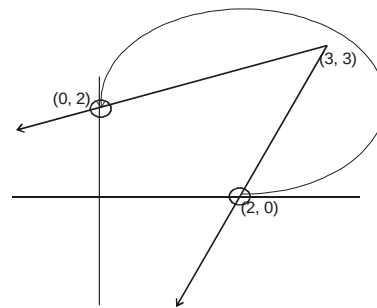
$$100 = f^2(100) - f^2(0) \Rightarrow f(0) = 0$$

$$\text{Also } n = f^2(n) - f^2(0) \Rightarrow f(n) = \sqrt{n}$$

$$f(9801) = 99$$

168. First note that $(3, 3)$ is in the interior of the arc, as

$$\arg\left(\frac{3+3i-2}{3+3i-2i}\right) = \arg\left(\frac{1+3i}{3+i}\right)$$



$$= \tan^{-1}3 - \tan^{-1}\frac{1}{3}$$

$$= \tan^{-1}\left(\frac{3-1/3}{2}\right)$$

$$= \tan^{-1}\left(\frac{8}{6}\right)$$

$$= \tan^{-1}\frac{4}{3} > \frac{\pi}{11}$$

Now : From the figure it is clear that if rays don't intersect the circle.

$$\text{then } \alpha \in \left[-\pi + \tan^{-1}\frac{1}{3}, -\pi + \tan^{-1}3\right]$$

$$\Rightarrow |a+b| = \left|-2\pi + \tan^{-1}3 + \tan^{-1}\frac{1}{3}\right|$$

$$= \left|-2\pi + \frac{\pi}{2}\right|$$

$$= \frac{3\pi}{2}$$

$$\Rightarrow 10|a+b| = 15\pi$$

$$\Rightarrow [10|a+b|] = [15\pi] = 47$$

$$169. f(x) = \ell_n \left(2 \sin x + \tan x - \frac{3x}{\pi} + 1\right) \quad x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$$

$$\text{consider } g(x) = 2 \sin x + \tan x - \frac{3x}{\pi} + 1$$

$$g'(x) = 2 \cos x + \sec^2 x - \frac{3}{\pi} > 0 \text{ for all } x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$$

$\therefore g(x)$ is monotonically increasing

$$g\left(\frac{\pi}{6}\right) = 2 \sin \frac{\pi}{6} + \tan \frac{\pi}{6} - \frac{3}{\pi} \cdot \frac{\pi}{6} + 1$$

$$= 1 + \frac{1}{\sqrt{3}} - \frac{1}{2} + 1$$

$$\therefore f\left(\frac{\pi}{6}\right) = \ln\left(\frac{3}{2} + \frac{1}{\sqrt{3}}\right)$$

$$f\left(\frac{\pi}{3}\right) = \ln(\sqrt{3} + \sqrt{3} - 1 + 1) = \ln 2\sqrt{3}$$

$$\therefore a + b = \ln\left(\frac{3}{2} + \frac{1}{\sqrt{3}}\right) + \ln 2\sqrt{3}$$

$$= \ln\left(\left(\frac{3}{2} + \frac{1}{\sqrt{3}}\right) 2\sqrt{3}\right) = \ln(3\sqrt{3} + 2) = \ln 7.2$$

$$\therefore 1 < a + b < 2$$

$$\therefore [a + b] = 1$$

170. $g'(x) = x[f'(2x^2 - 1) - f'(1 - x^2)]$

Case-I $x > 0$

For $g(x)$ to be strictly increasing $f'(2x^2 - 1) - f'(1 - x^2) > 0$

$$\Rightarrow f'(2x^2 - 1) > f'(1 - x^2) \Rightarrow 2x^2 - 1 > 1 - x^2 \Rightarrow 3x^2 > 2$$

$$\Rightarrow x^2 > \frac{2}{3} \quad \Rightarrow \quad x > \sqrt{\frac{2}{3}}$$

($\because x > 0$)

Case-II $x < 0$

For $g(x)$ to be strictly increasing $f'(2x^2 - 1) - f'(1 - x^2) < 0$

$$\Rightarrow f'(2x^2 - 1) < f'(1 - x^2) \Rightarrow 2x^2 - 1 < 1 - x^2 \Rightarrow 3x^2 < 2$$

$$\Rightarrow x^2 < \frac{2}{3} \Rightarrow -\sqrt{\frac{2}{3}} < x < \sqrt{\frac{2}{3}} \Rightarrow -\sqrt{\frac{2}{3}} < x < 0$$

Thus $g(x)$ strictly increases on the interval $\left(-\sqrt{\frac{2}{3}}, 0\right) \cup \left(\sqrt{\frac{2}{3}}, \infty\right)$

171. $3 + 4 \cos x = t \quad \Rightarrow \quad -4 \sin x dx = dt$

$$I = -\frac{1}{8} \int \frac{t-3}{t^3} dt$$

$$= -\frac{1}{8} \int \frac{1}{t^2} - \frac{3}{t^3} dt$$

$$= \frac{1}{8} \left(\frac{1}{t} - \frac{3}{2t^2} \right) + e$$

$$= \frac{2t-3}{16t^2} = \frac{8\cos x + 3}{16(3+4\cos x)^2} + e$$

$$\Rightarrow a = 3, b = 8, c = 16, d = 2$$

172. $f(x) = x^3 - x^2 + 100x + 1001$

$$f'(x) = 3x^2 - 2x + 100 > 0 \quad \forall x \in \mathbb{R}$$

$\therefore f(x)$ is increasing (strictly)

$$\therefore f\left(\frac{1}{1999}\right) > f\left(\frac{1}{2000}\right)$$

$$f(x+1) > f(x-1)$$

173.

	a_1	
	a_2	
a_5	a_3	a_6
	a_4	

$$(a_1 + a_2 + a_3 + a_4) = 21,$$

$$a_5 + a_3 + a_6 + a_7 = 21,$$

$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 = 35$$

$$\Rightarrow a_3 = 7, a_1 + a_2 + a_4 = 14, a_5 + a_6 + a_7 = 14$$

$$a_1, a_2, a_4 \in \{2, 4, 8\}, a_5, a_6, a_7 \in \{3, 5, 6\}$$

or

$$a_1, a_2, a_4 \in \{3, 5, 6\}, a_5, a_6, a_7 \in \{2, 4, 8\}$$

$$\text{Required number of ways} = (3! \times 3!) + (3! \times 3!) = 72$$

174. $f(x) = \sin(\ell n x) - \cos(\ell n x), x > 0$

$$= \sqrt{2} \sin\left(\ell n x - \frac{\pi}{4}\right)$$

$$f'(x) = \frac{\sqrt{2}}{x} \cos\left(\ell n x - \frac{\pi}{4}\right)$$

$$= \frac{\sqrt{2}}{x} \sin\left(\frac{\pi}{4} + \ell n x\right)$$

$$f'(x) \geq 0 \quad \Rightarrow \quad \sin\left(\frac{\pi}{4} + \ell n x\right) \geq 0$$

$$\Rightarrow 0 \leq \frac{\pi}{4} + \ell n x \leq \pi \Rightarrow -\frac{\pi}{4} \leq \ell n x \leq \frac{3\pi}{4}$$

$$\Rightarrow \mu - \lambda = \pi \quad \Rightarrow \quad -5 \cos(\mu - \lambda) = 5$$

175. $\int \frac{\sin 3x \cos 5x - \sin 3x \cos 4x}{2 \sin 3x \cos 3x + \sin 3x} dx$

$$= \frac{1}{2} \int \frac{(\sin 8x - \sin 2x) - (\sin 7x - \sin x)}{\sin 6x + \sin 3x} dx$$

$$= \frac{1}{2} \int \frac{2 \sin \frac{9x}{2} \cos \frac{7x}{2} - 2 \sin \frac{9x}{2} \cos \frac{5x}{2}}{2 \sin \frac{9x}{2} \cos \frac{3x}{2}} dx$$

$$= \frac{1}{2} \int -\frac{2 \sin 3x \sin \frac{x}{2}}{\cos \frac{3x}{2}} dx = -\int 2 \sin \frac{3x}{2} \sin \frac{x}{2} dx$$

$$= \int (\cos 2x - \cos x) dx = \frac{\sin 2x}{2} - \sin x + C$$

$$a = 2$$

176. $f'(x) = 12 \cos^3 x (-\sin x) + 30 \cos^2 x (-\sin x) + 12 \cos x (-\sin x)$
 $= -3 \sin 2x (2 \cos^2 x + 5 \cos x + 2)$
 $= -3 \sin 2x (2 \cos x + 1) (\cos x + 2)$

When, $f'(x) = 0 \Rightarrow \sin 2x = 0 \Rightarrow x = 0, \frac{\pi}{2}, \pi$

or, $2 \cos x + 1 = 0 \Rightarrow x = \frac{2\pi}{3}$

or, $\cos x + 2 = 0$

(not possible)

Sign scheme for $f'(x)$ in $[0, \pi]$ is as below.

$$\begin{array}{ccccccc} 0 & -ve & \pi/2 & +ve & 2\pi/3 & -ve & \pi \end{array}$$

$$f(x) \text{ decreases on } \left(0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \pi\right)$$

$$\text{and increases on } \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$$

177. Number of selections of 7 digits out of the digit 1,2,3,...,9 = 9C_7
 Number of digits out of these 7 selected digits excluding the greatest digit = 6

These 6 digits can be divided in two group each having 3

$$\text{digits} = \frac{6!}{3!3!2!} = {}^6C_3 \times \frac{1}{2!}$$

But the 3 digits on one side can go on the other side

$\therefore$ Required number of ways =

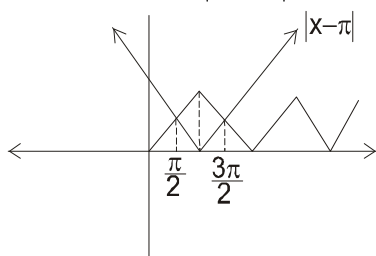
$${}^9C_7 \cdot {}^6C_3 \cdot \frac{2}{2!} = {}^9C_7 \cdot {}^6C_3 = {}^9C_2 \cdot {}^6C_3 = 720$$

178. $b = 40 - 12\pi$

$c = 3\pi - 10$

$$\cos^{-1} \cos x \geq |x + a - b - c|$$

$$\Rightarrow \cos^{-1} \cos x \geq |x - \pi|$$



Integral value of x satisfying given equations = 2,3,4

179. $(n^3 - 1)^3 = (n + 1)^3(n - 1)^3$
 $(n + 1)^3 - (n - 1)^3 = 6n^2 + 2$

$$\frac{3n^2 + 1}{(n^2 - 1)^3} = \frac{1}{2} \frac{6n^2 + 2}{(n^2 - 1)^3} = \frac{1}{2} \left(\frac{(n + 1)^3 - (n - 1)^3}{(n + 1)^3(n - 1)^3} \right)$$

$$= \frac{1}{2} \left(\frac{1}{(n - 1)^3} - \frac{1}{(n + 1)^3} \right)$$

S =

$$\frac{1}{2} \left[\left(\frac{1}{1^3} - \frac{1}{3^3} \right) + \left(\frac{1}{2^3} - \frac{1}{4^3} \right) + \left(\frac{1}{3^3} - \frac{1}{5^3} \right) + \left(\frac{1}{4^3} - \frac{1}{6^3} \right) + \dots \right]$$

$$= \frac{1}{2} \left(1 + \frac{1}{8} \right) = \frac{9}{16} \Rightarrow 16S = 9 \Rightarrow 16 = \frac{9}{S}$$

180. $\frac{A}{B} = \lim_{n \rightarrow \infty} \left(\frac{(2n + 1)(2n + 2)(2n + 3) \dots (2n + n)}{(n + 1)(n + 2)(n + 3) \dots (n + n)} \right)^{1/n}$

$$\ln(A/B) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ln \left(\frac{2 + \frac{r}{n}}{1 + r/n} \right)$$

$$\int_0^1 \ln \left(\frac{(2 + x)}{(1 + x)} \right) dx = \frac{27}{16}$$

$\therefore A = 27$ & $B = 16$

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